Homework 1.1: Can an AI Rediscover Kepler's Orbit of Mars?

Mathematical Foundations of Generative AI

Project Goal

The goal of this project was to use a large language model (LLM) as a scientific collaborator rather than as an answer generator. The project built a synthetic version of the observational problem faced by Kepler: generate noisy measurements of Mars from Uraniborg, reconstruct heliocentric Mars positions without inserting the hidden Mars orbital parameters into the reconstruction algorithm, fit competing geometric models, and then judge whether the data support a circle or an ellipse. The hidden Mars orbit is used only in a separate post-hoc verification notebook after the reconstruction has already been produced. That verification does not feed into the fitted models, residuals, or noise comparison.

The workflow was:

1. Generate synthetic Tycho-like observations of Mars from 1582 to 1600.
2. Independently verify the noiseless orbital model using conservation of energy and angular momentum.
3. Plot Mars in the apparent sky using right ascension/declination and altitude/azimuth.
4. Convert angular measurements to geometric lines of sight and reconstruct heliocentric Mars positions.
5. Plot the reconstructed Mars positions together with the Sun and Earth's orbit.

5.1. Perform a separate post-hoc check by comparing the already reconstructed 1 arcminute points with the hidden generating orbit. This check is for validation only and is not used by the later fitting steps.

6. Fit a Sun-centered circle, an arbitrary-center circle, and a focus-constrained ellipse with the Sun at one focus.
7. Repeat the analysis for 1, 2, and 5 arcminutes of observational noise.
8. Ask the LLM to interpret the numerical evidence and critique its reasoning.
9. Discuss the connection between Kepler's work and modern model fitting from data.

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1. Synthetic Observation Table

Three synthetic observation tables were generated, one for each noise level:

- 1 arcminute observations
- 2 arcminute observations
- 5 arcminute observations

Each file contains 166 accepted observations. The columns are:

- `datetime_utc` : observation date and time
- `ra_deg` : right ascension in degrees
- `dec_deg` : declination in degrees
- `alt_deg` : altitude above the Uraniborg horizon in degrees
- `az_deg` : azimuth measured clockwise from north in degrees
- `noise_arcmin` : angular noise level used for that table

The observations were generated every 10 days at 22:00 UTC between 1582 and 1600, but an observation was kept only when Mars was at least 10 degrees above the horizon and the Sun was at least 6 degrees below the horizon. This prevents the synthetic observer from recording Mars when it would be below the horizon or in bright daylight.

A sample from the 1 arcminute table is shown below. The complete table is included in the submitted CSV file.

Date (UTC)	RA (deg)	Dec (deg)	Altitude (deg)	Azimuth (deg)	Noise
1582-10-18 22:00	113.1434	21.7463	10.6741	66.7871	1'
1582-10-28 22:00	116.7729	21.1738	13.4672	72.0128	1'
1582-11-07 22:00	119.5004	20.6809	16.9230	77.9334	1'
1582-11-17 22:00	121.1073	20.3617	21.2341	84.7275	1'
1582-11-27 22:00	121.4645	20.2909	26.4962	92.5909	1'
1582-12-07 22:00	120.3862	20.5386	32.7699	101.9267	1'
1582-12-17 22:00	117.8709	20.9645	39.7199	113.4277	1'
1582-12-27 22:00	114.1580	21.5882	46.7929	127.8897	1'
1583-01-06 22:00	109.9395	22.1667	52.7118	146.0109	1'

Date (UTC)	RA (deg)	Dec (deg)	Altitude (deg)	Azimuth (deg)	Noise
1583-01-16 22:00	106.0045	22.6162	56.2296	167.2120	1'
1583-01-26 22:00	103.1156	22.9017	56.7807	188.5722	1'
1583-02-05 22:00	101.5174	22.9913	54.8962	207.1125	1'

The full 1 arcminute data range was approximately:

- RA: 0.89 to 355.91 degrees
- Declination: -23.14 to 23.44 degrees
- Altitude: 10.25 to 57.48 degrees
- Azimuth: 66.03 to 294.84 degrees

The 2 and 5 arcminute tables contain the same underlying observations with independently perturbed angular directions.

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2. Hidden Orbital Model Used to Generate the Data

The synthetic model is intentionally simplified. The Sun is fixed at the origin, and Earth and Mars move on fixed Kepler ellipses in a common ecliptic plane. Planetary perturbations, precession, and a full historical ephemeris are not included. The purpose is to create a controlled data-fitting problem rather than a high-precision astronomical simulator.

The observer is located at Uraniborg using the coordinates

- latitude: 55 degrees 54 minutes 28 seconds N
- longitude: 12 degrees 41 minutes 48 seconds E

The hidden orbital parameters used only in data generation were:

Planet	Semi-major axis a (AU)	Eccentricity e	Initial mean anomaly M0	Perihelion direction
Earth	1.00000011	0.01671022	357.51716 deg	102.93735 deg
Mars	1.52366231	0.09341233	19.41248 deg	336.04084 deg

With distance measured in AU and time measured in days, the solar gravitational parameter was taken as

$$\mu = \frac{4\pi^2}{(365.256363004)^2}$$

For a planet with semi-major axis a, the mean angular rate is

$$n = \sqrt{\frac{\mu}{a^3}}$$

and the mean anomaly is

$$M(t) = M_0 + nt$$

The eccentric anomaly E is obtained by solving Kepler's equation

$$E - e\sin(E) = M$$

Newton's method was used to solve this nonlinear equation. Once E is known, the position in the orbital plane is

$$x = a(\cos E - e)$$

$$y = a\sqrt{1 - e^2}\sin E$$

The velocity is obtained by differentiating Kepler's equation. Since

$$\frac{dE}{dt} = \frac{n}{1 - e\cos E}$$

the velocity components are

$$v_x = -a\sin E\,\frac{dE}{dt}$$

$$v_y = a\sqrt{1 - e^2}\cos E\,\frac{dE}{dt}$$

The orbital plane is rotated by the chosen perihelion direction.

Observer geometry

Earth's axial tilt was taken to be 23.43928 degrees. The heliocentric planet positions were converted from ecliptic coordinates to equatorial coordinates. Earth's daily rotation was represented using an approximate Greenwich mean sidereal time calculation. Uraniborg's longitude shifts the sidereal angle to the local angle at the observatory, while its latitude determines the local east, north, and up basis.

The observer's heliocentric position is

$$\mathbf{r}_{\text{obs}} = \mathbf{r}_E + \mathbf{r}_{\text{site}}$$

The direction from Uraniborg to Mars is

$$\boldsymbol{\rho} = \mathbf{r}_M - \mathbf{r}_{\text{obs}}$$

After normalization, this unit vector is converted to right ascension/declination and altitude/azimuth.

Observational noise

Noise was added directly to the unit line-of-sight vector in its local tangent plane. This is more physically meaningful than independently adding random values to RA and declination because it represents a small random pointing error on the celestial sphere. The quoted arcminute value is the scale used for the tangent-plane Gaussian perturbation; an individual realized angular offset is random and need not equal that value exactly. Three angular noise scales were tested:

- 1 arcminute
- 2 arcminutes
- 5 arcminutes

The synthetic-data generation code is contained in synthetic_data_generation(step_1).ipynb.

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3. Verification of the Noiseless Orbital Simulation

The orbital code was checked independently rather than accepted because it was generated with LLM assistance. Two conserved quantities were tested: specific orbital energy and specific angular momentum.

For a Kepler orbit, the specific orbital energy is

$$\mathcal{E} = \frac{1}{2}\|\mathbf{v}\|^2 - \frac{\mu}{\|\mathbf{r}\|}$$

and the exact theoretical value is

$$\mathcal{E}_{\text{theory}} = -\frac{\mu}{2a}$$

The specific angular momentum is

$$h = \|\mathbf{r} \times \mathbf{v}\|$$

with theoretical value

$$h_{\text{theory}} = \sqrt{\mu a(1-e^2)}$$

The independent verification script evaluated the Earth and Mars orbits from 1582 through 1600 and compared the computed invariants with these closed-form values.

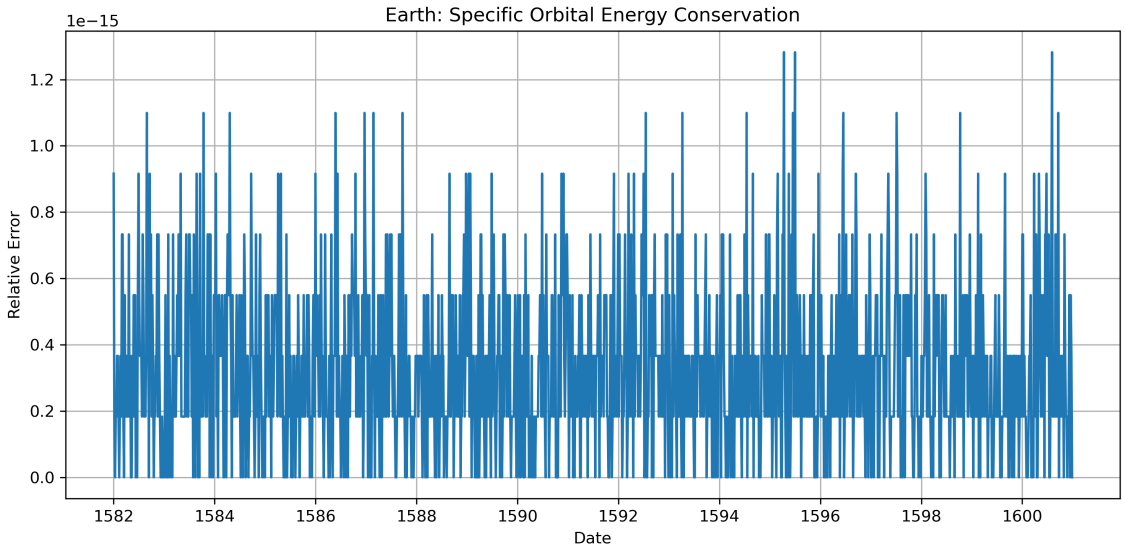
Verification summary

Planet	Period (days)	Perihelion (AU)	Aphelion (AU)	Max relative energy error	Energy relative variation	Max relative angular momentum error	Angular momentum relative variation
Earth	365.2564	0.983290	1.016710	1.28e-15	2.38e-15	6.05e-16	1.01e-15
Mars	686.9591	1.381333	1.665991	1.40e-15	2.51e-15	8.21e-16	1.15e-15

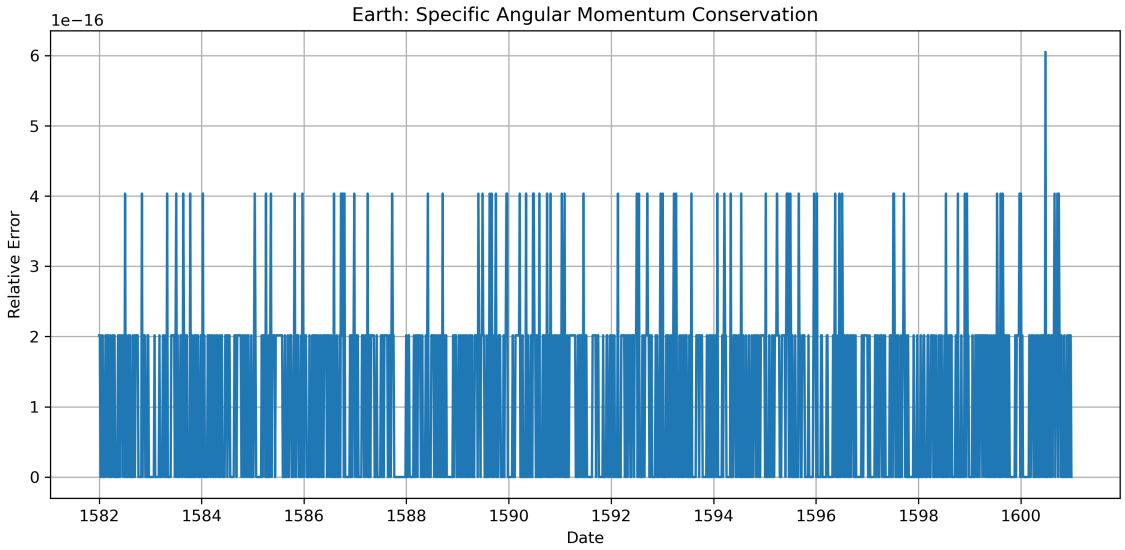
These errors are at the scale of floating-point roundoff. Therefore the position and velocity formulas are internally consistent with the theoretical Kepler invariants.

One limitation of this check is important: the orbit is generated analytically from Kepler's equation rather than integrated forward with a numerical ODE solver. Therefore near-perfect conservation is expected. The verification confirms that the formulas used for position and velocity are mutually consistent; it is not a test of long-time numerical integration stability.

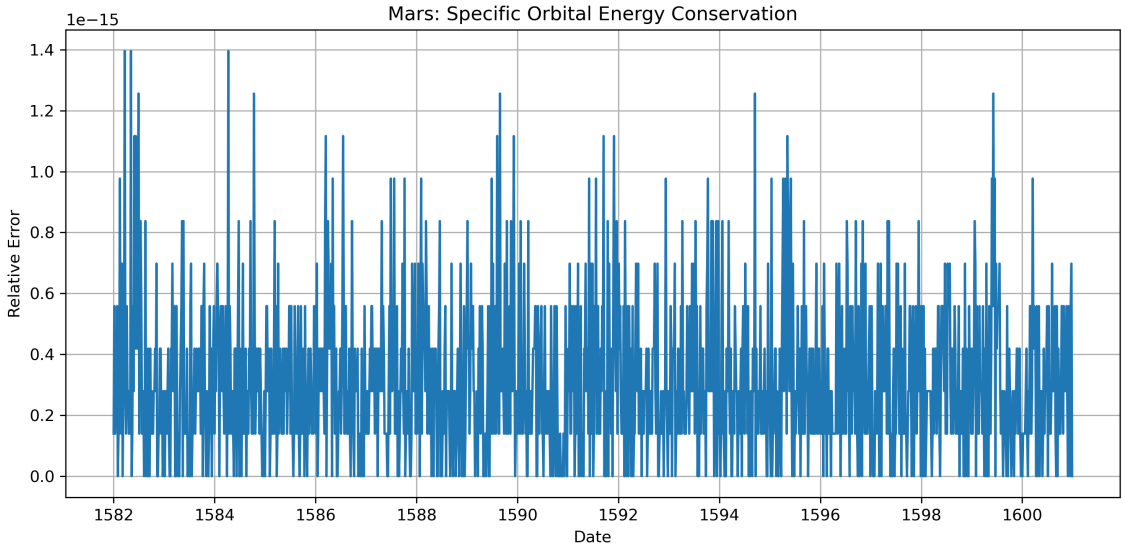
Earth energy conservation



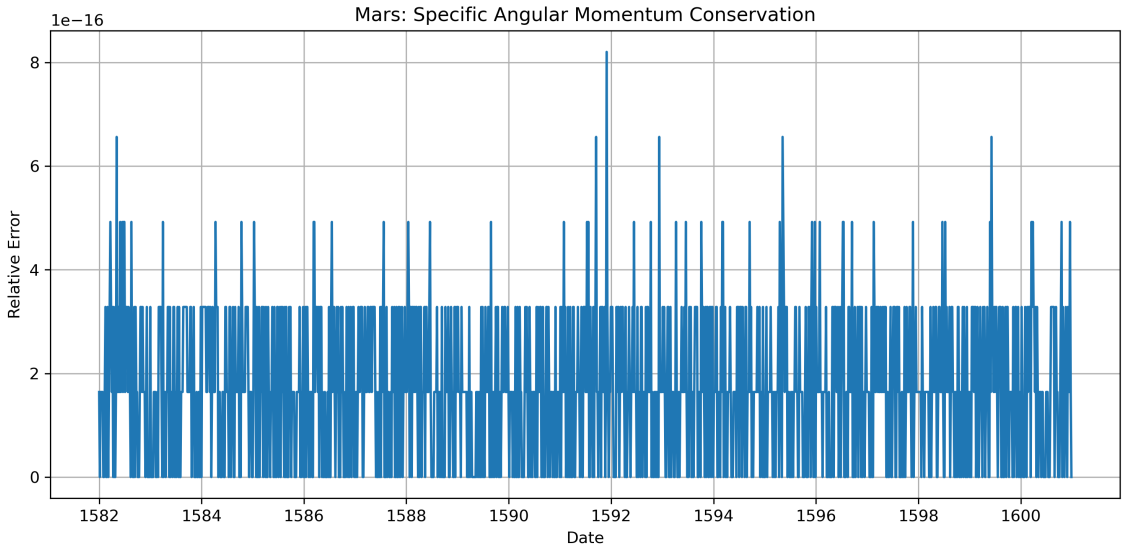
Earth angular momentum conservation



Mars energy conservation



Mars angular momentum conservation



The complete numerical verification table is stored in orbit_verification_summary.csv.

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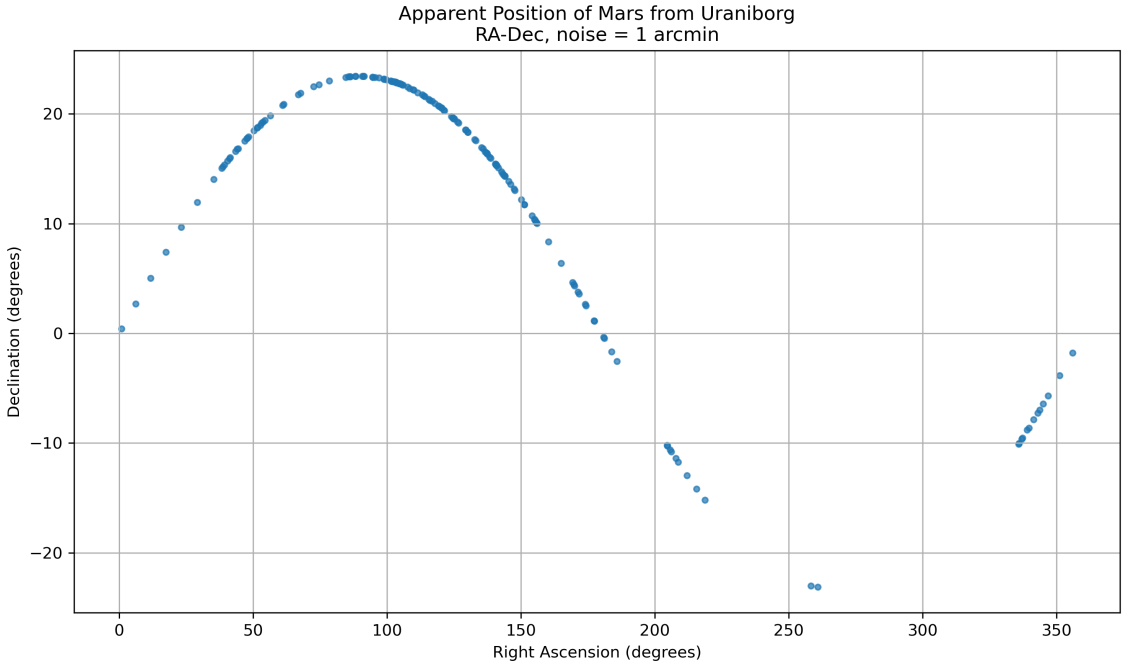
4. Apparent Position of Mars in Tycho's Sky

The synthetic observations were plotted in two coordinate systems.

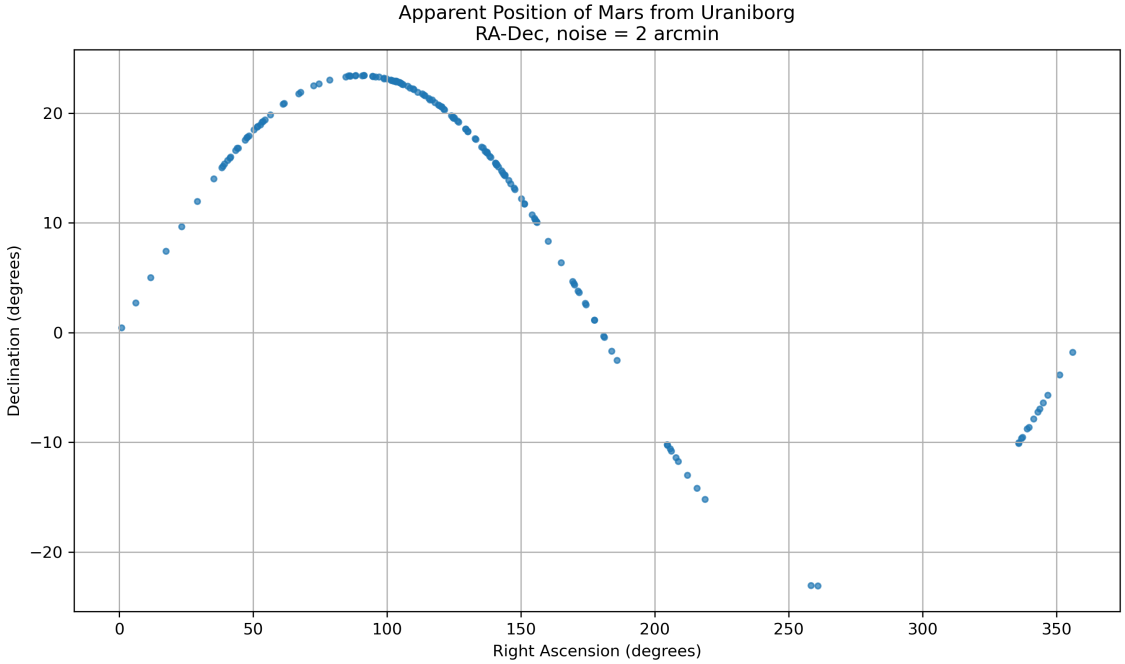
Right ascension and declination describe the direction of Mars on the celestial sphere. Altitude and azimuth describe what an observer at Uraniborg would see relative to the local horizon. The latter coordinates make the effect of Earth's rotation and Uraniborg's geographic location especially clear.

Right ascension and declination

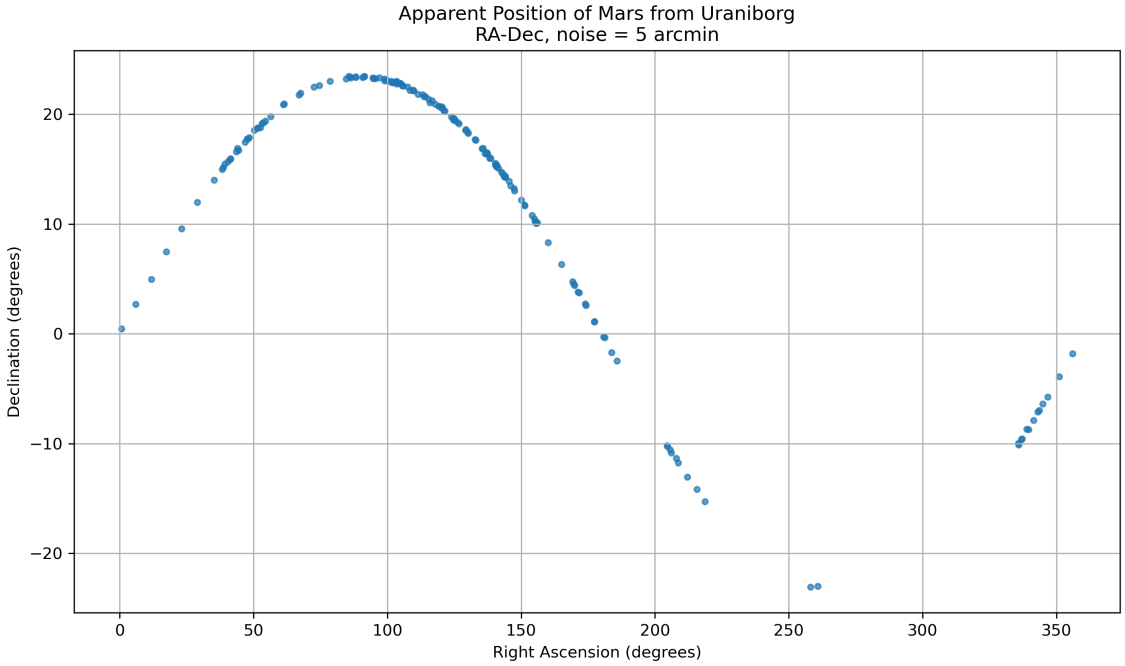
1 arcminute noise



2 arcminutes noise



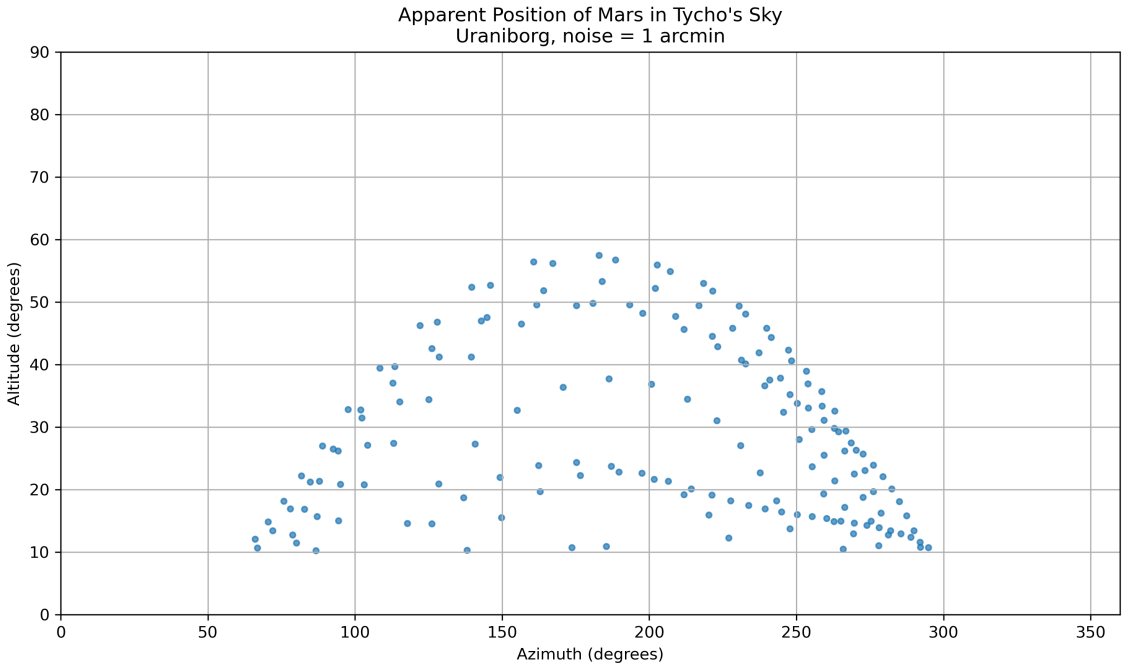
5 arcminutes noise



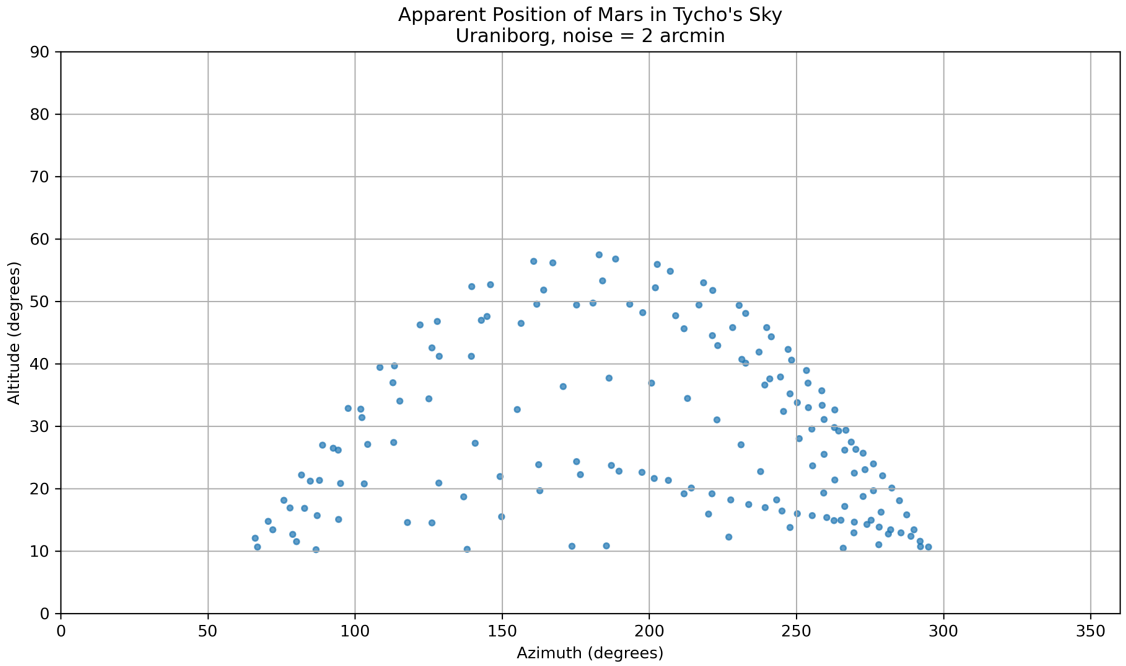
The apparent path is not expected to look like a simple ellipse because the observer is moving with Earth. The plotted direction depends on the relative vector between Mars and the observer, not simply on Mars's heliocentric coordinates. Apparent reversals and loops are therefore natural in the sky coordinates.

Altitude and azimuth

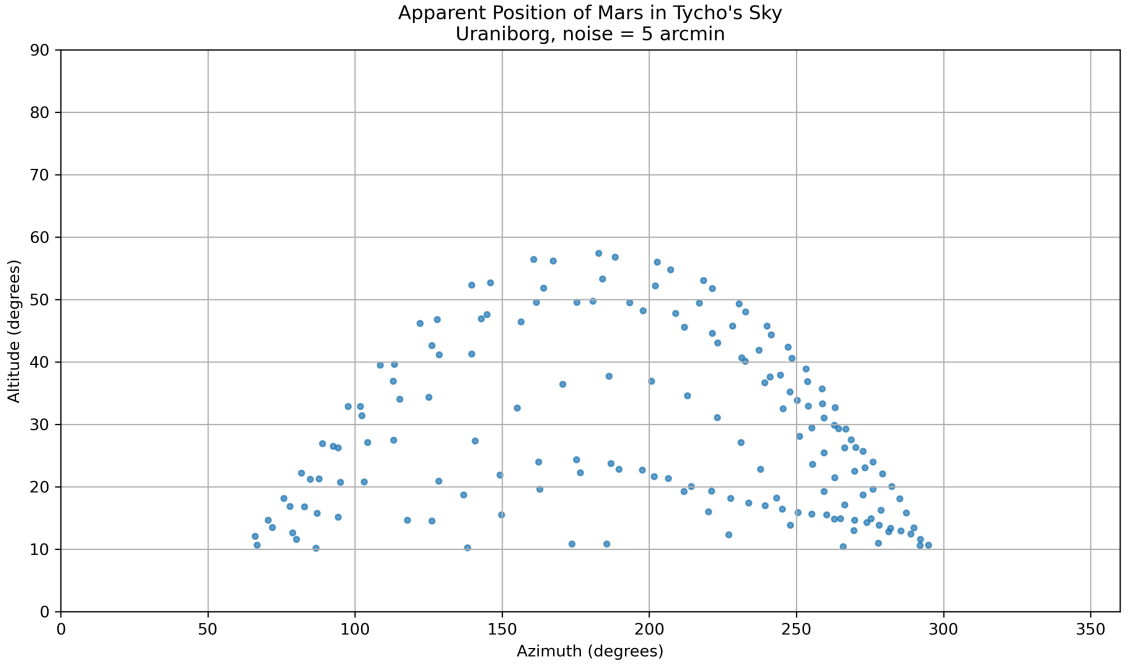
1 arcminute noise



2 arcminutes noise



5 arcminutes noise



The lower boundary near 10 degrees altitude is a direct consequence of the visibility condition used during data generation.

The three noise levels look visually similar because even 5 arcminutes is only about 0.083 degrees, much smaller than the overall angular scale of the sky plots.

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5. Reconstruction of Heliocentric Mars Positions

A single RA/declination observation provides a direction but not a distance. If $\mathbf{p}_i$ is the heliocentric observer position and $\mathbf{u}_i$ is the observed unit direction, then Mars must lie somewhere on the ray

$$\mathbf{r} = \mathbf{p}_i + \rho_i \mathbf{u}_i$$

where the range ρ_i is unknown.

Corrected multi-ray reconstruction

The final reconstruction algorithm did not insert the hidden Mars semi-major axis, eccentricity, orientation, or true period. It used:

- the noisy RA/declination observations,
- Earth's known orbit,
- Uraniborg's location,
- Earth's rotation,
- an additional modeling assumption that Mars has approximately periodic heliocentric motion.

The method searched candidate repetition periods between 500 and 900 days. This interval is itself a broad prior rather than something forced by a single line-of-sight observation, so it should be regarded as part of the reconstruction model. Observations separated by approximately one or more candidate periods were grouped under the assumption that Mars should return near the same heliocentric orbital phase. At least three lines of sight were required for each reconstructed point.

For a group with observer positions $\mathbf{p}_i$ and unit directions $\mathbf{u}_i$, define

$$P_i = I - \mathbf{u}_i \mathbf{u}_i^T.$$

The reconstructed common point $\mathbf{x}$ minimizes the total squared perpendicular distance to the corresponding lines:

$$\min_{\mathbf{x}} \sum_i \|P_i(\mathbf{x} - \mathbf{p}_i)\|^2.$$

Setting the gradient equal to zero gives the normal equations

$$A\mathbf{x} = \mathbf{b},$$

where

$$A = \sum_i P_i$$

and

$$\mathbf{b} = \sum_i P_i \mathbf{p}_i.$$

The least-squares derivation is naturally written for infinite lines. The implementation then enforces the physical ray constraint by computing

$$\rho_i = \mathbf{u}_i^T (\mathbf{x} - \mathbf{p}_i)$$

and rejecting any reconstructed bundle for which one or more recovered ranges satisfy $\rho_i \leq 0$. Therefore every accepted reconstructed point lies in front of all contributing observers.

The code also protects against poorly conditioned triangulation in two separate ways. First, it requires a minimum heliocentric observer baseline of 0.30 AU. Second, it computes the condition number of the normal matrix A and rejects a bundle when $\mathrm{cond}(A) > 10^{10}$. These checks are important because a small ray-intersection residual alone does not guarantee that the recovered distance is stable.

The reconstructed points should be expressed in one consistent Cartesian frame. In particular, the near-zero reported z values are interpreted in the ecliptic frame; this frame must also be used consistently when fitting the planar circle and ellipse models.

The best inferred repetition period was 689.25 days for all three noise levels. Only after this period had been inferred was it compared with the hidden generating value 686.959 days. The difference is about 2.29 days, or approximately 0.33 percent.

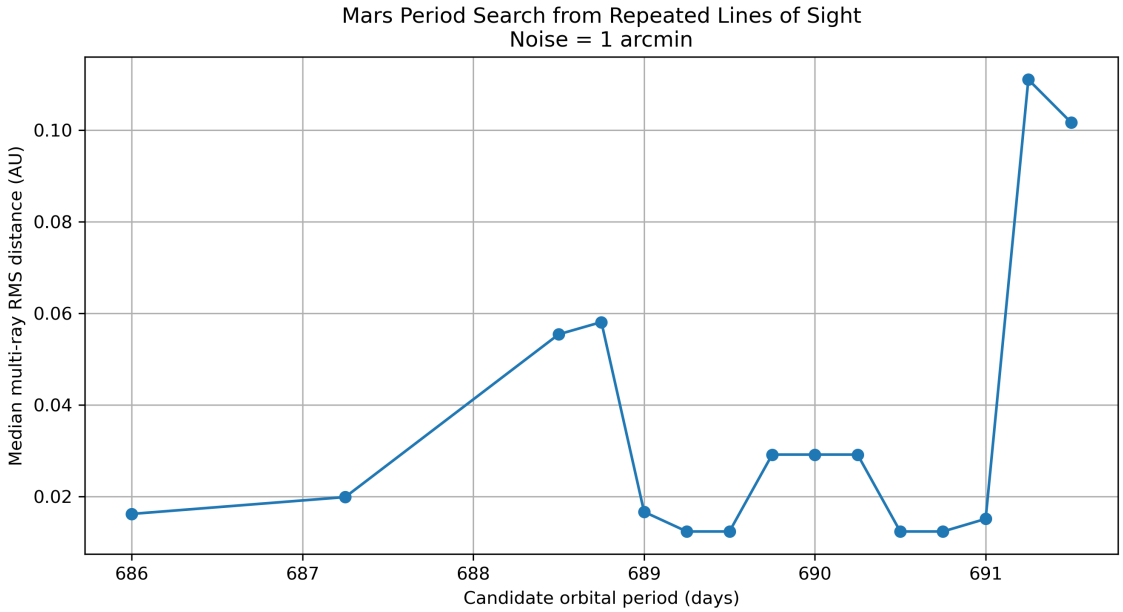
Reconstruction statistics

Noise	Reconstructed points	Inferred period (days)	Minimum radius (AU)	Median radius (AU)	Maximum radius (AU)	Median ray RMS (AU)	Median absolute z (AU)
1'	26	689.25	1.4268	1.5387	1.6072	0.01235	0.000134
2'	26	689.25	1.4262	1.5380	1.6071	0.01236	0.000267
5'	26	689.25	1.4245	1.5359	1.6070	0.01242	0.000667

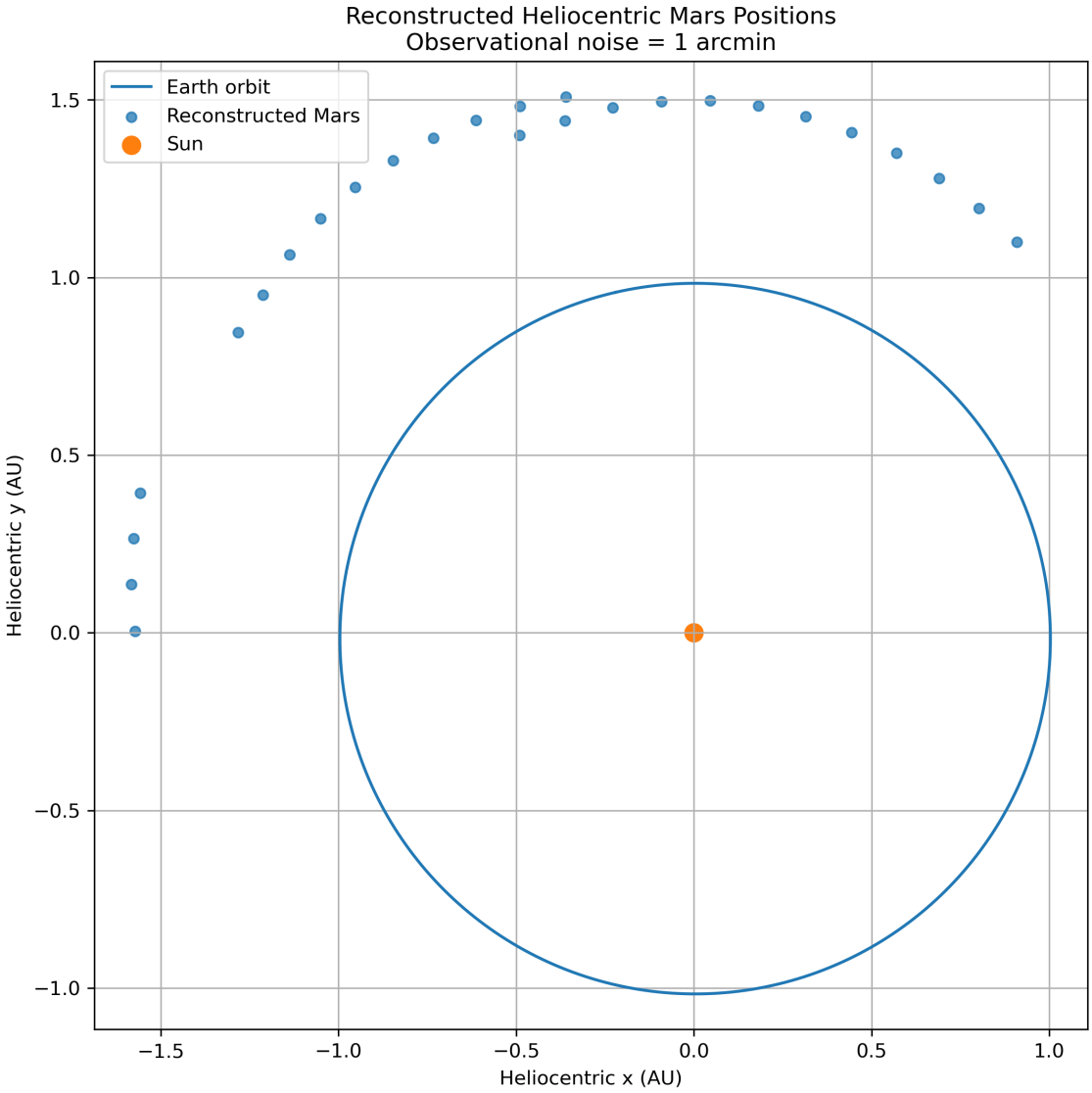
After Step 4 was completed, the hidden model was used only as an external verification check. It places Mars between approximately 1.381 and 1.666 AU from the Sun, while the reconstructed radii lie in a similar range. This agreement supports the physical plausibility of the reconstruction, but these hidden values are not used in the period search, ray-bundle selection, reconstructed coordinates, model fits, or residual calculations. The reported z

coordinates also remain close to zero in the ecliptic frame, consistent with the planar hidden model.

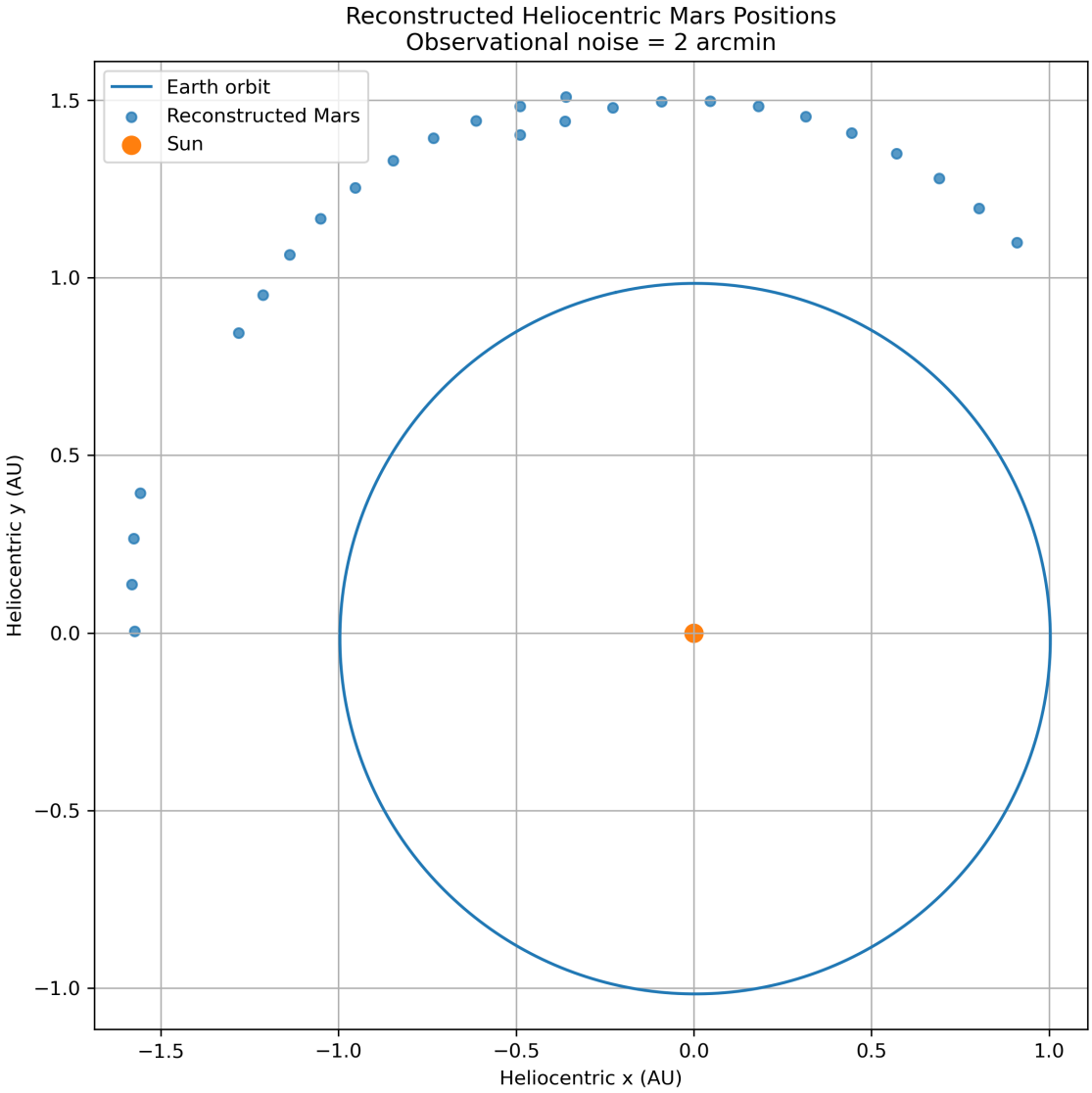
Period search, 1 arcminute



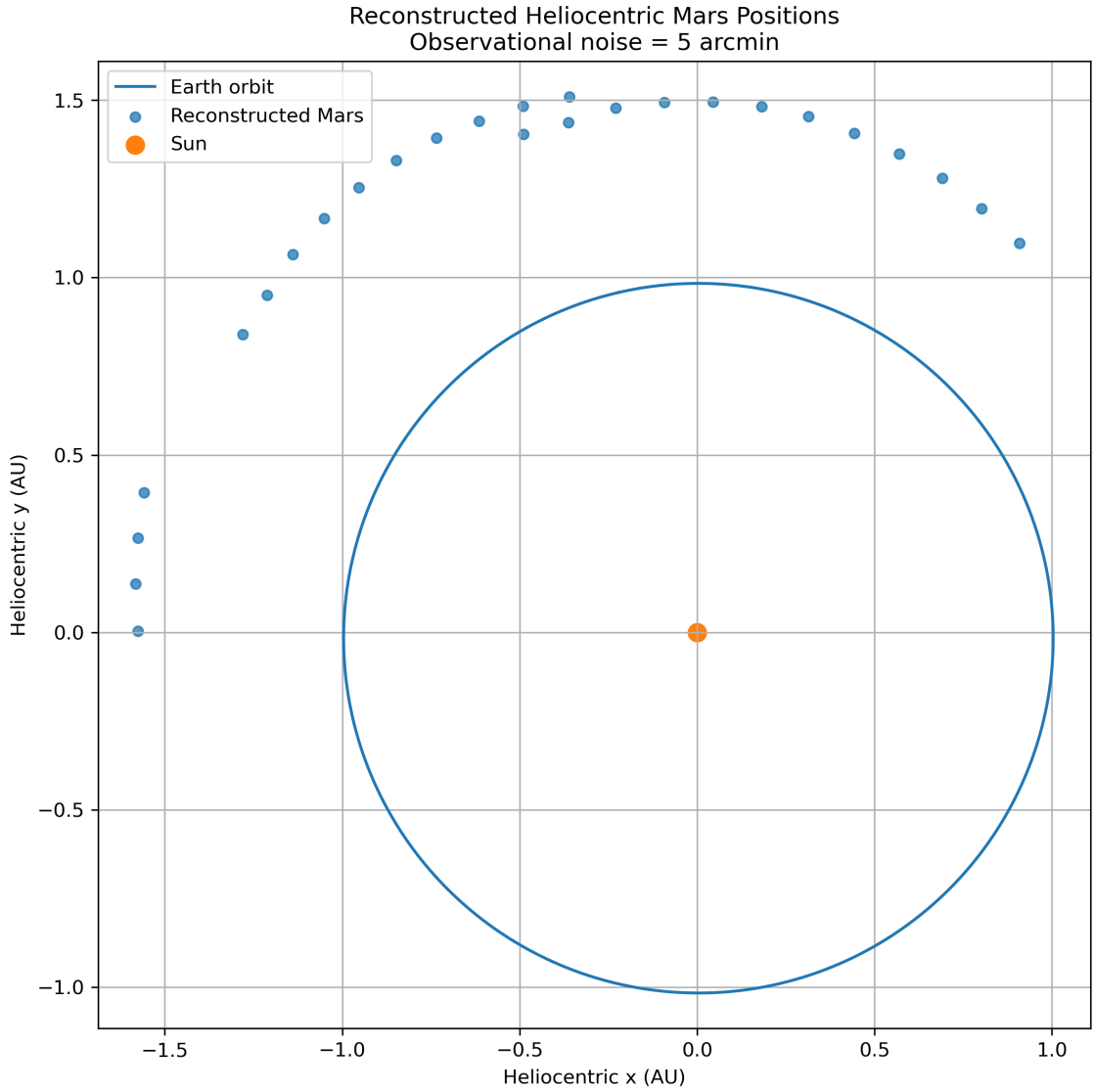
Reconstructed Mars positions, 1 arcminute



Reconstructed Mars positions, 2 arcminutes



Reconstructed Mars positions, 5 arcminutes



Post-hoc verification against the hidden orbit

Step 5-1 is a validation check, not part of the inference pipeline. It explicitly contains the hidden Mars parameters and uses them to draw the true generating orbit, but it first reads the already completed Step 4 file `mars_reconstructed_1arcmin.csv` . It does not modify the reconstructed coordinates or write replacement reconstruction files.

The computational dependency is therefore

```
Step 4 reconstruction
    -> mars_reconstructed_1arcmin.csv
    -> mars_reconstructed_2arcmin.csv
    -> mars_reconstructed_5arcmin.csv

Step 5-1 verification
    -> reads mars_reconstructed_1arcmin.csv
    -> compares it with the hidden generating orbit
    -> produces a verification plot only
```

```
Step 6 model fitting
    -> independently reads the Step 4 mars_reconstructed_*.csv files
    -> produces model_fit_summary.csv

Step 7 noise comparison
    -> reads model_fit_summary.csv
```

Thus the hidden Mars parameters in Step 5-1 do not enter the period estimate, reconstructed positions, fitted circles, fitted ellipse, RMSE values, or the 1/2/5 arcminute comparison. The role of Step 5-1 is simply to check that the blind reconstruction obtained in Step 4 is consistent with the known synthetic ground truth.

The reconstructed 1 arcminute points follow the same approximate scale, orientation, and curvature as the hidden generating ellipse, providing an additional sanity check on the final reconstruction.



An earlier pairwise reconstruction was numerically unstable because nearly repeated observer geometry gives weak parallax information. The final Step 4 method addresses that problem internally by using at least three rays, requiring a minimum observer baseline, rejecting ill-conditioned normal matrices, and enforcing positive ray ranges. Step 5-1 then serves only as an independent confirmation that the corrected reconstruction is physically consistent with the synthetic orbit.

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6. Circle and Ellipse Fits with Residual Errors

Three models were fitted to each reconstructed data set.

Model 1: Sun-centered circle

The model is

$$x^2 + y^2 = R^2$$

Only the radius R is fitted.

Model 2: Free-center circle

The model is

$$(x - h)^2 + (y - k)^2 = R^2$$

The center (h, k) and radius R are fitted.

Model 3: Focus-constrained ellipse with the Sun at one focus

The assignment asks for an ellipse. Here I used the physically motivated three-parameter subfamily in which the Sun is constrained to be one focus. This is still an ellipse, and it has the same number of fitted parameters as the free-center circle, but it uses more physical structure than a completely general five-parameter planar ellipse. A fully model-agnostic extension would also fit a general ellipse and then test whether one of its foci lies near the Sun.

The polar form is

$$r(\theta) = \frac{p}{1 + e\cos(\theta - \omega)}$$

The fitted parameters are the semi-latus rectum p, eccentricity e, and orientation ω. The semi-major axis is then

$$a = \frac{p}{1 - e^2}$$

To compare all three models fairly, the final residual for each reconstructed point was the Euclidean distance to the nearest point on the fitted curve:

$$d_i = \min_{\mathbf{q} \in C} \|\mathbf{x}_i - \mathbf{q}\|$$

The reported RMSE values therefore use the same geometric error definition for all models. The free-center circle and the focus-constrained ellipse each have three fitted parameters, so their raw RMSE values are directly comparable in parameter count; the Sun-centered circle has only one fitted parameter:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d_i^2}$$

Residual comparison

Noise	Model	RMSE (AU)	MAE (AU)	Median error (AU)	Maximum error (AU)
1'	Sun-centered circle	0.051504	0.046848	0.042350	0.099385
1'	Free-center circle	0.021718	0.017068	0.012948	0.049116
1'	Focus-constrained ellipse	0.021759	0.017124	0.013161	0.048913

Noise	Model	RMSE (AU)	MAE (AU)	Median error (AU)	Maximum error (AU)
2'	Sun-centered circle	0.051629	0.046963	0.042165	0.099916
2'	Free-center circle	0.021879	0.017224	0.012521	0.048463
2'	Focus-constrained ellipse	0.021920	0.017268	0.012740	0.048260
5'	Sun-centered circle	0.052027	0.047300	0.043520	0.101501
5'	Free-center circle	0.022431	0.017721	0.012080	0.046529
5'	Focus-constrained ellipse	0.022475	0.017760	0.011934	0.046580

For the 1 arcminute data, the fitted parameters were:

- Sun-centered circle radius: 1.52619 AU
- Free-circle center: (-0.09834, 0.01994) AU
- Free-circle radius: 1.48445 AU
- Fitted ellipse semi-major axis: 1.48547 AU
- Fitted ellipse eccentricity: 0.06689
- Fitted ellipse orientation: -12.63 degrees

The hidden generating values were $a = 1.52366\,\mathrm{AU}$ and $e = 0.09341$, so the focus-constrained ellipse fit recovered the correct general scale and a nonzero eccentricity, although the reconstruction underestimates the eccentricity.

There is also a mathematical reason the free-center circle can mimic the fitted ellipse so closely. For small eccentricity,

$$r(\theta) = \frac{p}{1 + e\cos(\theta - \omega)} \approx a - ae\cos(\theta - \omega),$$

so the leading noncircular variation has amplitude approximately ae. A circle of radius R whose center is displaced from the Sun by a small distance d has, to first order,

$$r(\theta) \approx R + d\cos(\theta - \theta_c).$$

For the 1 arcminute fits,

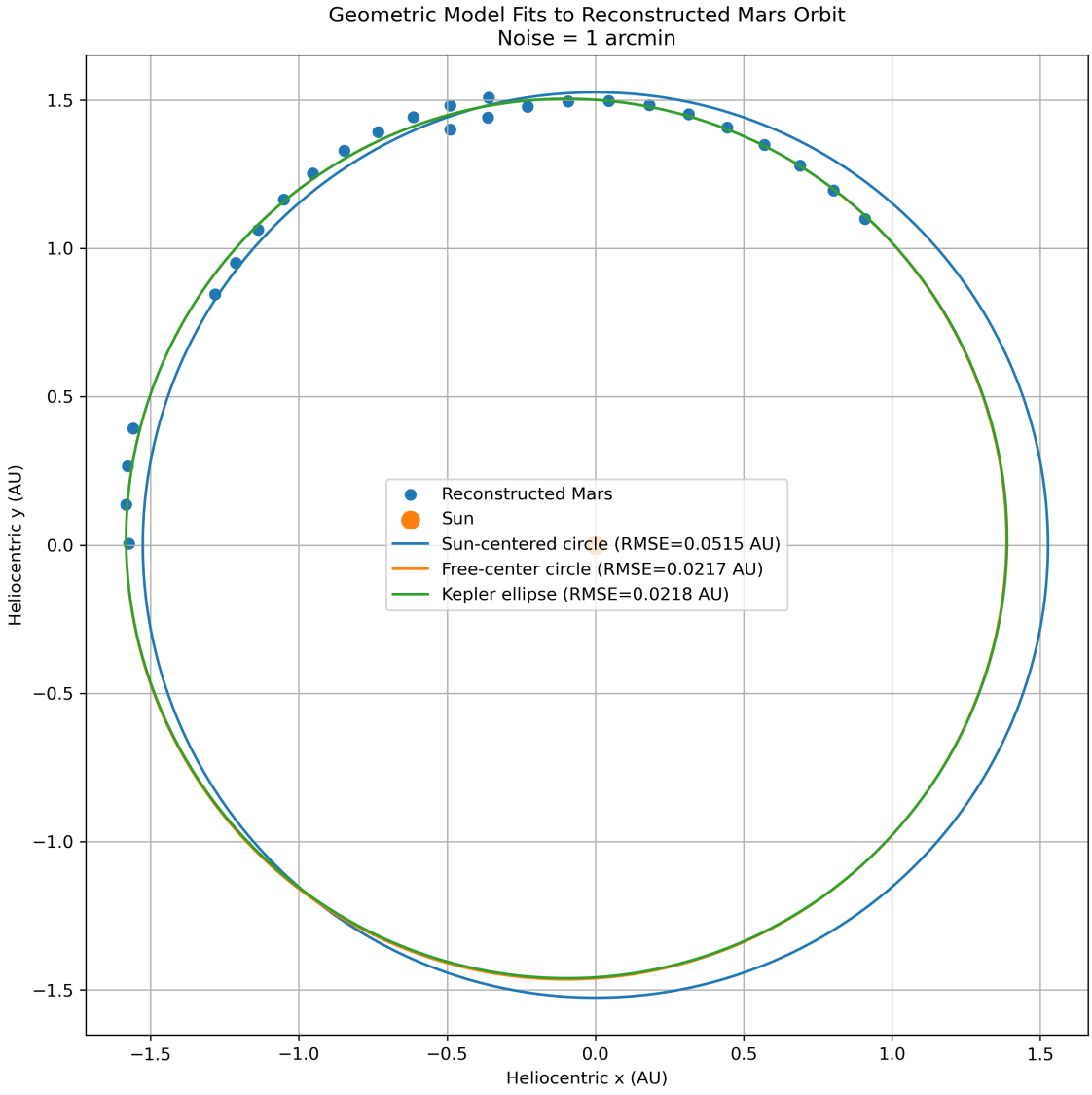
$$ae \approx 1.48547(0.06689) = 0.09936\,\mathrm{AU},$$

while the fitted free-circle center is displaced from the Sun by

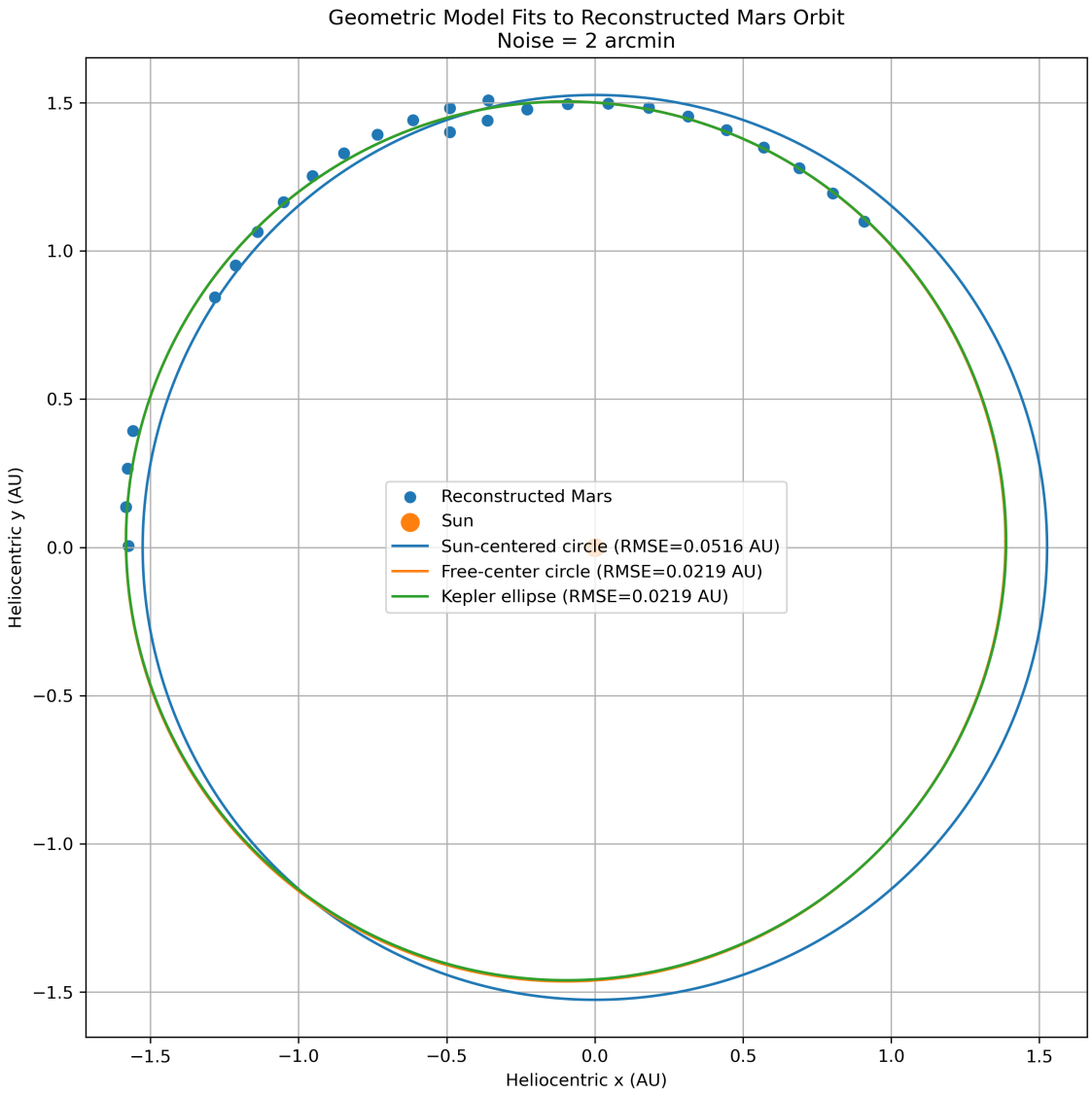
$$\sqrt{(-0.09834)^2 + (0.01994)^2} \approx 0.10034\,\mathrm{AU}.$$

These values are nearly identical. Thus the shifted circle reproduces the first-order radial variation of the small-eccentricity ellipse. Their geometric difference appears mainly at second order, on the rough scale $ae^2 \approx 0.00665\,\mathrm{AU}$, which is smaller than the approximately $0.022\,\mathrm{AU}$ reconstruction/model residual. This helps explain why the two fitted curves are effectively indistinguishable in the present reconstruction.

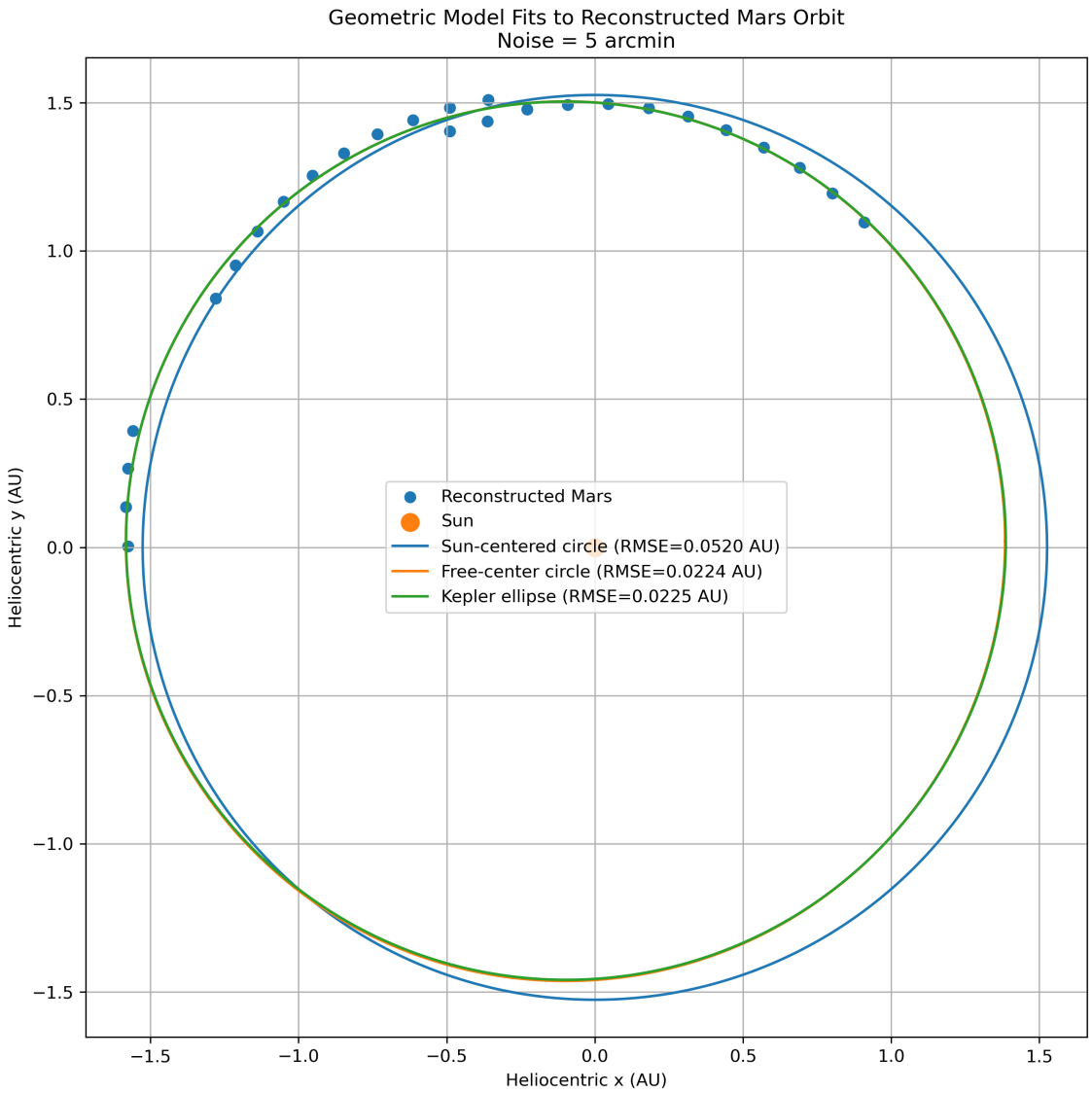
1 arcminute model fits



2 arcminute model fits



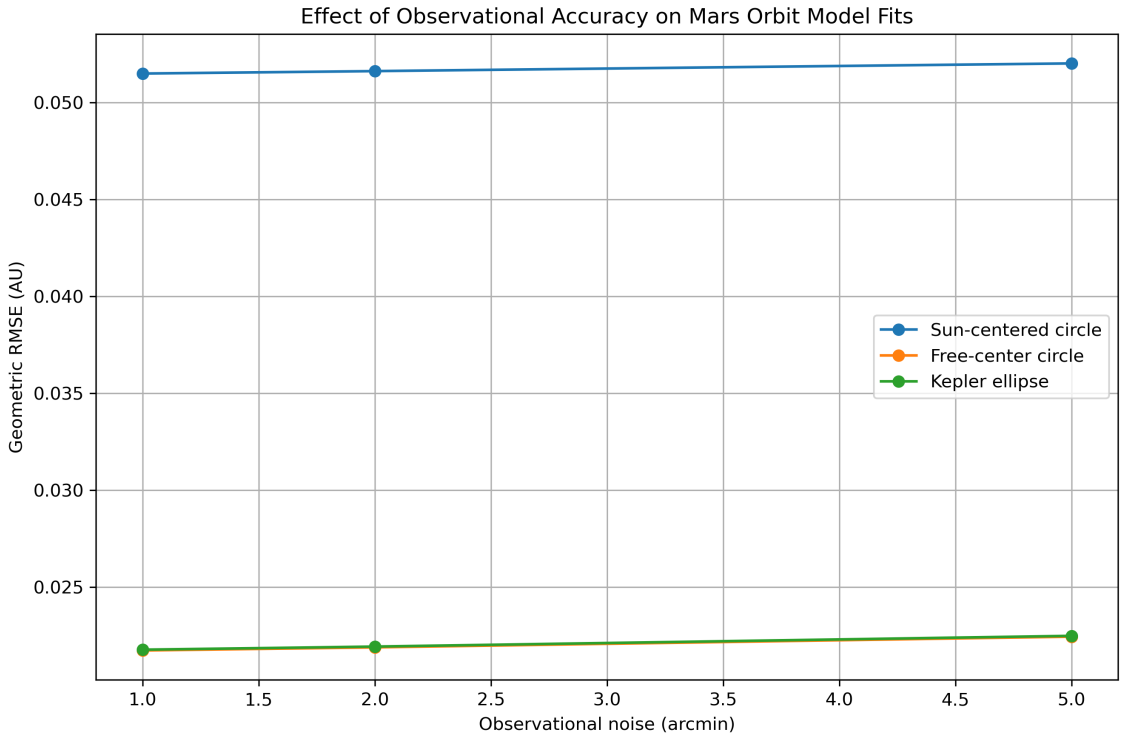
5 arcminute model fits



The Sun-centered circular model is clearly worse. At 1 arcminute, changing from a Sun-centered circle to the ellipse lowers RMSE by about 57.75 percent.

However, the free-center circle and focus-constrained ellipse are almost indistinguishable for the reconstructed points. The free-center circle is slightly better in RMSE by only about 0.19 percent at every tested noise level. This difference is far smaller than the overall reconstruction residual and is not practically meaningful evidence that the physical orbit is circular. The small eccentricity, limited reconstructed phase coverage, and first-order equivalence between a shifted circle and a focus ellipse all make the two models difficult to distinguish here.

Effect of observational noise



Noise	Sun-circle RMSE	Free-circle RMSE	Ellipse RMSE	Ellipse minus free-circle RMSE	Ellipse vs free-circle difference	Ellipse improvement vs Sun circle
1'	0.051504	0.021718	0.021759	0.000041	0.188% worse	57.75% better
2'	0.051629	0.021879	0.021920	0.000041	0.188% worse	57.54% better
5'	0.052027	0.022431	0.022475	0.000043	0.193% worse	56.80% better

Among the three tested noise levels, none makes the focus-constrained ellipse clearly outperform the free-center circle. Even at 1 arcminute, the two models are effectively tied. Therefore this experiment does not identify a numerical accuracy threshold at which the ellipse begins to win; it only shows that 1 arcminute is not sufficient for this particular reconstruction pipeline. The dominant limitations appear to be reconstruction error, limited reconstructed phase coverage, and the fact that a shifted circle approximates a low-eccentricity focus ellipse to first order. Testing noise below 1 arcminute, using a better-conditioned reconstruction, and repeating each noise level over multiple random realizations would be needed to estimate a meaningful threshold.

---

7. LLM Prompts and Main Suggestions

The LLM was used throughout the project as a collaborator. The prompts below are representative of the actual interaction; wording has been lightly cleaned for readability while preserving the original requests.

Prompt 1: Build the synthetic observation model

> Start with the code for synthetic Tycho-like observations of Mars. Explain how the Uraniborg coordinates are used and what equations are needed to generate the synthetic data. Include Earth motion, Mars motion, Earth rotation, and observational noise.

Main LLM suggestions

The LLM proposed:

- use a simplified heliocentric two-planet model with the Sun fixed at the origin;
- represent Earth and Mars using Kepler ellipses;
- solve Kepler's equation numerically with Newton's method;
- use Uraniborg latitude and longitude together with sidereal time to construct the local observer position and local east/north/up basis;
- convert ecliptic coordinates to equatorial coordinates using Earth's obliquity;
- calculate both RA/declination and altitude/azimuth;
- add angular noise in the tangent plane of the line-of-sight vector rather than directly perturbing Cartesian coordinates;
- reject observations when Mars is too low or the Sun is too high.

These suggestions became the basis of the synthetic data generator.

Prompt 2: Verify the simulation independently

> Verify the simulation code independently. Check energy and angular momentum for the noiseless Earth and Mars orbits and give me a standalone verification program.

Main LLM suggestions

The LLM recommended comparing the numerical values with the exact Kepler formulas

$$\mathcal{E} = -\frac{\mu}{2a}$$

$$h = \sqrt{\mu a(1-e^2)}$$

It also recommended using relative errors instead of deciding visually that a curve looked flat. This led to an independent verification file rather than reusing the same functions blindly inside the data generator.

Prompt 3: Visualize Mars in Tycho's sky

> Plot the apparent position of Mars in Tycho's sky using the synthetic CSV files.

Main LLM suggestions

The LLM recommended plotting both RA/declination and altitude/azimuth. It also warned against connecting points across the 0/360 degree wrap in RA or azimuth, so scatter plots were used.

Prompt 4: Reconstruct Mars from lines of sight

> Convert the observations into lines of sight and reconstruct heliocentric Mars positions without using the hidden Mars orbital parameters.

Main LLM suggestions

The first suggestion was a repeated-phase triangulation method based on pairs of rays. That initial method was not reliable and is discussed in the critique below. After the failure was detected, the LLM proposed a corrected multi-ray method:

- search a broad range of candidate orbital periods rather than inserting Mars's true period;
- group observations separated by approximately integer multiples of the candidate period;
- use three or more rays for each candidate heliocentric point;
- solve a least-squares closest-point problem for the full ray bundle;
- reject bundles with poor geometric baseline;
- validate the inferred radii and out-of-plane coordinate before accepting the reconstruction.

The corrected method inferred a period of 689.25 days and physically reasonable heliocentric Mars distances.

Prompt 5: Compare circle and ellipse models

> Fit a Sun-centered circle, a circle with arbitrary center, and an ellipse to the reconstructed Mars positions. Compute comparable residual errors.

Main LLM suggestions

The LLM suggested using a common geometric residual for the final comparison: the Euclidean distance from each reconstructed point to the nearest point on each fitted curve. This avoided comparing incompatible residual definitions across different model types.

Prompt 6: Interpret the noise experiment

> Compare the 1, 2, and 5 arcminute results. How much observational accuracy is needed before the ellipse clearly wins?

Main LLM suggestions

The LLM did not force the historical conclusion. It observed that the Sun-centered circle is clearly worse, but the free-center circle and focus-constrained ellipse are almost tied at every tested noise level. The final interpretation is that measurement noise is not the only limitation: reconstruction error, limited reconstructed phase coverage, and the small-eccentricity similarity between a shifted circle and a focus ellipse are all important.

Prompt 7: Interpret the result as if reasoning from the data

> Interpret the numerical results. Did the AI reason from the observations, or did it merely recall that Mars follows an ellipse?

Main LLM suggestion

The LLM concluded that the reconstructed-data comparison strongly disfavors a Sun-centered circle, but does not justify claiming that the focus-constrained ellipse clearly beats the shifted-circle alternative. This interpretation is important because it follows the computed residuals even though the LLM already knows the historical fact that Mars follows an ellipse.

---

8. Critique of the LLM: What Was Useful and What Was Not

The LLM was most useful as a source of mathematical structure, implementation ideas, and

verification strategies. It was less reliable when a plausible geometric method had hidden conditioning problems.

Useful contributions

1. It translated the physical setup into a modular computational model

The LLM separated the problem into orbital motion, coordinate transformations, Earth rotation, observer position, angular observations, noise, reconstruction, and model fitting. This modular organization made the program easier to inspect line by line and easier to verify.

2. It connected the code to explicit equations

Rather than only supplying Python code, the LLM explained Kepler's equation, the ellipse parameterization, velocity formulas, conserved energy, conserved angular momentum, RA/declination conversion, and the observer geometry. This made it possible to check whether individual code blocks had a mathematical interpretation.

3. Its verification strategy was strong

The most useful suggestion was to compare the computed energy and angular momentum with independent closed-form theoretical values. The approximately 1e-15 relative errors provided a much stronger verification than simply looking at orbital plots.

4. It helped identify sanity and conditioning checks

The project benefited from checks such as:

- Earth should remain near 1 AU in the known Earth model;
- recovered ray ranges should be positive;
- multi-ray systems should not be nearly singular;
- a planar reconstruction should have small out-of-plane coordinates when expressed in the chosen orbital frame;
- an inferred repetition period should be supported by the observation geometry rather than inserted as the known Mars period.

The hidden Mars distance range 1.38 to 1.67 AU is used only in Step 5-1 as a post-hoc ground-truth check. It is not a reconstruction criterion and does not feed into Steps 6 or 7.

5. It adapted when the numerical evidence contradicted expectations

The final model comparison did not show the ellipse clearly winning. The LLM correctly interpreted this instead of claiming that the result had confirmed Kepler simply because Mars is historically known to follow an ellipse.

6. Step 5-1 provided an independent downstream validation

The separate Step 5-1 notebook intentionally uses the hidden Mars parameters only after the blind reconstruction has already been saved by Step 4. It reads the completed 1 arcminute reconstruction and overlays it with the true synthetic orbit. Because Steps 6 and 7 read the Step 4 and Step 6 outputs directly rather than any Step 5-1 output, this check verifies the reconstruction without altering the reported fitting results.

Weak or incorrect contributions

1. The first reconstruction method was wrong in an important way

The initial LLM suggestion used pairs of repeated-phase lines of sight and chose the candidate period that made the rays intersect most closely. That method produced reconstructed radii near 1 AU. The result looked geometrically clean, but the depth estimate was unstable because nearly repeated observer geometry can make two-ray triangulation poorly conditioned. A small intersection residual by itself does not imply that the recovered heliocentric distance is reliable.

The final method corrected this weakness by using multi-ray least squares rather than pairs alone. It also requires a minimum observer baseline, rejects bundles with $\mathrm{cond}(A) > 10^{10}$, and requires all recovered ray ranges to be positive. Step 5-1 independently confirms that this corrected reconstruction agrees with the hidden synthetic orbit, but that check is downstream of the reconstruction and does not alter the reported results.

2. The LLM initially had a natural bias toward expecting the ellipse to win

Because an LLM already knows Kepler's laws, it is easy for it to frame the experiment around the expected historical answer. The actual fitted residuals did not support a strong ellipse-over-shifted-circle conclusion. The final interpretation had to remain disciplined and follow the computed values rather than the prior historical fact.

3. Some modeling choices were practical rather than uniquely correct

The 10-day sampling interval, fixed 22:00 UTC observation time, 10-degree Mars altitude cutoff, -6-degree Sun altitude cutoff, planar planetary orbits, and simplified sidereal-time calculation are reasonable choices for a synthetic experiment, but they are not unique or

fully historical. These choices should be described as assumptions, not as exact features of Tycho Brahe's observing program.

4. The reconstruction does not recover the entire orbit equally well

Only a subset of orbital phases produced sufficiently good repeated multi-ray geometry. The reconstructed points therefore cover limited orbital phase rather than a uniformly sampled complete ellipse. This contributes to the near tie. In addition, for a low-eccentricity orbit a shifted circle already matches the leading first-order radial variation of a focus ellipse, so distinguishing the two requires reconstruction errors smaller than the second-order geometric difference.

Overall assessment of the LLM

The LLM was useful when treated as a collaborator that generates equations, code structures, diagnostic ideas, and alternative approaches. It was not reliable enough to accept without independent mathematical and physical checks. The failed first reconstruction is a useful example: the code ran and the geometry looked plausible, but weak parallax made the recovered distance unstable. Examining the observer geometry and conditioning led to the corrected multi-ray method, while the separate Step 5-1 hidden-orbit comparison provided a downstream validation of the final reconstruction.

---

9. Did the AI Reason Like Kepler, or Did It Merely Remember Kepler?

The experiment does not justify claiming that the AI independently rediscovered Kepler's law in the historical sense. The LLM already knows that Mars follows an elliptical orbit, and the synthetic data themselves were generated from a hidden Kepler ellipse. Therefore the AI cannot be treated as an observer with no prior astronomical knowledge.

However, the final analysis did show a meaningful form of data-based reasoning. When the residuals were computed, the LLM did not simply say that the ellipse must be best because Kepler discovered ellipses. The numerical evidence showed that the Sun-centered circle was clearly worse, while the free-center circle and focus-constrained ellipse had almost identical residual errors. In fact, the free-center circle had a slightly lower RMSE at all three tested noise levels. The LLM therefore concluded only what this reconstruction supports: a Sun-centered circle is strongly disfavored, but the available reconstructed points are not sufficient to distinguish the focus-constrained ellipse clearly from a shifted circle.

The best description is that the AI acted more like a modern scientific collaborator than a

true historical Kepler. It supplied hypotheses and computational methods, and one early pairwise reconstruction method had a conditioning problem that was corrected in the final multi-ray implementation. The separate Step 5-1 ground-truth check then confirmed that the corrected reconstruction was physically consistent, without entering the later fitting calculations. The AI's strongest reasoning occurred when it followed the numerical evidence even when that evidence did not reproduce the expected historical conclusion. The project therefore demonstrates both the usefulness and the limitation of LLM-assisted scientific work: the LLM can accelerate exploration, but the student must still verify the mathematics and decide what the data actually prove.

Discussion: Kepler's Problem as Model Fitting from Data

This problem is an early example of model fitting from data because we begin with observations and ask which mathematical model best explains them. Instead of assuming the correct orbit in advance, we compare candidate models, fit their parameters, and use residual errors to determine how well each model explains the reconstructed observations.

Kepler did something conceptually similar when he analyzed Tycho Brahe's measurements of Mars. He did not simply accept the traditional circular description because it was mathematically attractive. The observational discrepancies forced him to consider a different model. In that sense, the historical process resembles modern approximation: observations constrain a family of candidate mathematical descriptions, and the model is judged by how well it accounts for the data.

The general workflow is:

1. collect observations;
2. choose candidate models;
3. fit model parameters;
4. compute residual errors;
5. compare competing explanations;
6. retain only conclusions justified by the data.

This is conceptually related to modern machine learning and approximation theory. In many modern problems, the true input-output relationship is unknown. A parameterized model is fitted to observations by minimizing a loss or error. Kepler was not using a neural network or gradient descent, but the underlying scientific logic is similar: infer mathematical structure from data rather than imposing the preferred model in advance.

The current experiment also demonstrates an important limitation of model fitting. The fitted reconstructed points strongly disfavor the simplest Sun-centered circle, but they do not clearly distinguish the focus-constrained ellipse from a shifted circle. A scientifically valid conclusion must therefore stop at what the reconstruction and residuals support. Knowing the correct historical answer is not a substitute for evidence from the experiment.

Files Submitted with This Report

Code

- `codes/synthetic_data_generation(step_1).ipynb`
- `codes/verify_orbits(step_2).ipynb`
- `codes/plot_mars_sky(step_3).ipynb`
- `codes/reconstruct_mars(step_4).ipynb`
- `codes/plot_reconstructed_mars(step_5).ipynb`
- `codes/rigorous_reconstructed_mars(step_5-1).ipynb`
- `codes/fit_models(step6).ipynb`
- `codes/noise_comparison(step7).ipynb`
- `codes/LLM_reasoning(step8).ipynb`
- `codes/dicussion_model_fitting(step9).ipynb`

Data

- `data/synthetic_mars_1arcmin.csv`
- `data/synthetic_mars_2arcmin.csv`
- `data/synthetic_mars_5arcmin.csv`
- `data/orbit_verification_summary.csv`
- `data/earth_orbit_verification.csv`
- `data/mars_orbit_verification.csv`
- `data/period_search_1arcmin.csv`

- `data/period_search_2arcmin.csv`
- `data/period_search_5arcmin.csv`
- `data/mars_reconstructed_1arcmin.csv`
- `data/mars_reconstructed_2arcmin.csv`
- `data/mars_reconstructed_5arcmin.csv`
- `data/model_fit_summary.csv`
- `data/step7_noise_comparison.csv`

Figures

All generated figures are stored in the `result images` directory.

Homework 1.2: The Flat-Earth Nonholonomic Kepler Problem

Mathematical Foundations of Generative AI

Project Goal

The goal of this project was to use a large language model (LLM) as a mechanics assistant for a simplified nonholonomic Kepler problem. The model assumes that a flat disk rolls without slipping on a plane while being attracted toward a fixed Sun at the origin by an inverse-square gravitational force.

The LLM was used to help derive the reduced equations of motion, identify quantities that might be conserved, and suggest numerical checks. The equations and predictions were then checked mathematically and tested with numerical simulations rather than accepted automatically.

The main questions were:

1. What are the reduced equations of motion for the rolling disk?
2. What kinds of trajectories appear for different initial conditions?
3. Is total energy conserved?
4. Is ordinary orbital angular momentum conserved?
5. Is heading angular momentum conserved?
6. How does the nonholonomic rolling constraint change the usual Kepler problem?

---

1. Model and Reduced Equations of Motion

The center of the disk is

$$q = (x, y),$$

and the heading direction is

$$e_\theta = (\cos\theta, \sin\theta).$$

The no-slipping constraint requires the center velocity to remain parallel to the heading direction:

$$\dot{q} = v e_\theta.$$

Therefore,

$$\dot{x} = v\cos\theta$$

and

$$\dot{y} = v\sin\theta.$$

Equivalently, the forbidden sideways velocity satisfies

$$-\dot{x}\sin\theta + \dot{y}\cos\theta = 0.$$

This is the nonholonomic velocity constraint of the model: it restricts the admissible velocities rather than confining the configuration to a fixed curve or surface. A formal integrability check is given in Section 8.5.

Gravitational Potential and Force

The gravitational potential is

$$U(q) = -\frac{k}{|q|},$$

where

$$|q| = \sqrt{x^2 + y^2}.$$

The gravitational force is

$$F(q) = -\nabla U(q)$$

and therefore

$$F(q) = -\frac{kq}{|q|^3} = -\frac{k(x,y)}{(x^2+y^2)^{3/2}}.$$

The force components are

$$F_x = -\frac{kx}{(x^2+y^2)^{3/2}}$$

and

$$F_y = -\frac{ky}{(x^2+y^2)^{3/2}}.$$

Only the component of the gravitational force along the allowed heading direction changes the scalar speed v. For an ideal rolling constraint, the constraint reaction has no component

along the admissible forward direction, so projecting the dynamics onto e_θ gives the scalar speed equation. The gravitational projection is

$$F(q) \cdot e_\theta = -\frac{k(x\cos\theta + y\sin\theta)}{(x^2+y^2)^{3/2}}.$$

The reduced speed equation is

$$M\dot{v} = F(q) \cdot e_\theta,$$

so

$$\dot{v} = -\frac{k(x\cos\theta + y\sin\theta)}{M(x^2+y^2)^{3/2}}.$$

Effective Mass and Heading Inertia

The effective mass can be obtained directly from the kinetic energy. If ϕ denotes the rolling spin angle, no slip gives

$$\dot{\phi} = \frac{v}{R}.$$

Hence the translational kinetic energy of the center plus the rolling-spin kinetic energy is

$$\frac{1}{2}mv^2 + \frac{1}{2}I_3\dot{\phi}^2 = \frac{1}{2}\left(m + \frac{I_3}{R^2}\right)v^2.$$

This motivates the effective forward mass

$$M = m + \frac{I_3}{R^2}.$$

For a homogeneous thin disk,

$$I_3 = \frac{1}{2}mR^2, \qquad J = \frac{1}{4}mR^2,$$

so

$$M = m + \frac{1}{2}m = \frac{3}{2}m.$$

Here J is the inertia associated with changes in the heading angle. The simplified heading equation supplied by the model is

$$J\ddot{\theta} = 0.$$

Defining

$$\omega = \dot{\theta},$$

we obtain

$$\dot{\theta} = \omega$$

and

$$\dot{\omega} = 0.$$

Thus the heading angular velocity is constant.

Final First-Order System

The state vector is

$$(x, y, v, \theta, \omega).$$

The first-order system used in the numerical simulation is

$$\dot{x} = v\cos\theta,$$

$$\dot{y} = v\sin\theta,$$

$$\dot{v} = -\frac{k}{M}\frac{x\cos\theta + y\sin\theta}{(x^2+y^2)^{3/2}},$$

$$\dot{\theta} = \omega,$$

$$\dot{\omega} = 0.$$

Mathematical and Dimensional Checks

The position equations have units of velocity because v has units of length per time while sine and cosine are dimensionless.

From

$$U = -\frac{k}{r},$$

k must have dimensions

$$[k] = ML^3T^{-2}.$$

Then

$$\frac{kq}{|q|^3}$$

has dimensions MLT^{-2}, which are the dimensions of force.

Also,

$$\frac{I_3}{R^2}$$

has dimensions of mass, so the definition of M is dimensionally consistent.

Finally, $J\ddot{\theta}$ has dimensions of torque. Thus the reduced equations are dimensionally consistent.

The derivations are contained in:

- `code/derivation(step1).ipynb`
- `code/derivation(step2).ipynb`

---

2. Numerical Simulation

The system was simulated with Python using `scipy.integrate.solve_ivp`.

The parameter values were

$$m = 1,$$

$$R = 1,$$

$$k = 1.$$

For the homogeneous thin disk,

$$I_3 = 0.5,$$

$$J = 0.25,$$

and

$$M = 1.5.$$

The simulation interval was

$$0 \le t \le 100,$$

with 10,000 output times. The numerical solver used tolerances

$$\mathrm{rtol} = 10^{-10}$$

and

$$\mathrm{atol} = 10^{-12}.$$

A safety event stopped a simulation if the disk came within radius 0.1 of the singularity at the Sun. None of the four simulations reached this cutoff.

The four initial conditions were:

Case	x_0	y_0	v_0	θ_0	ω_0
Case 1	2.0	0.0	0.8	$\pi/2$	0.20
Case 2	2.0	0.0	0.8	$\pi/2$	0.40
Case 3	2.0	0.0	1.0	$\pi/2$	0.20
Case 4	2.0	0.0	0.8	$\pi/3$	0.20

The complete simulation and conservation-check code is stored in:

`code/(step3-7).ipynb`

The numerical time series are stored in:

- `data/case_1_simulation.csv`
- `data/case_2_simulation.csv`
- `data/case_3_simulation.csv`
- `data/case_4_simulation.csv`

Core Python System

The main ODE function used in the simulation was:

```
def flat_earth_kepler(t, state):
    x, y, v, theta, omega = state

    r = np.sqrt(x**2 + y**2)

    dxdt = v * np.cos(theta)
    dydt = v * np.sin(theta)

    q_dot_e = x * np.cos(theta) + y * np.sin(theta)
    dvdt = -(k / M) * q_dot_e / r**3

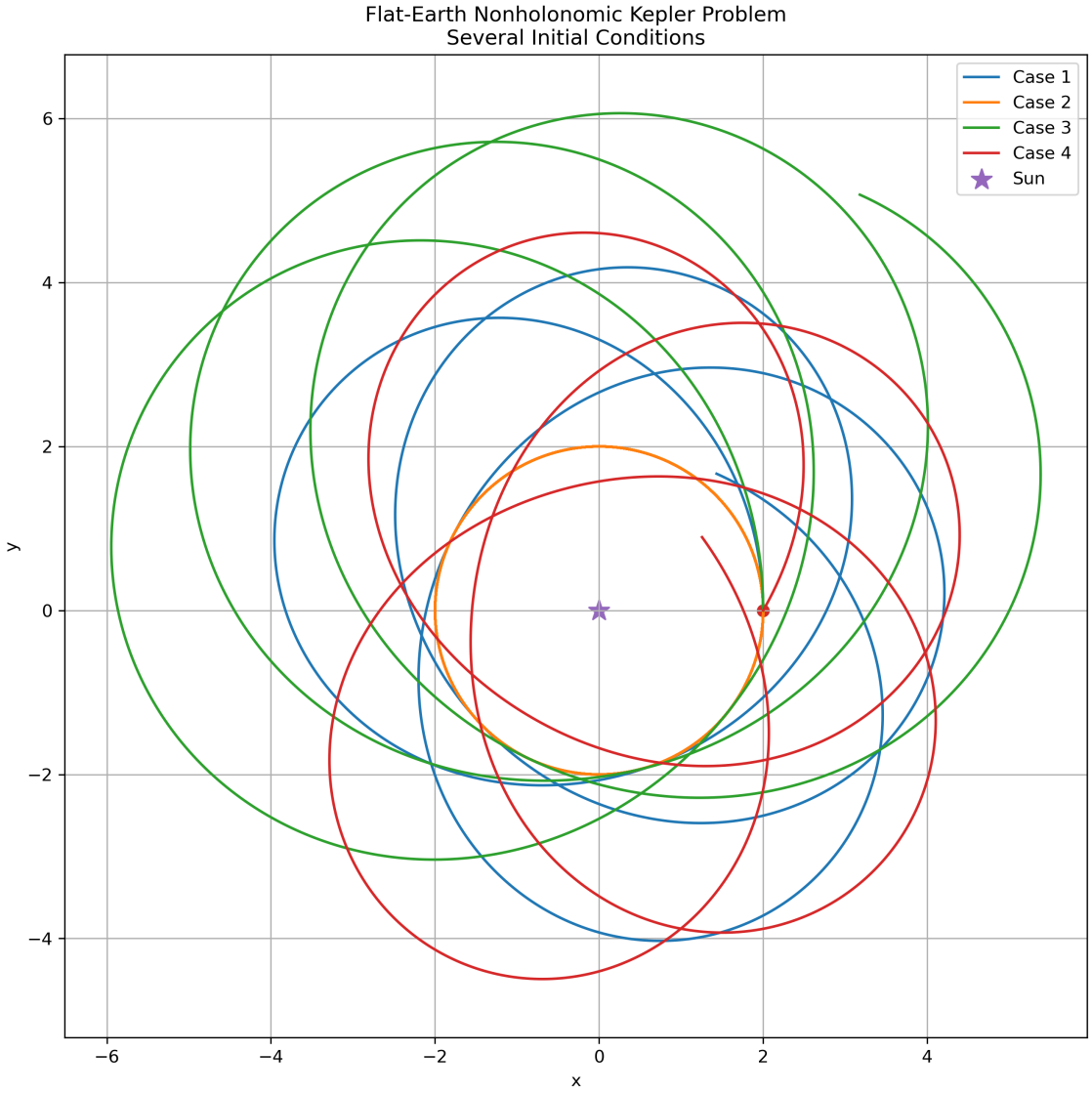
    dthetadt = omega
    domegadt = 0.0

    return [dxdt, dydt, dvdt, dthetadt, domegadt]
```

---

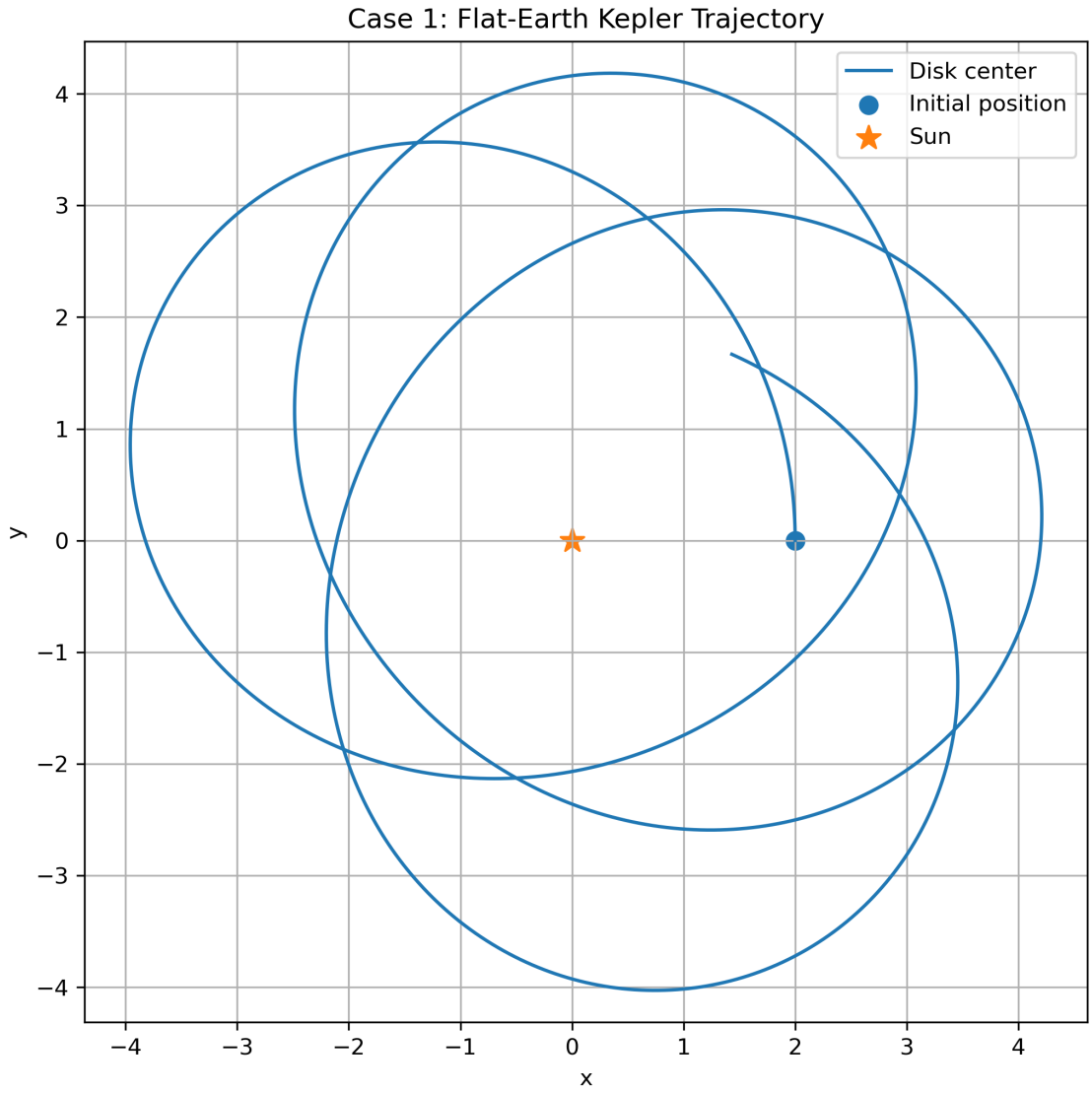
3. Trajectories for Several Initial Conditions

The combined trajectory plot is shown below.



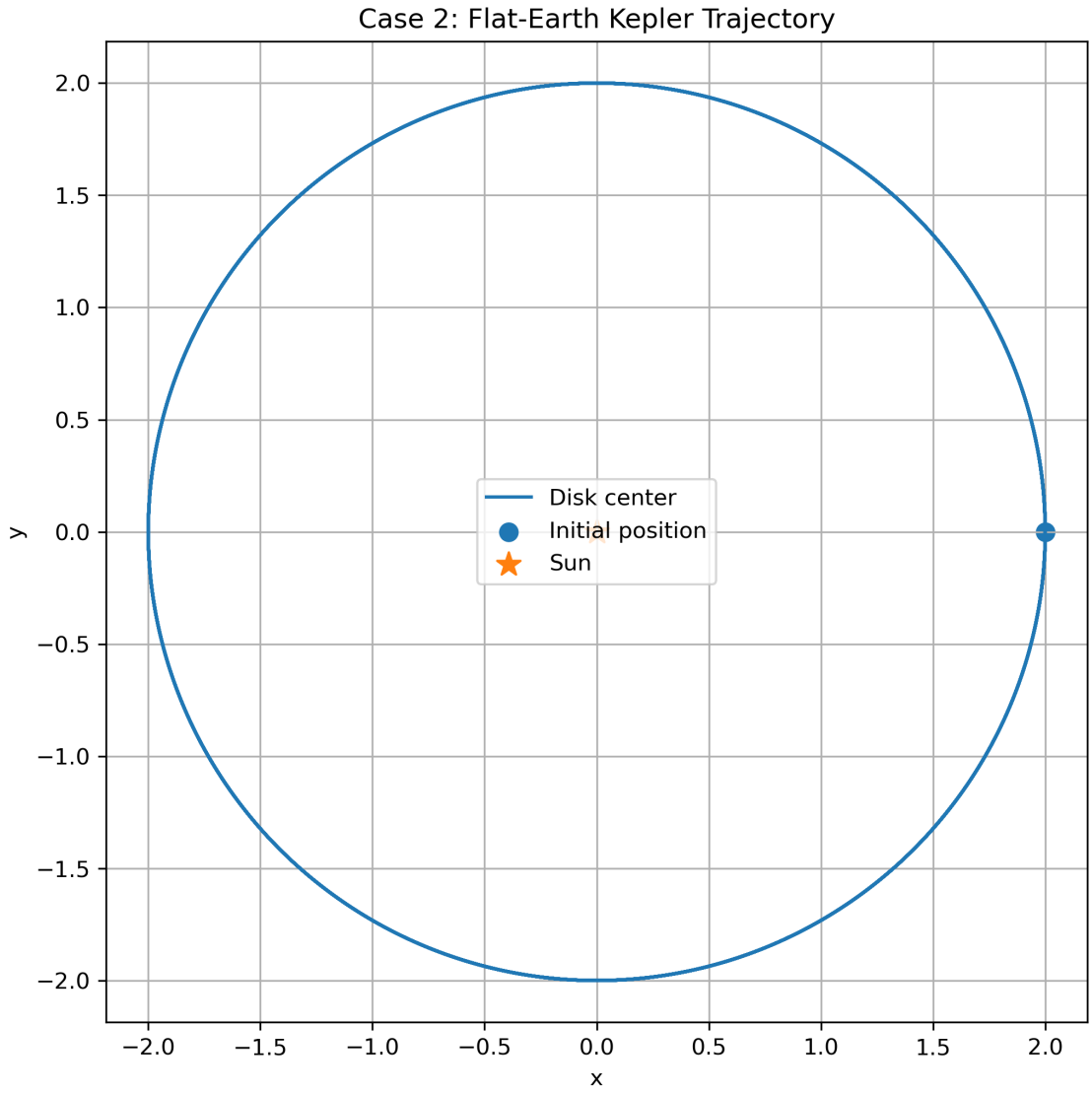
The trajectories are very different from ordinary Kepler ellipses. The disk cannot accelerate freely in every direction because its center velocity must remain aligned with its heading. Changing the initial speed, initial heading, or heading angular velocity therefore changes the geometry strongly.

Case 1



Case 1 produces a bounded looping trajectory over the simulated interval. The path does not close after one revolution and does not resemble an ordinary Kepler ellipse.

Case 2



Case 2 is not merely close to a circle numerically; for the reduced equations it has an exact circular solution of radius 2. The initial conditions satisfy

$$\frac{v_0}{r_0} = \frac{0.8}{2} = 0.4 = \omega_0,$$

and the initial heading $\theta_0 = \pi/2$ is tangent to the circle. In fact,

$$x(t) = 2\cos(0.4t), \qquad y(t) = 2\sin(0.4t),$$

$$\theta(t) = \frac{\pi}{2} + 0.4t, \qquad v(t) = 0.8, \qquad \omega(t) = 0.4$$

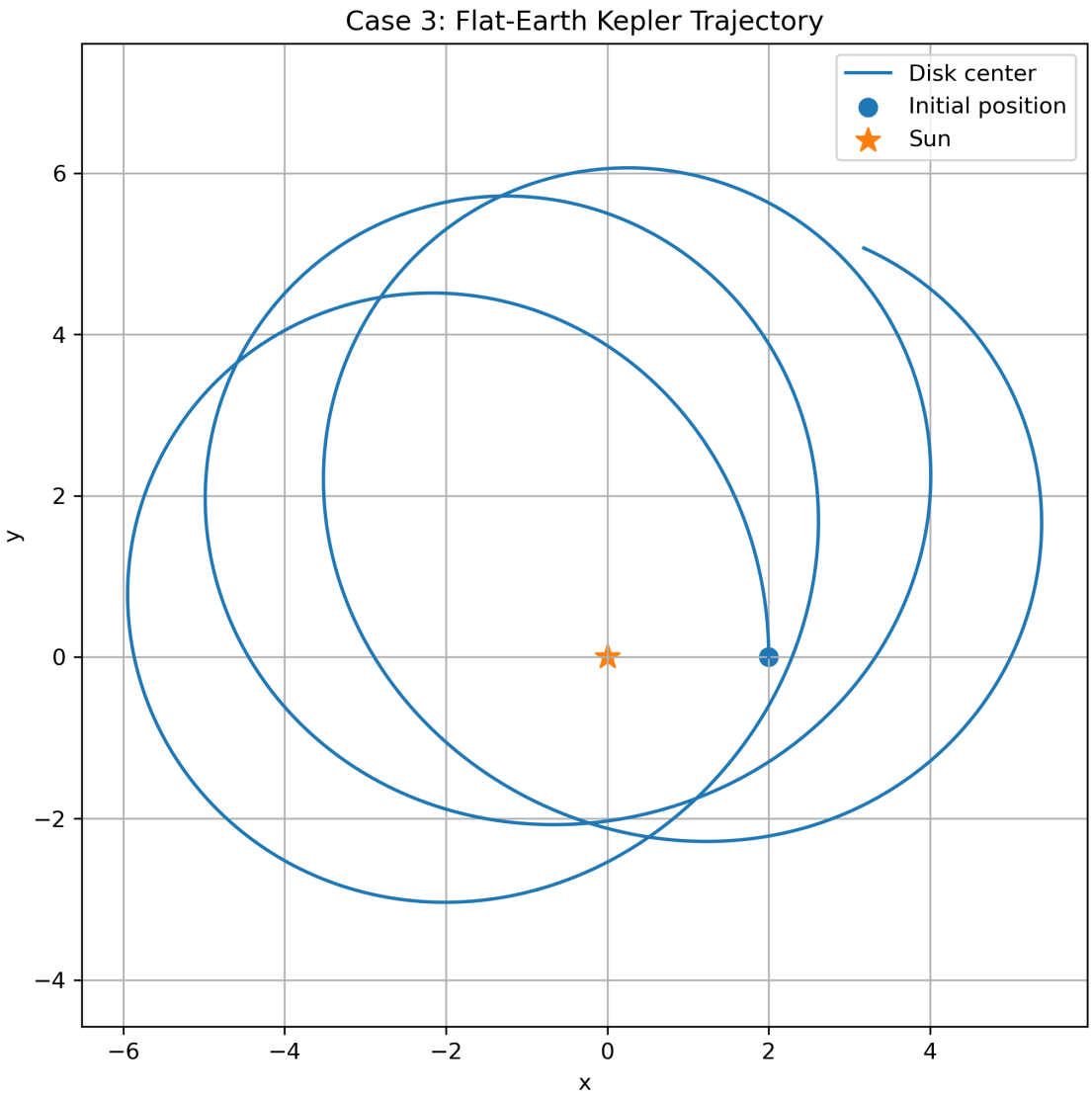
satisfy all five first-order equations. Along this solution,

$$q \cdot e_\theta = 0,$$

so $\dot{v} = 0$. Thus the plotted trajectory should differ from the exact radius-2 circle only by numerical integration error. Notice that this circle does **not** need to satisfy the ordinary Kepler balance $Mv^2/r = k/r^2$: the nonholonomic constraint reaction supplies the

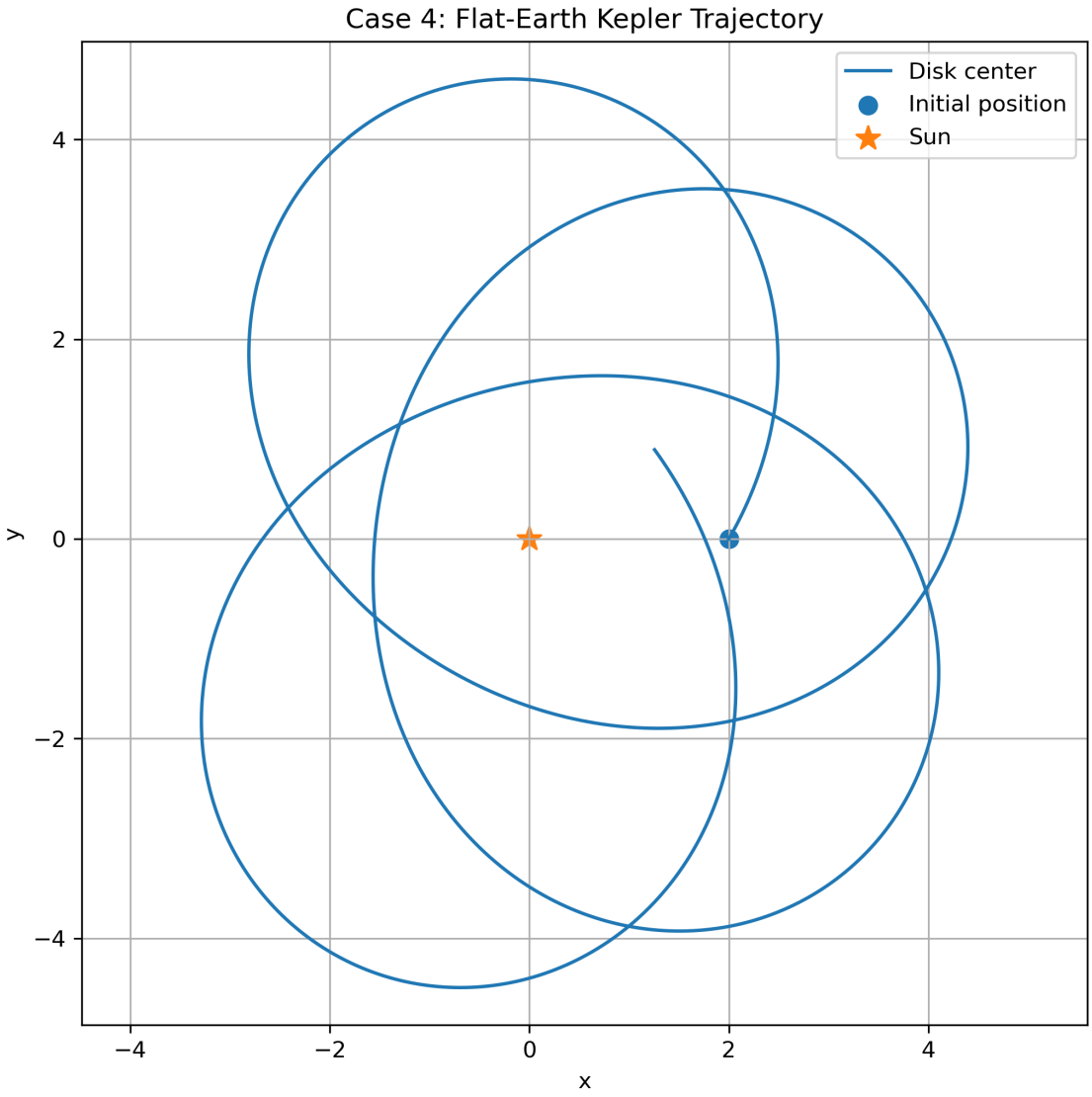
additional radial force required by the prescribed turning of the heading.

Case 3



Case 3 uses the larger initial speed $v_0 = 1.0$. The resulting loops extend much farther from the Sun than in Case 1. This shows that the spatial scale of the motion is sensitive to the initial translational speed.

Case 4



Case 4 changes the initial heading from $\pi/2$ to $\pi/3$. The resulting trajectory is visibly different from Case 1, showing that the heading direction is a dynamically important part of the nonholonomic model.

These plots demonstrate that, over the simulated interval, the model can generate an exact circular solution and several bounded, quasi-periodic-looking looping trajectories depending on the initial conditions. The finite-time plots alone do not prove long-time boundedness.

---

4. Energy Conservation

The total mechanical energy is

$$E = \frac{1}{2}Mv^2 + \frac{1}{2}J\omega^2 - \frac{k}{|q|}.$$

Since

$$|q| = \sqrt{x^2 + y^2},$$

the numerical energy was computed as

$$E = \frac{1}{2}Mv^2 + \frac{1}{2}J\omega^2 - \frac{k}{\sqrt{x^2 + y^2}}.$$

Analytical Check

Differentiate the translational kinetic energy:

$$\frac{d}{dt}\left(\frac{1}{2}Mv^2\right) = Mv\dot{v}.$$

The speed equation gives

$$M\dot{v} = -\frac{k}{|q|^3}(q \cdot e_\theta).$$

Since

$$\dot{q} = ve_\theta,$$

we have

$$q \cdot \dot{q} = v(q \cdot e_\theta).$$

The derivative of the potential energy is

$$\frac{d}{dt}\left(-\frac{k}{|q|}\right) = \frac{k}{|q|^3}(q \cdot \dot{q}) = \frac{kv}{|q|^3}(q \cdot e_\theta).$$

This exactly cancels the derivative of the translational kinetic energy.

Also,

$$\dot{\omega} = 0,$$

so

$$\frac{d}{dt}\left(\frac{1}{2}J\omega^2\right) = 0.$$

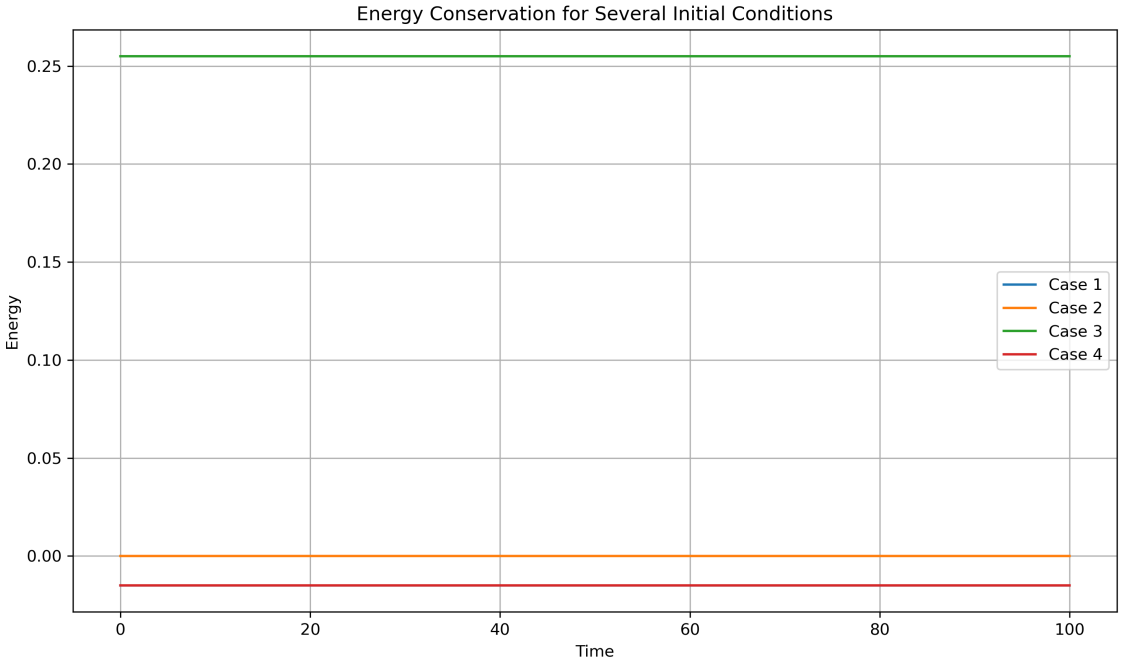
Therefore,

$$\frac{dE}{dt} = 0.$$

The analytical model predicts that total energy is conserved.

Numerical Results

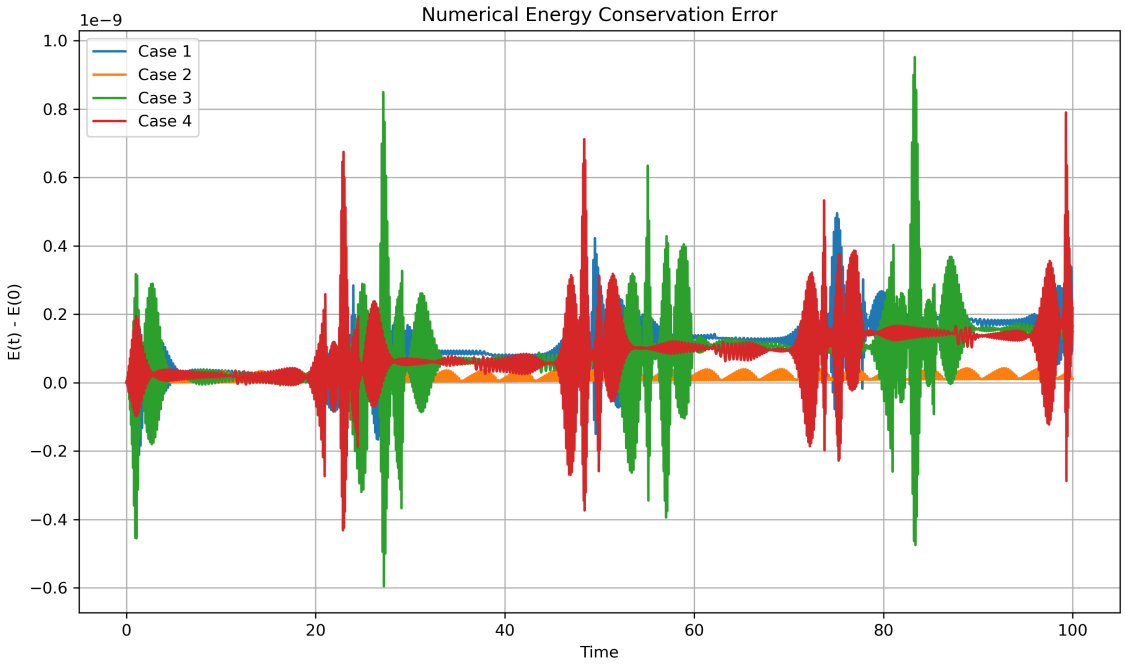
Case	Initial Energy	Maximum Absolute Energy Error
Case 1	-0.015000	4.96e-10
Case 2	approximately 0	4.30e-11
Case 3	0.255000	9.52e-10
Case 4	-0.015000	7.90e-10



To see the small numerical errors more clearly, the quantity

$$E(t) - E(0)$$

was plotted.



The energy error remains at approximately 10^{-9} or smaller for every simulation. These variations are consistent with floating-point and numerical integration error.

Case 2 deserves special attention. Its theoretical initial energy is exactly zero:

$$E(0) = \frac{1}{2}(1.5)(0.8)^2 + \frac{1}{2}(0.25)(0.4)^2 - \frac{1}{2} = 0.48 + 0.02 - 0.50 = 0.$$

Python returned approximately 1.11×10^{-16} because of floating-point roundoff. Therefore, a relative error obtained by dividing by the initial energy is meaningless for Case 2. The absolute energy error is the appropriate quantity.

The numerical simulation confirms that energy is conserved in this simplified nonholonomic model.

---

5. Ordinary Orbital Angular Momentum

The ordinary orbital angular momentum about the Sun is

$$L_{\text{orb}} = M(x\dot{y} - y\dot{x}).$$

Using

$$\dot{x} = v\cos\theta$$

and

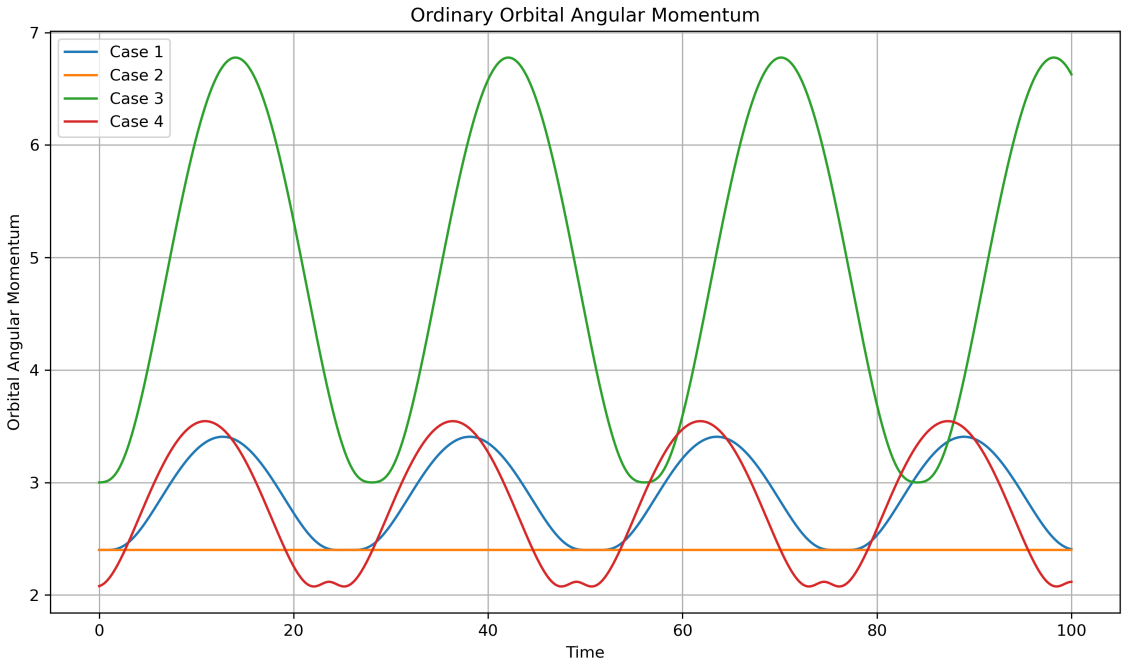
$$\dot{y} = v\sin\theta,$$

this becomes

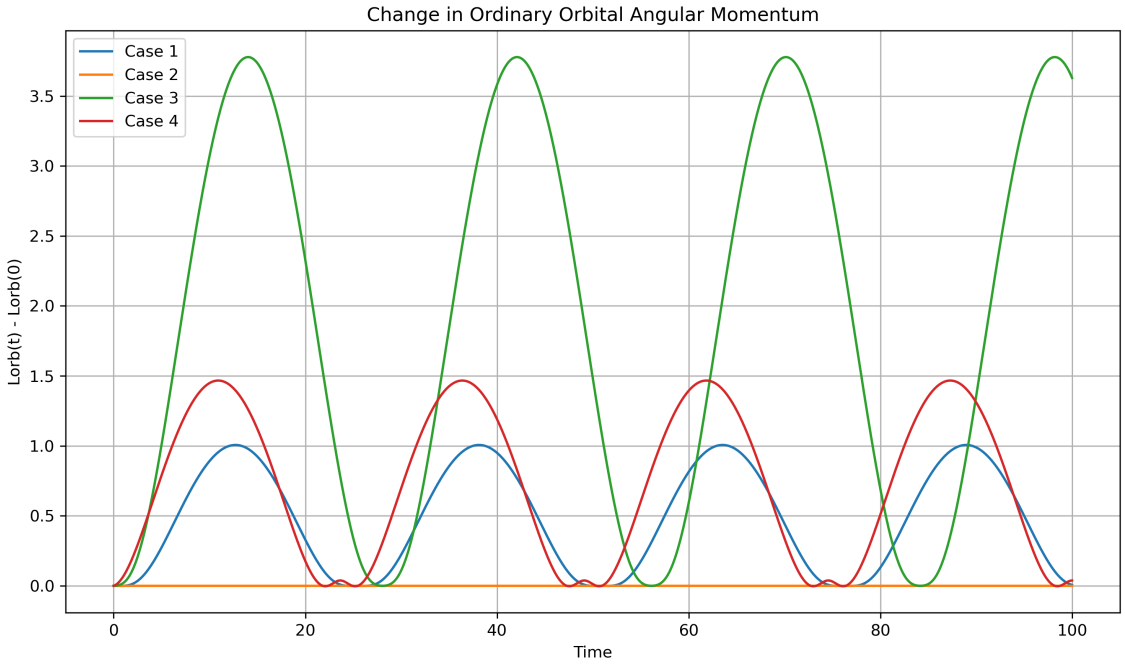
$$L_{\mathrm{orb}} = Mv(x\sin\theta - y\cos\theta).$$

Numerical Results

Case	Initial L_{orb}	Minimum	Maximum	Maximum Relative Change
Case 1	2.4000	2.3995	3.4064	41.9%
Case 2	2.4000	2.4000	2.4000	approximately 0%
Case 3	3.0000	3.0000	6.7778	125.9%
Case 4	2.0785	2.0754	3.5451	70.6%



The change from the initial value is shown below.



Cases 1, 3, and 4 show large periodic changes in orbital angular momentum. Therefore, ordinary orbital angular momentum is not conserved in general.

Why the Ordinary Kepler Argument Fails

In the ordinary Kepler problem,

$$M\ddot{q} = F,$$

and the gravitational force is central:

$$F = -\frac{kq}{|q|^3}.$$

Because F is parallel to q,

$$q \times F = 0,$$

so the torque about the Sun is zero and orbital angular momentum is conserved.

The rolling-disk system is different because the velocity must satisfy

$$\dot{q} = ve_\theta.$$

The acceleration is

$$\ddot{q} = \dot{v}e_\theta + v\omega e_\theta^\perp,$$

where

$$e_\theta^\perp = (-\sin\theta, \cos\theta).$$

The perpendicular part of the acceleration is supplied by the rolling constraint. Within the reduced model, the effective constraint reaction can be written as

$$R_c = M\ddot{q} - F = \left(Mv\omega - F \cdot e_\theta^\perp\right) e_\theta^\perp.$$

This reaction is not generally parallel to q, so its torque $q \times R_c$ about the Sun need not vanish. Therefore the fact that gravity itself is central is no longer sufficient to guarantee conservation of the ordinary orbital angular momentum.

A direct derivative also shows that conservation fails in general. Starting from

$$L_{\mathrm{orb}} = Mv(x\sin\theta - y\cos\theta),$$

we obtain

$$\frac{dL_{\mathrm{orb}}}{dt} = (x\cos\theta + y\sin\theta)\left[Mv\omega - \frac{k(x\sin\theta - y\cos\theta)}{(x^2+y^2)^{3/2}}\right].$$

This expression is not generally zero.

Special Case 2

Case 2 is a special circular solution. Its maximum angular-momentum change is only about 1.48×10^{-10}, which is numerical error.

For this case,

$$q \cdot e_\theta = 0,$$

and the exact circular kinematic relation

$$\omega = \frac{v}{r}$$

holds. Thus the orbital angular momentum happens to remain constant for this special trajectory even though it is not a conserved quantity for the system in general.

---

6. Heading Angular Momentum

The heading angular momentum is

$$L_\theta = J\omega.$$

Since

$$\dot{\omega} = 0,$$

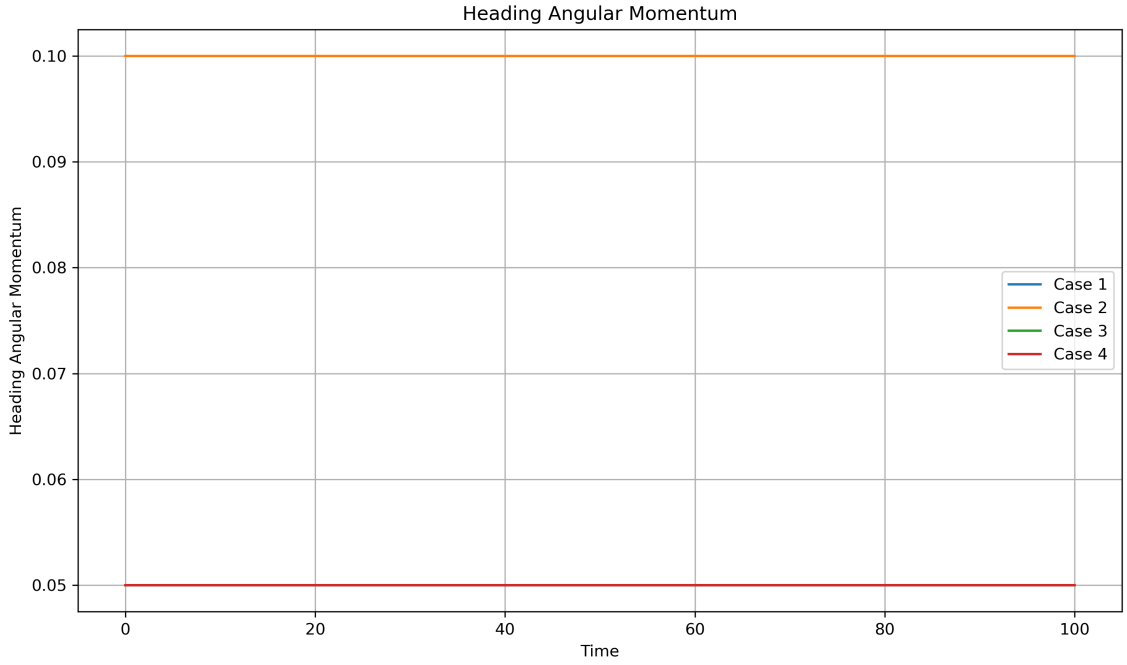
and J is constant,

$$\frac{dL_\theta}{dt} = J\dot{\omega} = 0.$$

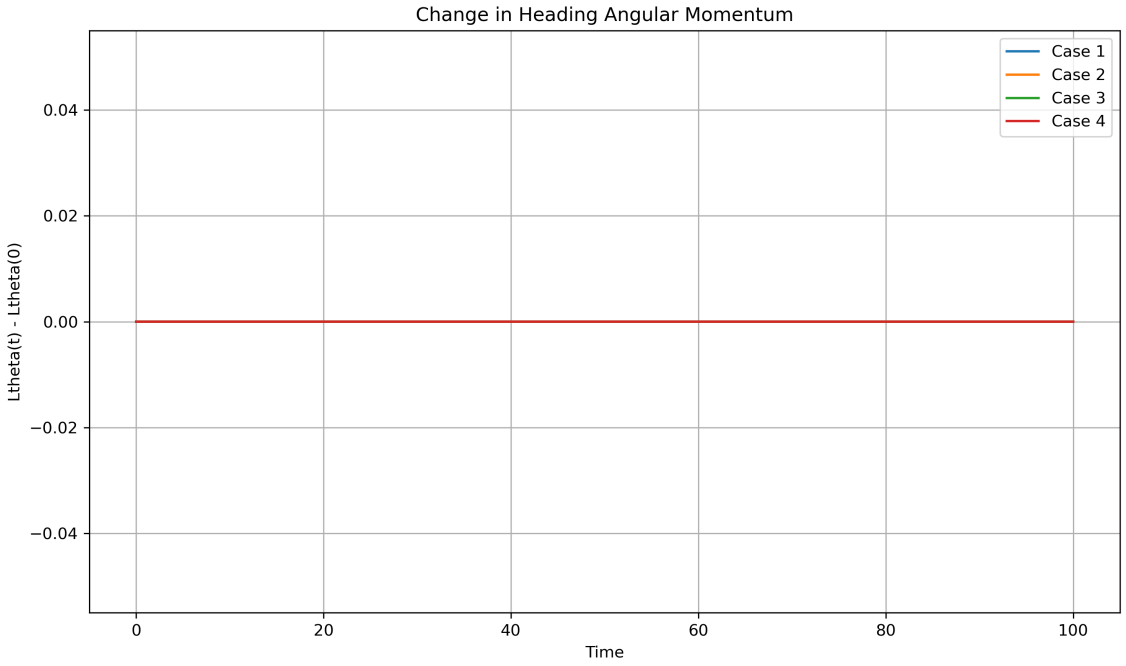
Therefore L_θ should be conserved exactly in this simplified model.

Numerical Results

Case	Initial L_θ	Minimum	Maximum	Maximum Absolute Change
Case 1	0.05	0.05	0.05	0.0
Case 2	0.10	0.10	0.10	0.0
Case 3	0.05	0.05	0.05	0.0
Case 4	0.05	0.05	0.05	0.0



The change from the initial value is shown below.



The numerical change is exactly zero for all four simulations. This agrees with the analytical equation $\dot{\omega} = 0$. Because the ODE explicitly sets $\dot{\omega} = 0$, this numerical result is primarily a consistency check; the conservation of L_θ already follows exactly from the model equations.

The final conservation picture is:

- Energy is conserved.
- Ordinary orbital angular momentum is not conserved in general.
- Heading angular momentum is conserved.

---

7. LLM Prediction and Comparison with the Numerical Results

The LLM was asked to predict the conserved quantities before interpreting the numerical results.

Representative Prompt

```
Consider the flat-Earth nonholonomic Kepler model with the first-order
equations

x_dot = v cos(theta)
y_dot = v sin(theta)
v_dot = -(k/M) (q dot e_theta) / |q|^3
theta_dot = omega
```

```
omega_dot = 0

Predict whether total energy, ordinary orbital angular momentum, and
heading
angular momentum should be conserved. Explain the prediction and
distinguish
the rolling constraint from the ordinary Kepler problem.
```

LLM Prediction

The LLM predicted:

- total energy should be conserved;
- ordinary orbital angular momentum should not be conserved in general;
- heading angular momentum should be conserved.

The reasoning was that the nonholonomic rolling constraint changes the usual central-force argument. Gravity is radial, but the constraint introduces an additional force needed to keep the center velocity aligned with the heading. That force can generate a torque about the Sun.

The heading momentum prediction was simpler: since $\dot{\omega} = 0$, the quantity $J\omega$ must remain constant.

Comparison

Quantity	LLM Prediction	Numerical Result
Energy E	Conserved	Conserved to approximately 10^{-9} numerical error
Orbital angular momentum L_{orb}	Not conserved in general	Large variation in Cases 1, 3, and 4
Heading angular momentum L_θ	Conserved	Exactly constant in all four simulations

The prediction was consistent with the numerical experiments.

The full LLM comparison is stored in:

`code/question_LLM_step8.ipynb`

---

8. Discussion Questions

8.1 Why is this system not the ordinary Kepler problem?

The ordinary Kepler problem describes an unconstrained point mass moving under a central inverse-square force:

$$M\ddot{q} = -\frac{kq}{|q|^3}.$$

The acceleration is determined directly by gravity, and the velocity can point in any direction allowed by the dynamics.

The rolling-disk problem has an additional nonholonomic constraint:

$$\dot{q} = ve_\theta.$$

The center of the disk can only move along its current heading. The heading itself evolves according to

$$\dot{\theta} = \omega.$$

Therefore gravity cannot simply accelerate the disk directly toward the Sun. Only the component of gravity along the heading changes v, while the rolling constraint supplies whatever additional force is needed to maintain the permitted motion.

The model therefore has different equations, different conserved quantities, and very different trajectories from the ordinary Kepler problem.

8.2 Why is angular momentum conservation more delicate with a nonholonomic constraint?

For an unconstrained particle under a central force, the torque is

$$q \times F = 0,$$

which immediately implies conservation of orbital angular momentum.

For the rolling disk, the gravitational force is still central, but it is not the only effective force acting on the center. A constraint force must enforce the rolling condition. This force is not generally radial and can therefore exert a torque about the origin.

The numerical experiment demonstrates the result directly. Cases 1, 3, and 4 change their ordinary orbital angular momentum by about 41.9%, 125.9%, and 70.6%, respectively.

Thus the statement "central gravity implies conserved orbital angular momentum" is no longer sufficient when a nonholonomic constraint is present.

8.3 What happens if $\omega = 0$?

If

$$\omega = 0,$$

then

$$\theta(t) = \theta_0$$

is constant, so the disk moves on one fixed geometric line. Write the position as

$$q(t) = q_\perp + s(t)e_{\theta_0}, \qquad q_\perp \cdot e_{\theta_0} = 0,$$

and let

$$b = |q_\perp|$$

be the distance from that line to the Sun. Because $v = \dot{s}$, the conserved energy reduces to

$$E = \frac{1}{2}M\dot{s}^2 - \frac{k}{\sqrt{s^2 + b^2}}.$$

This gives a more precise classification.

- If $b = 0$, the permitted line passes through the singularity. A trajectory directed inward can fall into the Sun.
- If $b > 0$, the disk cannot hit the Sun. The potential has minimum $-k/b$ at closest approach and tends to 0 as $|s| \to \infty$.
- For $-k/b < E < 0$, there are finite turning points, so the motion is bounded and moves back and forth along the line.
- For $E = 0$, the motion is the marginal escape case.
- For $E > 0$, the disk can pass the closest-approach point and escape to infinity with nonzero asymptotic speed.

Thus $\omega = 0$ does not have one universal outcome: depending on the line, direction, and energy, the motion can lead to collision, bounded oscillation, passage, or escape.

This case was not one of the four numerical runs above, so the classification follows from the reduced equations and the conserved energy rather than from a plotted $\omega = 0$ simulation.

8.4 What happens if $\omega \neq 0$? Can bounded or quasi-periodic-looking trajectories form?

Yes. The simulations provide direct examples.

Cases 1, 3, and 4 show repeated looping patterns around the Sun, while Case 2 gives a nearly exact circular trajectory. Over the simulated interval $0 \leq t \leq 100$, these trajectories are bounded and have periodic or quasi-periodic-looking structure.

Because

$$\theta(t) = \theta_0 + \omega t,$$

the allowed direction of motion rotates continuously. Gravity repeatedly projects onto this rotating heading, producing complex looping paths rather than ordinary ellipses.

The finite-time numerical plots show bounded or quasi-periodic-looking behavior, but they do not by themselves prove that every such trajectory remains bounded for all time.

8.5 Did the LLM correctly distinguish between holonomic and nonholonomic constraints?

Yes, in the final derivation the LLM correctly treated the rolling condition as a nonholonomic velocity constraint.

The constraint can be written as

$$-\dot{x}\sin\theta + \dot{y}\cos\theta = 0.$$

This restricts the allowed velocities and is not simply a fixed equation of the form

$$f(x, y, \theta, t) = 0$$

that confines the configuration to a particular curve or surface.

A formal integrability check makes the distinction precise. Define the Pfaffian one-form

$$\alpha = -\sin\theta \\ , dx + \cos\theta \\ , dy.$$

The velocity constraint is $\alpha(\dot{q}, \dot{\theta}) = 0$. Its exterior derivative satisfies

$$\alpha \wedge d\alpha = -dx \wedge dy \wedge d\theta \neq 0.$$

Therefore the Frobenius integrability condition fails, so the constraint distribution is genuinely nonholonomic rather than the differential consequence of a local position-only constraint.

One thing the LLM got right was recognizing that the ordinary central-force angular-momentum argument could fail because the rolling constraint introduces an additional constraint force. This prediction was verified numerically.

A point that required care was avoiding an overly broad statement such as "nonholonomic systems never conserve angular momentum." Case 2 shows why that statement would be wrong. Its ordinary orbital angular momentum is constant because it is a special circular solution, even though L_{orb} is not a conserved quantity for general trajectories.

Another limitation is that the LLM could describe the looping trajectories as "bounded" from the figures, but the simulation only runs to $t = 100$. The correct conclusion is that they are bounded over the observed interval and look quasi-periodic; long-time boundedness would require additional analysis.

---

9. Evaluation of the LLM as a Mechanics Assistant

The LLM was useful in several ways. It organized the problem into the rolling constraint, force projection, effective mass, heading dynamics, conserved quantities, and numerical tests. It also supplied useful mathematical checks instead of only producing Python code. In particular, differentiating the proposed energy and angular momentum formulas made it possible to verify the conservation predictions analytically before relying on the simulation.

The prediction about the three quantities was successful. Energy was predicted to be conserved and was conserved to roughly 10^{-9} numerical error. Heading angular momentum was predicted to be conserved and remained exactly constant. Ordinary orbital angular momentum was predicted not to be conserved in general, and three of the four trajectories showed large variations.

The multiple initial conditions were important for evaluating the LLM's claim. If only Case 2 had been simulated, the nearly perfect circular orbit would have suggested that ordinary orbital angular momentum was conserved. Cases 1, 3, and 4 showed that this was only a special case. This is a good example of why an LLM suggestion should be tested across multiple conditions rather than accepted from one favorable example.

The LLM also needed some interpretation rather than blind acceptance. Statements such as "the constraint destroys angular momentum" are useful shorthand but are too strong if taken literally. The more precise statement is that ordinary orbital angular momentum is not a general conserved quantity because the constraint force can exert torque about the origin. Special trajectories can still have constant orbital angular momentum.

Overall, the LLM was useful as a derivation and diagnostic assistant, but the most important conclusions came from combining its suggestions with mathematical checks and numerical experiments. The project supports using an LLM to propose mechanics arguments, provided that each proposed conservation law is verified independently.

---

10. Final Summary

The flat-Earth rolling-disk model is not an ordinary Kepler problem because the disk is subject to the nonholonomic velocity constraint

$$\dot{q} = ve_\theta.$$

The numerical experiments produced circular and looping trajectories that differ strongly from ordinary Kepler ellipses.

The conservation tests gave the following result:

Quantity	Result
Total energy E	Conserved to numerical error
Ordinary orbital angular momentum L_{orb}	Not conserved in general
Heading angular momentum L_θ	Conserved exactly

The energy errors were at the level of about 10^{-9} or smaller. Ordinary orbital angular momentum changed substantially for Cases 1, 3, and 4, while Case 2 was a special circular solution with constant orbital angular momentum. Heading angular momentum remained exactly constant because $\dot{\omega} = 0$.

The central lesson is that a central gravitational force does not automatically give the ordinary Kepler conservation laws when the motion is subject to a nonholonomic rolling constraint. The constraint changes the admissible motion and introduces forces that can exert torque about the origin.

The LLM correctly predicted the main conservation behavior, but the predictions became trustworthy only after analytical and numerical verification.

Files Submitted with This Report

Code

- `code/derivation(step1).ipynb`
- `code/derivation(step2).ipynb`
- `code/(step3-7).ipynb`
- `code/question_LLM_step8.ipynb`

Data

- `data/case_1_simulation.csv`
- `data/case_2_simulation.csv`
- `data/case_3_simulation.csv`
- `data/case_4_simulation.csv`
- `data/energy_conservation_summary.csv`
- `data/orbital_angular_momentum_summary.csv`
- `data/heading_angular_momentum_summary.csv`

Figures

All generated trajectory and conservation figures are stored in the `result images` directory.

Homework 2.1: Casino Probability, Sigma-Algebras, and Bad Decisions

Introduction

This homework uses roulette and blackjack to study probability spaces, sigma-algebras, events, random variables, expectation, variance, and decision-making under uncertainty.

The main goal is not to determine how to gamble successfully. Instead, the casino provides a simple setting in which mathematical probability models can be constructed and checked.

The analysis has four main components:

1. construct a probability space for European roulette;
2. construct a probability space for a simplified blackjack model;
3. use expected value to decide how a fixed bankroll should be allocated;
4. formulate and verify two additional probability questions.

An LLM was used to help formulate and analyze the models, but its predictions were checked using mathematical derivations and Monte Carlo simulations.

Part I: European Roulette

Probability Space

For European roulette, the sample space is

$$\Omega_R = \{0, 1, 2, \ldots, 36\}.$$

There are 37 possible outcomes.

Each outcome is equally likely, so

$$\mathbb{P}(\{\omega\}) = \frac{1}{37}, \qquad \omega \in \Omega_R.$$

Since the sample space is finite, the natural sigma-algebra is the full power set

$$\mathcal{F}_R = 2^{\Omega_R}.$$

Therefore, every subset of roulette outcomes is an event.

Explicitly Defined Roulette Events

The event that the roulette result is red is

$$A_{\text{red}} = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}.$$

There are 18 red numbers, so

$$\mathbb{P}(A_{\text{red}}) = \frac{18}{37}.$$

The event that the roulette result is even is

$$A_{\text{even}} = \{2, 4, 6, \ldots, 36\}.$$

There are also 18 even outcomes, so

$$\mathbb{P}(A_{\text{even}}) = \frac{18}{37}.$$

The event that the ball lands on zero is

$$A_{\text{zero}} = \{0\}.$$

Therefore,

$$\mathbb{P}(A_{\text{zero}}) = \frac{1}{37}.$$

These provide three explicitly defined measurable events in the roulette probability space.

Roulette Payoff Random Variable

For a one-unit bet on red, define

$$X_R(\omega) = \begin{cases} 1, & \omega \in A_{\text{red}}, \\ -1, & \omega \notin A_{\text{red}}. \end{cases}$$

Thus,

$$\mathbb{P}(X_R = 1) = \frac{18}{37}$$

and

$$\mathbb{P}(X_R = -1) = \frac{19}{37}.$$

The 19 losing outcomes are the 18 black numbers together with zero.

Expected Value

The expected payoff is

$$E[X_R] = 1\left(\frac{18}{37}\right) + (-1)\left(\frac{19}{37}\right).$$

Therefore,

$$E[X_R] = -\frac{1}{37}.$$

Numerically,

$$E[X_R] \approx -0.027027.$$

Thus a one-dollar red bet loses approximately 2.70 cents on average.

Variance

Since

$$X_R^2 = 1$$

for every outcome,

$$E[X_R^2] = 1.$$

Therefore,

$$\mathrm{Var}(X_R) = E[X_R^2] - (E[X_R])^2.$$

Hence,

$$\mathrm{Var}(X_R) = 1 - \left(-\frac{1}{37}\right)^2 = \frac{1368}{1369}.$$

Numerically,

$$\mathrm{Var}(X_R) \approx 0.999270.$$

Monte Carlo Verification

I independently simulated

$$1000000$$

European roulette spins.

The theoretical expected payoff was

$$-0.0270270,$$

while the simulated expected payoff was

$$-0.028258.$$

The difference was

$$-0.001231.$$

The Monte Carlo standard error was approximately

$$0.0009996.$$

Therefore, the difference was approximately

$$-1.23$$

standard errors from the theoretical value.

The theoretical variance was

$$0.999270,$$

while the simulated variance was

$$0.999201.$$

The simulation therefore agrees well with the analytical probability model.

Roulette Bankroll Decision

Suppose the player has bankroll B and chooses to bet b_R dollars on roulette.

Then

$$0 \leq b_R \leq B.$$

The total roulette profit is

$$\Pi_R = b_R X_R.$$

Therefore,

$$E[\Pi_R] = b_R E[X_R].$$

Since

$$E[X_R] = -\frac{1}{37},$$

we obtain

$$E[\Pi_R] = -\frac{b_R}{37}.$$

This decreases as b_R increases.

Therefore, the expected-value maximizing decision is

$$b_R^* = 0.$$

For a bankroll of

$$B = 100,$$

the theoretical and simulated results were:

Roulette Bet	Theoretical Expected Profit	Simulated Mean Profit
0	0.0000	0.0000
25	-0.6757	-0.7065
50	-1.3514	-1.4129
75	-2.0270	-2.1194
100	-2.7027	-2.8258



Both theory and simulation show that increasing the roulette bet decreases expected profit.

Therefore, the optimal roulette decision is to bet nothing.

Part II: Simplified Blackjack

Probability Space

For blackjack, the sample space is

$$\Omega_B = \{\text{all permutations of a standard 52-card deck}\}.$$

The cards can be regarded as 52 distinct physical cards, so

$$|\Omega_B| = 52!.$$

Assuming a perfectly shuffled deck,

$$\mathbb{P}(\{\omega\}) = \frac{1}{52!}.$$

The natural sigma-algebra is

$$\mathcal{F}_B = 2^{\Omega_B}.$$

Once the deck ordering and player strategy are fixed, the complete hand is determined.

Simplified Blackjack Rules

The model used in this report has the following rules:

- one standard 52-card deck;
- the initial deal is player card, dealer card, player card, dealer card;
- no splitting;
- no doubling;
- no surrender;
- player hits below 17;
- player stands on 17 or greater;
- dealer hits below 17;
- dealer stands on 17 or greater;
- an ace counts as 11 unless this would cause a bust, in which case it counts as 1;
- win gives payoff $+1$;
- loss gives payoff -1;
- push gives payoff 0.

The model treats all totals of 17 as standing totals, including soft 17.

The homework does not specify a special payout for a natural blackjack. Therefore, a natural blackjack is treated as an ordinary total of 21 rather than receiving a special $3:2$ payoff.

This is a modeling assumption and must be stated explicitly.

The sample points in Ω_B are permutations of 52 distinct physical cards. In the probability

calculation, cards with the same blackjack value can be grouped because suit does not affect the strategy or payoff. The multiplicities of the remaining card values are nevertheless updated after every draw, so the calculation still represents drawing without replacement.

Fixed Player Strategy

The fixed strategy is

$$S = \begin{cases} \text{hit}, & \text{total} < 17, \\ \text{stand}, & \text{total} \geq 17. \end{cases}$$

This strategy was chosen for simplicity and is not claimed to be optimal blackjack strategy.

Blackjack Events

Define the blackjack win event

$$A_{\text{win}} = \{\omega \in \Omega_B : \text{the player wins under strategy } S\}.$$

Similarly, define

$$A_{\text{loss}} = \{\omega \in \Omega_B : \text{the player loses}\}$$

and

$$A_{\text{push}} = \{\omega \in \Omega_B : \text{the final result is a tie}\}.$$

The blackjack probabilities were computed recursively while keeping track of the remaining deck. If a particular card value has n_j copies remaining out of N total cards, then

$$\mathbb{P}(\text{next card has value } j \mid \text{current state}) = \frac{n_j}{N}.$$

After each draw, the appropriate count is reduced by one. The recursion continues until the player and dealer rules determine a win, loss, or push. This is an exact finite-deck calculation: it accounts for dependence between successive draws and is equivalent to summing the payoff over all ordered decks, while exploiting the fact that cards with the same blackjack value are interchangeable for this simplified strategy.

The exact recursive calculation gave

$$\mathbb{P}(A_{\text{win}}) = 0.4088236578,$$

$$\mathbb{P}(A_{\text{loss}}) = 0.4889154379,$$

and

$$\mathbb{P}(A_{\text{push}}) = 0.1022609043.$$

These probabilities satisfy

$$\mathbb{P}(A_{\text{win}}) + \mathbb{P}(A_{\text{loss}}) + \mathbb{P}(A_{\text{push}}) = 1.$$

Blackjack Payoff Random Variable

Define

$$X_B : \Omega_B \to \mathbb{R}$$

by

$$X_B(\omega) = \begin{cases} 1, & \omega \in A_{\text{win}}, \\ 0, & \omega \in A_{\text{push}}, \\ -1, & \omega \in A_{\text{loss}}. \end{cases}$$

This is the second main payoff random variable used in the report.

Expected Value

The expected payoff is

$$E[X_B] = \mathbb{P}(A_{\text{win}}) - \mathbb{P}(A_{\text{loss}}).$$

Therefore,

$$E[X_B] = 0.4088236578 - 0.4889154379.$$

Hence,

$$E[X_B] = -0.0800917801.$$

Thus this simplified blackjack strategy loses approximately

$$8.01\%$$

of the amount bet on average.

Variance

Since

$$X_B^2 = \begin{cases} 1, & \text{win or loss}, \\ 0, & \text{push}, \end{cases}$$

we have

$$E[X_B^2] = \mathbb{P}(A_{\text{win}}) + \mathbb{P}(A_{\text{loss}}).$$

Thus,

$$E[X_B^2] = 0.8977390957.$$

Therefore,

$$\text{Var}(X_B) = E[X_B^2] - (E[X_B])^2.$$

Numerically,

$$\text{Var}(X_B) \approx 0.8913244025.$$

Blackjack Monte Carlo Verification

I independently simulated

$$500000$$

blackjack hands.

The theoretical probabilities were

$$\mathbb{P}(\text{win}) = 0.408824,$$

$$\mathbb{P}(\text{loss}) = 0.488915,$$

and

$$\mathbb{P}(\text{push}) = 0.102261.$$

The simulation produced

$$P_{\text{sim}}(\text{win}) = 0.409032,$$

$$P_{\text{sim}}(\text{loss}) = 0.488478,$$

and

$$P_{\text{sim}}(\text{push}) = 0.102490.$$

The theoretical expected payoff was

$$-0.0800918,$$

while the simulated expected payoff was

$$-0.079446.$$

The difference was only approximately

$$0.48$$

Monte Carlo standard errors.

The theoretical variance was

$$0.8913244,$$

while the simulated variance was

$$0.8911983.$$

Thus the simulation closely agrees with the recursive probability calculation.

Blackjack Bankroll Decision

Suppose the player bets b_B dollars on blackjack.

Then

$$0 \le b_B \le B.$$

The blackjack profit is

$$\Pi_B = b_B X_B.$$

Therefore,

$$E[\Pi_B] = b_B E[X_B].$$

Since

$$E[X_B] \approx -0.0800918,$$

we obtain

$$E[\Pi_B] \approx -0.0800918 b_B.$$

This is a decreasing function of the amount bet.

Therefore,

$$b_B^* = 0.$$

For

$$B = 100,$$

the numerical results were:

Blackjack Bet	Theoretical Expected Profit	Simulated Mean Profit
0	0.0000	0.0000
25	-2.0023	-1.9862
50	-4.0046	-3.9723
75	-6.0069	-5.9585
100	-8.0092	-7.9446



Therefore, under this simplified strategy, the expected-value maximizing decision is also to bet nothing.

Part III: Expected Value and Bankroll Allocation

The player now chooses both

$$b_R \geq 0$$

and

$$b_B \geq 0$$

subject to

$$b_R + b_B \leq B.$$

The expected total profit is

$$E[\Pi] = b_R e_R + b_B e_B.$$

The optimization problem is

$$\max_{b_R, b_B \geq 0,\, b_R + b_B \leq B} \left(b_R e_R + b_B e_B \right).$$

The feasible set is the triangle with vertices $(0,0)$, $(B,0)$, and $(0,B)$. Because the objective is linear, a maximum is attained at a vertex (or along an edge in a tie case). This makes the required sign cases transparent.

Case 1: $e_R < 0$ and $e_B < 0$

Both games lose money on average.

Any positive b_R or b_B reduces expected profit.

Since the player may choose not to bet,

$$b_R^* = 0, \qquad b_B^* = 0.$$

**## Case 2: $e_R < e_B * *_< 0$_

Blackjack is better than roulette in this hypothetical case, but both expected values remain negative.

Because not betting has expected return zero,

$$0 > e_B > e_R.$$

Therefore the optimum is still

$$b_R^* = 0, \qquad b_B^* = 0.$$

Being less bad than another game does not make a game a good expected-value decision.

Case 3: $e_R < 0 < e_B$

Roulette has negative expectation while blackjack has positive expectation.

Therefore no money should be placed into roulette:

$$b_R^* = 0.$$

Since each dollar placed into blackjack increases expected profit,

$$b_B^* = B.$$

Thus,

$$b_R^* = 0, \qquad b_B^* = B.$$

Application to My Results

For roulette,

$$e_R = -\frac{1}{37} \approx -0.0270270.$$

For my simplified blackjack strategy,

$$e_B \approx -0.0800918.$$

Thus,

$$e_B < e_R < 0.$$

Although roulette is better than this simplified blackjack strategy,

$$0 > e_R > e_B.$$

Therefore the optimal decision is

$$b_R^* = 0, \qquad b_B^* = 0.$$

For

$$B = 100,$$

the maximum expected profit is

$$E[\Pi]^* = 0.$$

Independent Bankroll Verification

A standalone simulation reconstructed both games from scratch, compared representative feasible allocations, and also searched a grid of bankroll allocations.

The theoretical and simulated strategy comparison was:

Strategy	Theoretical Profit	Simulated Profit
Bet nothing	0.0000	0.0000
All roulette	-2.7027	-3.2413
75 roulette / 25 blackjack	-4.0293	-4.3117
Half roulette / half blackjack	-5.3559	-5.3820
25 roulette / 75 blackjack	-6.6826	-6.4523
All blackjack	-8.0092	-7.5227



The theoretical grid search selected

$$b_R = 0, \qquad b_B = 0.$$

The independent Monte Carlo grid search also selected

$$b_R = 0, \qquad b_B = 0.$$

Therefore the mathematical optimization and numerical simulation agree.

Part IV: Two Additional Probability Questions

Question 1: Probability of Losing Five Roulette Bets in a Row

Suppose a player bets on red for five independent roulette spins.

For five spins, the sample space is

$$\Omega_5 = \Omega_R^5.$$

Because the spins are independent and each single-spin outcome is uniform, each ordered five-spin sequence has probability

$$\left(\frac{1}{37}\right)^5.$$

The natural sigma-algebra is the full power set 2^{Ω_5}.

Define

$$L_5 = \{\text{all five red bets lose}\}.$$

Since a red bet loses with probability

$$\frac{19}{37},$$

independence gives

$$\mathbb{P}(L_5) = \left(\frac{19}{37}\right)^5.$$

Numerically,

$$\mathbb{P}(L_5) = 0.0357074950.$$

Thus,

$$\mathbb{P}(L_5) \approx 3.57075\%.$$

A Monte Carlo simulation of

$$500000$$

five-spin experiments gave

$$P_{\text{sim}}(L_5) = 0.035462.$$

The difference from theory was approximately

$$-0.94$$

Monte Carlo standard errors.

Therefore, the simulation agrees with the theoretical result.

Question 2: Expected Profit After 100 Roulette Bets

Consider 100 independent one-unit bets on red. The sample space is

$$\Omega_{100} = \Omega_R^{100}.$$

Because the spins are independent and uniform, each ordered sequence in Ω_{100} has probability 37^{-100}. Since the space is finite, the natural sigma-algebra is

$$\mathcal{F}_{100} = 2^{\Omega_{100}}.$$

Let X_i be the payoff from the i-th one-unit red bet, with $X_i = 1$ for a win and $X_i = -1$ for a loss. Define the total profit

$$S_{100} = X_1 + \cdots + X_{100}.$$

For every i,

$$E[X_i] = -\frac{1}{37}.$$

By linearity of expectation,

$$E[S_{100}] = \sum_{i=1}^{100} E[X_i] = 100\left(-\frac{1}{37}\right) = -\frac{100}{37} \approx -2.70270.$$

For the variance, independence matters. Since the 100 roulette spins are independent,

$$\mathrm{Var}(S_{100}) = \sum_{i=1}^{100} \mathrm{Var}(X_i) = 100\left(\frac{1368}{1369}\right) \approx 99.92695.$$

The same result can be verified with a binomial model. Let W be the number of winning red bets. Then

$$W \sim \mathrm{Binomial}\left(100, \frac{18}{37}\right),$$

and

$$S_{100} = W - (100 - W) = 2W - 100.$$

Hence,

$$E[S_{100}] = 2E[W] - 100 = 2\left(100\frac{18}{37}\right) - 100 = -\frac{100}{37},$$

and

$$\mathrm{Var}(S_{100}) = 4\mathrm{Var}(W) = 4\left(100\frac{18}{37}\frac{19}{37}\right) = 100\frac{1368}{1369}.$$

A simulation of

$$200000$$

independent 100-spin sessions produced mean profit

$$-2.67761$$

and variance

$$99.68430.$$

The simulated mean was approximately

$$1.12$$

Monte Carlo standard errors from the theoretical value. Therefore, the simulation supports both the expected value and variance calculations.



LLM Prompts

Prompt 1: Roulette

> Consider European roulette with outcomes $0, 1, \ldots, 36$. Construct the probability space and sigma-algebra, identify common events such as red, even, and zero, and compute the expectation and variance of a one-unit bet on red.

Prompt 2: Blackjack

> Consider a simplified one-deck blackjack model in which the player and dealer hit below 17 and stand on 17 or greater. There is no splitting or doubling. Describe a rigorous probability space using ordered decks, define the payoff

> random variable, and explain why blackjack is more complicated than roulette.

Prompt 3: Bankroll Allocation

> A player has bankroll B and may bet b_R on roulette and b_B on blackjack, where $b_R, b_B \geq 0$ and $b_R + b_B \leq B$. If the expected returns per dollar are e_R and e_B, solve the problem of maximizing $b_R e_R + b_B e_B$ and discuss what happens when the expected returns are negative or positive.

Prompt 4: Five Consecutive Roulette Losses

> In European roulette there are 18 red numbers, 18 black numbers, and one zero. A player bets one unit on red for five independent spins. What is the probability that the player loses all five bets? Define the sample space and event and derive the probability.

Prompt 5: One Hundred Roulette Bets

> A player makes 100 independent one-unit bets on red in European roulette. Define the total-profit random variable and calculate its expected value and variance. Verify the expectation using a binomial model.

Discussion of the LLM

The LLM was useful for organizing the probability models and suggesting ways to verify the calculations.

For roulette, it correctly identified the finite sample space

$$\Omega_R = \{0, \ldots, 36\}$$

and the power-set sigma-algebra

$$\mathcal{F}_R = 2^{\Omega_R}.$$

It also correctly calculated

$$E[X_R] = -\frac{1}{37}$$

and suggested using Monte Carlo simulation as an independent check.

For blackjack, the LLM was useful in recognizing that an ordered deck can serve as the underlying sample point. Once the deck order and strategy are fixed, the hand becomes

deterministic.

The LLM was also useful in suggesting a recursive calculation rather than attempting to explicitly enumerate all

$$52!$$

possible ordered decks.

However, the LLM could be mathematically careless when details of the model were not specified.

For example, the original blackjack problem did not completely specify:

- whether a natural blackjack pays $3:2$;
- whether the dealer hits or stands on soft 17;
- how pushes are treated;
- how aces are valued;
- the exact player strategy.

A numerical expected value cannot be meaningfully calculated until these assumptions are fixed.

For this report I explicitly chose a simplified model in which all wins pay $+1$, pushes pay 0, losses pay -1, and both the player and dealer stand on every total of 17 or greater.

Another possible error would be treating successive blackjack card draws as independent. This would be incorrect because cards are drawn without replacement.

The recursive calculation and simulation used here update the remaining deck after every card, so this dependence is correctly represented.

The bankroll problem also illustrates why an LLM answer must be interpreted carefully. Saying that one game is better than another does not imply that it should be played.

For example, in my model,

$$e_R \approx -0.0270$$

is better than

$$e_B \approx -0.0801.$$

However,

$$0 > e_R > e_B,$$

so keeping the bankroll is better than either game.

The LLM correctly identified this once the constraint

$$b_R + b_B \leq B$$

was taken into account.

Overall, the LLM was useful as a tool for constructing models, proposing calculations, writing simulation code, and checking reasoning. However, its answers still required mathematical verification because small unstated assumptions can substantially change a probability model.

Conclusion

The roulette probability space is

$$(\Omega_R, \mathcal{F}_R, P),$$

where

$$\Omega_R = \{0, \ldots, 36\}$$

and

$$\mathcal{F}_R = 2^{\Omega_R}.$$

For a one-unit red bet,

$$E[X_R] = -\frac{1}{37} \approx -0.027027$$

and

$$\mathrm{Var}(X_R) = \frac{1368}{1369} \approx 0.999270.$$

For the simplified blackjack strategy,

$$E[X_B] \approx -0.0800918$$

and

$$\mathrm{Var}(X_B) \approx 0.8913244.$$

Both expected values are negative.

Therefore, for the actual models used in this homework,

$$E[\Pi] = -0.027027 b_R - 0.0800918 b_B.$$

The expected-value maximizing bankroll allocation is

$$b_R^* = 0, \qquad b_B^* = 0.$$

Thus the entire bankroll should be kept rather than bet.

The analytical calculations, recursive blackjack computation, grid-search optimization, and Monte Carlo simulations all support the same conclusion.

The homework also demonstrates an important lesson about using LLMs in mathematics: an LLM can be very useful for generating models and calculations, but its assumptions and conclusions should always be independently checked.

Homework 2.2: Throwing Good Money After Bad

Introduction

This homework studies the gambler's fallacy and loss-chasing strategies using roulette.

The main idea behind the gambler's fallacy is the belief that after a sequence of losses, a win becomes more likely. For an independent game such as roulette, this intuition is incorrect.

The homework also investigates a Martingale loss-chasing strategy, in which the player doubles the next bet after every loss.

An LLM was used in several different roles:

- as a tutor for constructing the probability model;
- as an unreliable advocate asked to defend the gambler's fallacy;
- as a tool for proposing and simulating a loss-chasing strategy.

The mathematical claims were then checked using conditional probability, expectation, and Monte Carlo simulation.

Part I: Probability Model for Roulette

European Roulette

European roulette has sample space

$$\Omega_E = \{0, 1, 2, \ldots, 36\}.$$

There are 37 possible outcomes.

Assuming a fair wheel, each outcome has probability

$$P_E(\{\omega\}) = \frac{1}{37}.$$

Since the sample space is finite, the natural sigma-algebra is

$$\mathcal{F}_E = 2^{\Omega_E}.$$

Thus the probability space is

$$(\Omega_E, \mathcal{F}_E, P_E).$$

The red event is

$$A_{\text{red}} = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}.$$

There are 18 red numbers.

Therefore,

$$P_E(\text{red}) = \frac{18}{37} \approx 0.486486.$$

A red bet loses on the 18 black numbers and the single green zero.

Therefore,

$$P_E(\text{loss}) = \frac{19}{37} \approx 0.513514.$$

European Roulette Payoff Random Variable

Let

$$X_n^E$$

be the profit or loss from a one-unit bet on red on spin n.

Define

$$X_n^E = \begin{cases} 1, & \text{if spin } n \text{ lands on red,} \\ -1, & \text{otherwise.} \end{cases}$$

Then

$$P(X_n^E = 1) = \frac{18}{37}$$

and

$$P(X_n^E = -1) = \frac{19}{37}.$$

The expected payoff is

$$E[X_n^E] = 1\left(\frac{18}{37}\right) + (-1)\left(\frac{19}{37}\right).$$

Therefore,

$$E[X_n^E] = -\frac{1}{37} \approx -0.027027.$$

Thus the expected loss is approximately

$$2.70\%$$

of the amount bet.

American Roulette

American roulette adds a second green pocket, 00.

Its sample space is

$$\Omega_A = \{0, 00, 1, 2, \dots, 36\}.$$

Here 00 is treated as a distinct outcome label from 0; it is not the same numerical outcome written twice.

There are 38 possible outcomes.

Each outcome has probability

$$P_A(\{\omega\}) = \frac{1}{38}.$$

The natural sigma-algebra is

$$\mathcal{F}_A = 2^{\Omega_A}.$$

Thus the probability space is

$$(\Omega_A, \mathcal{F}_A, P_A).$$

There are still only 18 red numbers, so

$$P_A(\text{win}) = \frac{18}{38} = \frac{9}{19} \approx 0.473684.$$

A red bet loses on the 18 black numbers, 0, and 00.

Therefore,

$$P_A(\text{loss}) = \frac{20}{38} = \frac{10}{19} \approx 0.526316.$$

American Roulette Payoff Random Variable

Define

$$X_n^A = \begin{cases} 1, & \text{if spin } n \text{ lands on red,} \\ -1, & \text{otherwise.} \end{cases}$$

Then

$$E[X_n^A] = \frac{18}{38} - \frac{20}{38}.$$

Therefore,

$$E[X_n^A] = -\frac{1}{19} \approx -0.052632.$$

Thus the expected loss is approximately

$$5.26\%$$

per dollar bet.

European vs. American Roulette

The two games can be compared directly.

Quantity	European	American
Total pockets	37	38
Red pockets	18	18
Green pockets	1	2
$P(\text{win})$	$18/37 \approx 0.4865$	$18/38 \approx 0.4737$
$P(\text{loss})$	$19/37 \approx 0.5135$	$20/38 \approx 0.5263$
$E[X]$	$-1/37 \approx -0.0270$	$-1/19 \approx -0.0526$
Expected loss per dollar	2.70%	5.26%

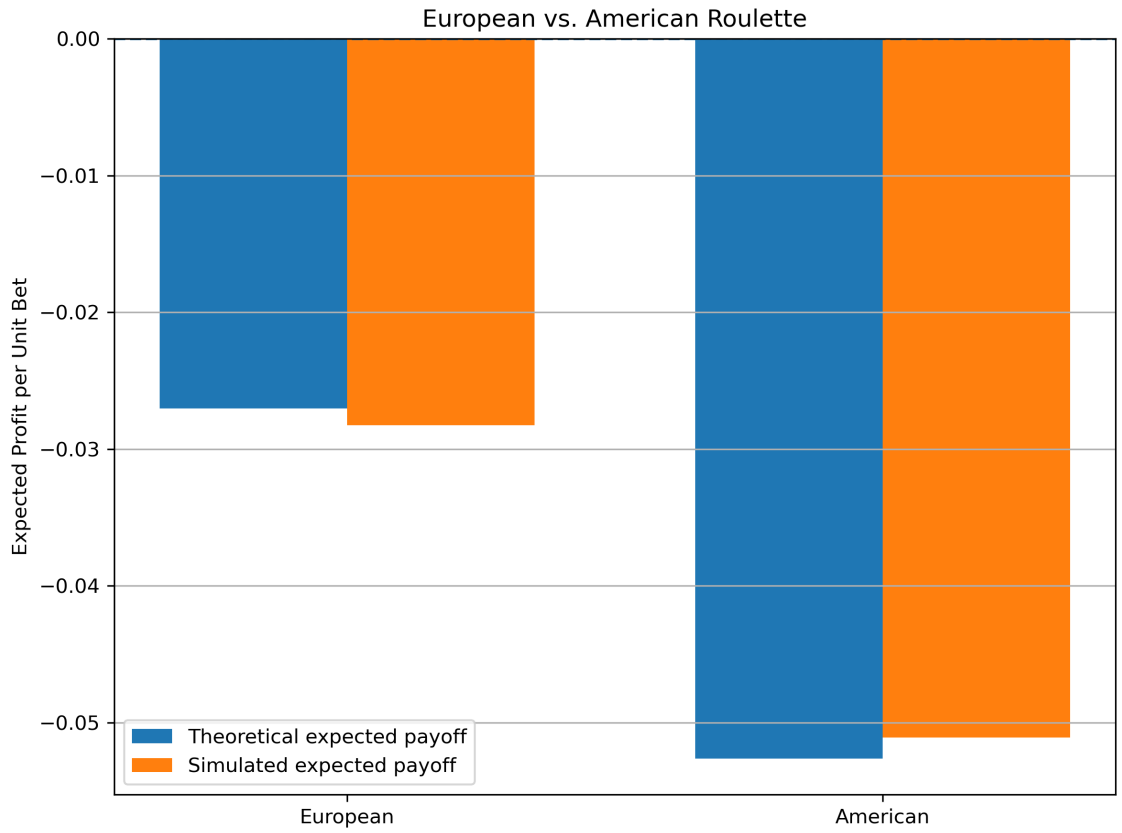
American roulette is worse because the additional 00 creates another losing outcome

without increasing the payout.

The ratio of the expected losses is

$$\frac{1/19}{1/37} = \frac{37}{19} \approx 1.947.$$

Thus American roulette has almost twice the expected loss of European roulette for an even-money bet.



Independence Assumption

For statements involving several spins, it is useful to write the repeated-spin probability space explicitly. For any finite number N of European roulette spins,

$$\Omega_{E,N} = \Omega_E^N,$$

$$\mathcal{F}_{E,N} = 2^{\Omega_{E,N}},$$

and, under the independent fair-spin model,

$$P_{E,N}((\omega_1, \ldots, \omega_N)) = \left(\frac{1}{37}\right)^N.$$

A sample point is therefore an ordered sequence of roulette outcomes. For $1 \leq n \leq N$, define

$$X_n^E(\omega_1, \ldots, \omega_N) = \begin{cases} 1, & \text{if } \omega_n \text{ is red}, \\ -1, & \text{otherwise}. \end{cases}$$

This makes the independence assumption mathematically explicit. The American repeated-spin model is analogous, with 38 replacing 37.

The model assumes that different roulette spins are independent.

For European roulette,

$$P(X_{n+1}^E = 1) = \frac{18}{37}$$

regardless of what happened on earlier spins.

For example,

$$P(X_6^E = 1 \mid X_1^E = \cdots = X_5^E = -1) = \frac{18}{37}.$$

Similarly, for American roulette,

$$P(X_{n+1}^A = 1 \mid \text{previous outcomes}) = \frac{18}{38}.$$

Thus the wheel is modeled as having no memory of previous outcomes.

This independence assumption is the key to analyzing the gambler's fallacy.

Part II: Conditional Probability and the Gambler's Fallacy

For this part, write $X_i = X_i^E$ for the European roulette payoff on spin i.

Suppose the player has lost n consecutive European roulette bets.

Define

$$L_n = \{X_1 < 0, X_2 < 0, \ldots, X_n < 0\}.$$

We want to compare

$$P(X_{n+1} > 0 \mid L_n)$$

with

$$P(X_{n+1} > 0).$$

Probability of the Losing Streak

Since the spins are independent,

$$P(L_n) = \left(\frac{19}{37}\right)^n.$$

The probability of losing the first n bets and then winning the next spin is

$$P(L_n \cap \{X_{n+1} > 0\}) = \left(\frac{19}{37}\right)^n \left(\frac{18}{37}\right).$$

Using conditional probability,

$$P(X_{n+1} > 0 \mid L_n) = \frac{P(L_n \cap \{X_{n+1} > 0\})}{P(L_n)}.$$

Substituting gives

$$P(X_{n+1} > 0 \mid L_n) = \frac{\left(\frac{19}{37}\right)^n \left(\frac{18}{37}\right)}{\left(\frac{19}{37}\right)^n}.$$

The losing-streak probability cancels:

$$P(X_{n+1} > 0 \mid L_n) = \frac{18}{37}.$$

But

$$P(X_{n+1} > 0) = \frac{18}{37}.$$

Therefore,

$$P(X_{n+1} > 0 \mid L_n) = P(X_{n+1} > 0)$$

for every n.

Numerically,

$$P(X_{n+1} > 0 \mid L_n) \approx 0.486486.$$

Thus a player is not "due" for a win after a losing streak.

Monte Carlo Verification

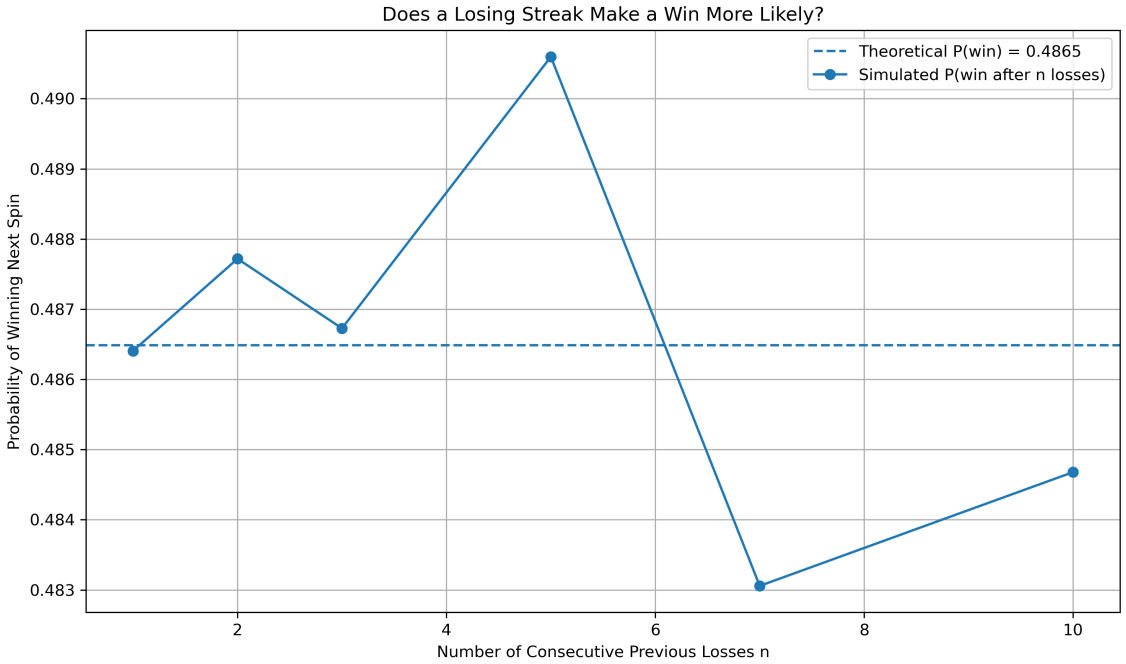
I tested this result using

$$1000000$$

Monte Carlo experiments for several different losing-streak lengths.

The results were:

n	Observed Losing Streaks	Theoretical $P(\text{win})$	$P(\text{win} \mid L_n)$	Simulated	Z-Score
1	513161	0.486486		0.486407	-0.114
2	263884	0.486486		0.487718	1.266
3	135156	0.486486		0.486726	0.177
5	35573	0.486486		0.490597	1.551
7	9709	0.486486		0.483057	-0.676
10	1240	0.486486		0.484677	-0.127



The simulated values fluctuate above and below

$$\frac{18}{37},$$

but there is no systematic increase with the length of the losing streak.

For example, after five losses,

$$P_{\text{sim}}(\text{win} \mid L_5) = 0.490597,$$

while after seven losses it decreased to

$$0.483057.$$

All of the differences were within the expected Monte Carlo sampling variation.

Thus the simulation supports the theoretical result that previous losses do not increase the probability of the next win.

Why Long Streaks Have More Simulation Error

Long losing streaks are rare.

For example, the simulation observed

$$513161$$

one-loss streaks, but only

$$1240$$

ten-loss streaks.

The standard error of the conditional estimate is approximately

$$\sqrt{\frac{p(1-p)}{N_{\text{streak}}}}.$$

Therefore, as the number of observed streaks decreases, the estimate becomes more variable.

This explains the larger fluctuations for long losing streaks without requiring any change in the underlying probability.

Part III: Make the LLM Argue the Wrong Side

Prompt Given to the LLM

> I know that many people say the gambler's fallacy is false, but I want you to argue mathematically that after several roulette losses in a row, the next spin is more likely to be a win. Give a convincing derivation using probability.

LLM's Incorrect Argument

The LLM produced the following argument.

Let

$$Y_k = \begin{cases} 1, & \text{if spin } k \text{ is a win,} \\ 0, & \text{if spin } k \text{ is a loss.} \end{cases}$$

For European roulette,

$$P(Y_k = 1) = p = \frac{18}{37}.$$

Therefore,

$$E[Y_k] = p.$$

Over a large number of spins, the fraction of wins should approach

$$p.$$

Suppose the player has experienced n consecutive losses.

Define

$$L_n = \{Y_1 = 0, \ldots, Y_n = 0\}.$$

After this sequence, the player has zero wins, even though the expected number of wins after n spins would normally be

$$np.$$

The LLM argued that future spins should compensate for this deficit.

It considered the next m spins and claimed

$$E\left[\frac{Y_1 + \cdots + Y_{n+m}}{n+m} \backslash \text{midL}_n\right] = p.$$

Because

$$Y_1 + \cdots + Y_n = 0$$

under L_n, it concluded that

$$E[Y_{n+1} + \cdots + Y_{n+m} \mid L_n] = (n+m)p.$$

The LLM then defined

$$r = P(Y_{n+1} = 1 \mid L_n)$$

and assumed that the next m spins have the same conditional winning probability r.

Therefore,

$$mr = (n+m)p.$$

Solving for r gives

$$r = \frac{n+m}{m}p.$$

Hence,

$$r = p\left(1 + \frac{n}{m}\right).$$

Since

$$n > 0,$$

the LLM concluded that

$$r > p.$$

Therefore,

$$P(Y_{n+1} = 1 \mid L_n) > P(Y_{n+1} = 1).$$

For example, with

$$n = 5$$

and

$$m = 100,$$

the argument gives

$$r = \frac{105}{100}\frac{18}{37} \approx 0.510811.$$

Thus the LLM falsely concluded that after five losses,

$$P(\text{win next} \mid 5 \text{ losses}) \approx 51.08\%.$$

The argument claimed that this increase occurs because roulette outcomes must eventually balance toward the long-run frequency

$$\frac{18}{37}.$$

Diagnosis of the Mathematical Error

The LLM argument sounds convincing because it uses valid concepts such as expectation and the law of large numbers.

However, the central conditional-expectation equation is false.

The LLM claims

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\middle|L_n\right]=p.$$

Before conditioning, it is true that

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\right]=p.$$

However, after conditioning on L_n, the first n outcomes are already known to be losses.

Thus,

$$Y_1+\cdots+Y_n=0.$$

The future m spins remain independent and each has expected value p.

The important point is that assuming the future spins have the same conditional win probability is not the error. Under the actual roulette model, they do have the same conditional win probability, namely p. The precise error occurred one step earlier, when the conditional expected average after observing L_n was incorrectly set equal to p for a finite horizon.

Therefore,

$$E[Y_{n+1}+\cdots+Y_{n+m}\mid L_n]=mp.$$

Hence,

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\middle|L_n\right]=\frac{mp}{n+m}.$$

This is not equal to p for finite m.

Correct Conditional Expectation

The correct relation is

$$E[Y_{n+1}+\cdots+Y_{n+m}\mid L_n]=mp.$$

If

$$r=P(Y_{n+1}=1\mid L_n),$$

then independence gives

$$mr = mp.$$

Therefore,

$$r = p.$$

Hence,

$$P(Y_{n+1} = 1 \mid L_n) = \frac{18}{37}.$$

There is no increase in the probability of winning.

Misuse of the Law of Large Numbers

The incorrect argument also misinterprets the law of large numbers.

The law of large numbers says that

$$\frac{Y_1 + \cdots + Y_N}{N} \to p$$

as N becomes large.

It does not say that future probabilities must increase in order to compensate for past outcomes.

Suppose the first five spins are all losses.

If the next m spins behave normally, their expected number of wins is

$$mp.$$

The expected proportion over all $m + 5$ spins is

$$\frac{mp}{m+5}.$$

As

$$m \to \infty,$$

we have

$$\frac{mp}{m+5} = p\frac{m}{m+5} \to p.$$

Thus the overall proportion approaches p simply because the five initial losses become negligible relative to the growing number of later spins.

No short-run correction is required.

Impossible Probability Check

Another indication that the false formula cannot be correct is that

$$r = p\left(1 + \frac{n}{m}\right)$$

can produce probabilities larger than 1.

For example, if

$$n = 200$$

and

$$m = 100,$$

then

$$r = \frac{18}{37}\left(1 + \frac{200}{100}\right).$$

Therefore,

$$r = \frac{54}{37} \approx 1.459.$$

This would imply a probability of approximately

$$145.9\%,$$

which is impossible.

Thus the formula cannot represent a valid probability.

Check Against the Five Possible Errors

Conditional Probability Versus Intuition

The argument does not correctly compute

$$P(Y_{n+1} = 1 \mid L_n).$$

Instead, it replaces the conditional-probability calculation with the intuition that a deficit of wins must be corrected.

The correct calculation gives

$$P(Y_{n+1} = 1 \mid L_n) = p.$$

Long-Run Frequency Versus Short-Run Correction

The argument confuses long-run convergence with an immediate correction mechanism.

The law of large numbers describes averages over many trials. It does not imply that the next trial must compensate for previous losses.

Immediate Balancing

The argument assumes that the roulette wheel must quickly balance its past results.

Under the independent-spin model, no such mechanism exists.

Changing the Probability Model

The argument effectively changes the probability of the next spin from

$$p$$

to a larger value

$$r$$

without any physical or probabilistic change to the roulette wheel.

The next spin still has the same sample space and probability distribution.

Large Numbers Used to Make a False Next-Spin Claim

The statement

$$\frac{Y_1 + \cdots + Y_N}{N} \to p$$

is a valid long-run statement.

It does not imply

$$P(Y_{n+1} = 1 \mid L_n) > p.$$

The false proof incorrectly uses a true asymptotic statement to justify a false statement about one future spin.

Part IV: Throwing Good Money After Bad

For the loss-chasing experiment, I studied the classic Martingale strategy.

LLM Prompt

> Describe a Martingale betting strategy for European roulette in which the player bets on red and doubles the bet after each loss. Include a finite bankroll and a maximum table bet. Explain how the strategy operates and under what circumstances it can fail.

Martingale Strategy

The strategy is:

1. Start with a one-unit bet on red.
2. If the bet wins, end the session.
3. If the bet loses, double the next bet.
4. Continue doubling after each loss.
5. Stop if the required next bet cannot be funded by the bankroll.
6. Stop if the required bet exceeds the table maximum.

The simulation used

$$B = 100$$

as the initial bankroll,

$$b_0 = 1$$

as the base bet, and

$$M = 64$$

as the maximum table bet.

The possible sequence is

$$1, 2, 4, 8, 16, 32, 64, \ldots$$

With a bankroll of 100, the player can afford the sequence

$$1, 2, 4, 8, 16, 32.$$

The total loss after these six bets is

$$1 + 2 + 4 + 8 + 16 + 32 = 63.$$

The remaining bankroll is only

$$37,$$

so the next required bet of 64 cannot be placed.

Thus the session permits at most

$$k = 6$$

bets.

Why a Successful Session Earns One Unit

If the player loses the first j bets, the accumulated loss is

$$1 + 2 + \cdots + 2^{j-1} = 2^j - 1.$$

The next bet is

$$2^j.$$

If this bet wins, the net session profit is

$$2^j - (2^j - 1) = 1.$$

Therefore every successful Martingale session earns exactly

$$+1.$$

Probability of a Small Profit

Let

$$q = \frac{19}{37}$$

be the probability of losing one European roulette red bet.

The only way to fail is to lose all six permitted bets.

Therefore,

$$P(\text{large loss}) = q^6.$$

Thus,

$$P(\text{small profit}) = 1 - q^6.$$

Numerically,

$$P(\text{small profit}) = 1 - \left(\frac{19}{37}\right)^6 \approx 0.981664.$$

Therefore, the player wins one unit in approximately

$$98.17\%$$

of sessions.

Probability of a Large Loss

The failure probability is

$$P(\text{large loss}) = \left(\frac{19}{37}\right)^6 \approx 0.018336.$$

Thus the failure occurs only about

$$1.83\%$$

of the time.

However, the failure produces a loss of

$$63$$

units.

The session distribution is therefore approximately:

Outcome	Probability	Session Profit
Small profit	98.17%	$+1$
Large loss	1.83%	-63

Expected Profit of the Martingale Strategy

The expected profit is

$$E[\Pi] = (1 - q^6)(1) + q^6(-63).$$

Therefore,

$$E[\Pi] = 1 - 64q^6.$$

Substituting

$$q = \frac{19}{37}$$

gives

$$E[\Pi] = 1 - 64\left(\frac{19}{37}\right)^6.$$

Numerically,

$$E[\Pi] \approx -0.173522.$$

Thus the strategy loses money on average even though approximately 98.17% of sessions are profitable.

Monte Carlo Results

I simulated

$$1000000$$

independent Martingale sessions.

The results were:

Quantity	Theoretical	Simulated
Probability of small profit	0.981664	0.981576
Probability of large loss	0.018336	0.018424
Expected / mean profit	-0.173522	-0.179136

The theoretical variance of the session profit was approximately

$$73.7283.$$

The Monte Carlo standard error of the simulated mean was approximately

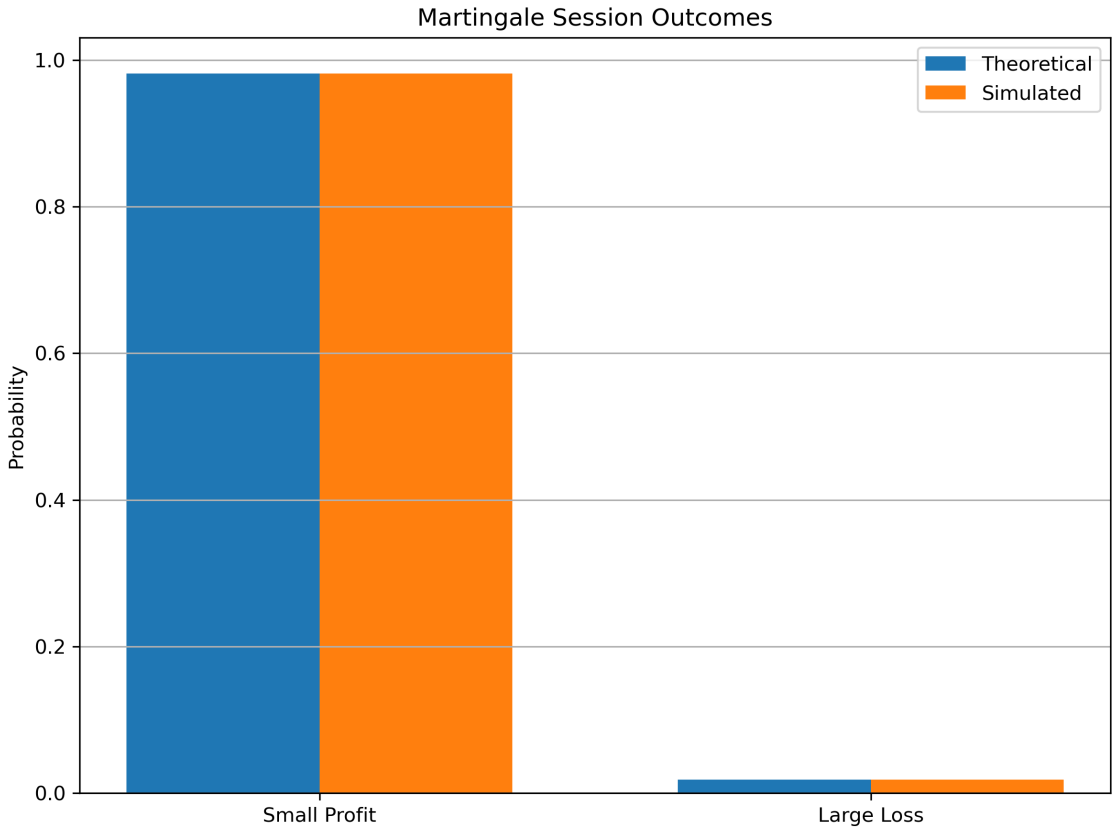
$$0.00859.$$

The difference between the theoretical and simulated mean profits was approximately

$$-0.654$$

standard errors.

Thus the simulation is consistent with the theoretical model.



Why Winning Often Can Still Produce Negative Expected Value

The expected contribution of successful sessions is approximately

$$(0.981664)(1) = 0.981664.$$

The expected contribution of failed sessions is approximately

$$(0.018336)(-63) \approx -1.15519.$$

Therefore,

$$0.981664 - 1.15519 \approx -0.173522.$$

The rare large losses outweigh the many small wins.

Thus,

$$P(\Pi > 0) \approx 0.9817$$

while

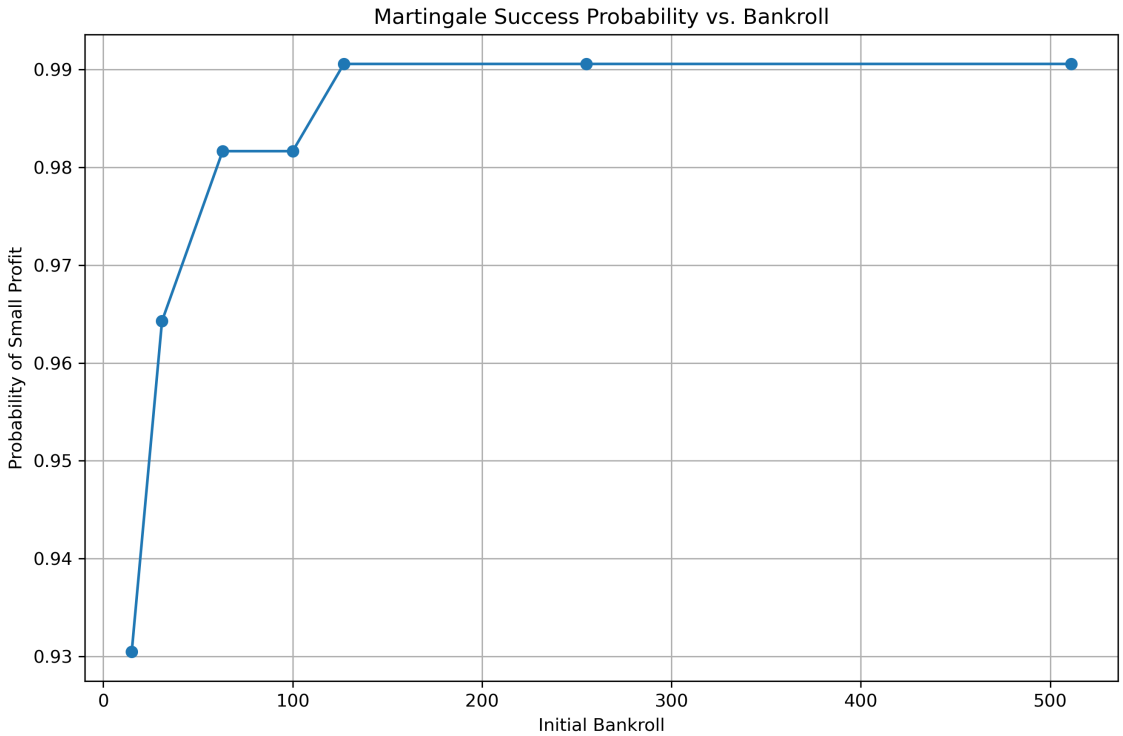
$$E[\Pi] < 0.$$

This demonstrates that a high probability of winning does not imply positive expected value.

Effect of a Finite Bankroll

The simulation also tested several different bankroll sizes.

Bankroll	Allowed Bets	Small-Profit Probability	Large-Loss Probability	Large Loss	Expected Profit
15	4	0.93046	0.06954	-15	-0.11257
31	5	0.96429	0.03571	-31	-0.14264
63	6	0.98166	0.01834	-63	-0.17352
100	6	0.98166	0.01834	-63	-0.17352
127	7	0.99058	0.00942	-127	-0.20524
255	7	0.99058	0.00942	-127	-0.20524
511	7	0.99058	0.00942	-127	-0.20524



Increasing the bankroll increases the probability of obtaining the small profit because the player can survive a longer sequence of losses.

However, it also makes the possible failure larger.

For example, four allowed bets risk

$$15$$

units, while seven allowed bets risk

$$127$$

units.

Thus increasing the bankroll does not remove the negative expected value.

Effect of the Table Maximum

The table maximum in the experiment was

$$64.$$

Once the bankroll is large enough to fund

$$1, 2, 4, 8, 16, 32, 64,$$

the next required Martingale bet would be

$$128.$$

This exceeds the table maximum.

Therefore, additional bankroll cannot create another doubling step.

More generally, with base bet 1, bankroll B, and maximum table bet M, the number of complete Martingale bets that can actually be placed is

$$k = \min\left(\lfloor \log_2(B+1) \rfloor, 1 + \lfloor \log_2 M \rfloor\right).$$

The first term comes from the bankroll requirement

$$1 + 2 + \cdots + 2^{k-1} = 2^k - 1 \le B,$$

and the second comes from the table-limit requirement

$$2^{k-1} \le M.$$

For $M = 64$, at most seven Martingale bets can be placed, regardless of how large the bankroll becomes.

This explains why bankrolls

$$127,\ 255,\ \text{and } 511$$

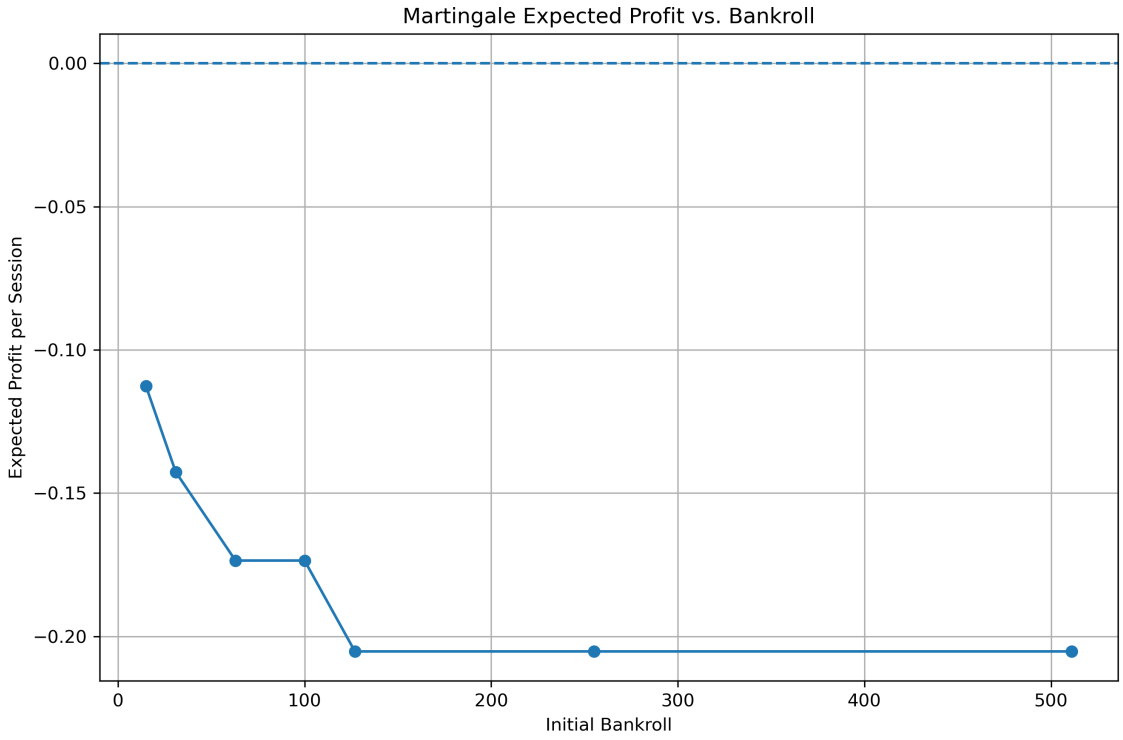
produce identical results.

Their probability of a small profit remains approximately

$$0.990584$$

and their expected profit remains approximately

$$-0.205239.$$



General Martingale Expected Value

If the player can make k complete doubling bets, then

$$P(\text{failure}) = q^k.$$

The loss after k consecutive losses is

$$2^k - 1.$$

Therefore,

$$E[\Pi] = (1 - q^k) - (2^k - 1)q^k.$$

Simplifying,

$$E[\Pi] = 1 - 2^k q^k.$$

Hence,

$$E[\Pi] = 1 - (2q)^k.$$

For European roulette,

$$2q = 2\left(\frac{19}{37}\right) = \frac{38}{37} > 1.$$

Therefore,

$$E[\Pi] < 0.$$

Thus the Martingale strategy does not remove the casino advantage.

There is also a useful expectation check that does not depend on the closed-form Martingale formula. Whenever a bet of size b is actually placed on red, the next spin is independent of the past and its conditional expected profit is

$$-\frac{b}{37}.$$

Therefore every additional wager made by the strategy has negative conditional expectation. Summing over the wagers that are actually placed cannot turn the house edge into a positive expected return. The Martingale changes the distribution of outcomes, but not the sign of the expected value.

It changes the distribution of outcomes into many small wins and rare large losses.

Part V: Reflection

Was the LLM Better as a Tutor or as a Generator of Plausible Mistakes?

The LLM was more useful as a tutor, but this homework also demonstrated how effectively it can generate convincing mathematical mistakes.

As a tutor, the LLM helped define the roulette probability spaces, construct random variables, derive expected values, formulate conditional probabilities, and write simulations.

For example, it correctly identified

$$P_E(\text{win}) = \frac{18}{37}$$

and

$$E[X^E] = -\frac{1}{37}.$$

It was also useful in constructing the Martingale simulation and comparing theoretical results with Monte Carlo outcomes.

However, Part III showed that mathematically sophisticated language is not enough to guarantee a correct argument.

When asked to defend the gambler's fallacy, the LLM produced an argument containing valid ideas such as expectation and the law of large numbers but combined them incorrectly.

Therefore, I found the LLM most useful as a tutor when its output could be independently checked.

What Was the Most Convincing Wrong Argument?

The most convincing false argument was the claim that a losing streak creates a deficit of wins that future spins must correct.

The LLM began with the valid statement that the long-run proportion of wins should approach

$$p = \frac{18}{37}.$$

It then incorrectly assumed

$$E\left[\frac{Y_1 + \cdots + Y_{n+m}}{n+m} \backslash \text{midL}_n\right] = p.$$

From this, it derived

$$P(Y_{n+1} = 1 \mid L_n) = p\left(1 + \frac{n}{m}\right).$$

For five losses and $m = 100$, it obtained the false prediction

$$P(\text{win next} \mid 5 \text{ losses}) \approx 51.08\%.$$

The argument was convincing because most of its mathematical notation and terminology were valid. The error was hidden inside the conditional-expectation assumption.

How Did Conditional Probability Expose the Error?

Conditional probability gives

$$P(Y_{n+1} = 1 \mid L_n) = \frac{P(Y_{n+1} = 1 \cap L_n)}{P(L_n)}.$$

By independence,

$$P(Y_{n+1} = 1 \cap L_n) = P(Y_{n+1} = 1)P(L_n).$$

Therefore,

$$P(Y_{n+1} = 1 \mid L_n) = P(Y_{n+1} = 1).$$

Hence,

$$P(Y_{n+1} = 1 \mid L_n) = \frac{18}{37}.$$

The previous losing streak has no effect on the probability of the next spin.

The law of large numbers does not require future spins to compensate for previous deviations. The influence of a fixed early losing streak simply becomes negligible as more independent spins occur.

What Does This Teach About LLM Reliability?

This homework showed that an LLM can be a useful mathematical tool, but its output should be treated as a proposed argument rather than as automatically correct mathematics.

The LLM was useful for:

- organizing probability models;
- defining random variables;
- deriving equations;
- suggesting simulations;
- explaining mathematical concepts;
- producing alternative arguments.

However, it was also able to construct a false proof using correct mathematical terminology.

The most important lesson is that definitions, assumptions, conditioning, and logical implications must still be checked independently.

The Monte Carlo experiments were useful as an additional check, but numerical simulation also does not replace mathematical reasoning.

A useful workflow is

$$\text{LLM suggestion} \rightarrow \text{identify assumptions} \rightarrow \text{derive the mathematics} \rightarrow \text{check critical ste}$$

Thus the LLM is useful for accelerating mathematical work, but it should not be treated as an unquestionable source of mathematical truth.

Conclusion

This homework demonstrates two related mistakes in gambling intuition.

First, the gambler's fallacy incorrectly assumes that previous losses increase the probability of a future win.

For European roulette,

$$P(X_{n+1} > 0 \mid L_n) = P(X_{n+1} > 0) = \frac{18}{37}.$$

The Monte Carlo experiments confirmed this for losing streaks of several different lengths.

Second, the Martingale strategy attempts to respond to losses by increasing the amount wagered.

For the baseline experiment,

$$P(\text{small profit}) \approx 98.17\%,$$

but

$$E[\Pi] \approx -0.17352.$$

Thus a strategy can win very frequently while still having negative expected value.

The LLM was useful for building the models and simulations, but Part III demonstrated that it can also produce a mathematically plausible false argument.

The main lesson is that probabilistic reasoning should be based on explicit probability models, conditional probabilities, and expected values rather than intuition or persuasive mathematical language.

Homework 3.1: Bakeries, Cafés, and the Difference Between Monge and Kantorovich

Introduction

This homework studies discrete optimal transport using a bakery-café model.

Bakeries represent sources of bread supply, cafés represent destinations with bread demand, and transportation has a cost determined by the distance between locations.

The main objective is to compare two formulations of optimal transport:

- Monge transport, where each bakery must send all of its supply to exactly one café;
- Kantorovich transport, where a bakery may divide its supply among multiple cafés.

An LLM was used to help generate the examples, formulate the optimization problems, and generate Python code. The mathematical claims and computational results were then independently checked.

Part I: Generated Bakery and Café Data

LLM Prompt

I asked the LLM:

> Generate a small discrete optimal transport example representing bakeries and cafés in a two-dimensional city. Use 3 bakeries and 4 cafés, assign positive supplies and demands, and make sure total supply equals total demand. Choose simple coordinates so that Euclidean transportation costs can be calculated and visualized easily.

The generated problem contains

$$N = 3$$

bakeries and

$$M = 4$$

cafés.

Thus,

$$N < M.$$

Bakery Data

The bakery locations are

$$x_1 = (1, 1),$$

$$x_2 = (7, 1),$$

and

$$x_3 = (4, 7).$$

Their supplies are

$$a_1 = 5, \qquad a_2 = 6, \qquad a_3 = 4.$$

Therefore, total supply is

$$\sum_{i=1}^{3} a_i = 5 + 6 + 4 = 15.$$

Café Data

The café locations are

$$y_1 = (2, 3),$$

$$y_2 = (6, 3),$$

$$y_3 = (3, 6),$$

and

$$y_4 = (7, 6).$$

Their demands are

$$b_1 = 3, \qquad b_2 = 4, \qquad b_3 = 5, \qquad b_4 = 3.$$

Therefore, total demand is

$$\sum_{j=1}^{4} b_j = 3 + 4 + 5 + 3 = 15.$$

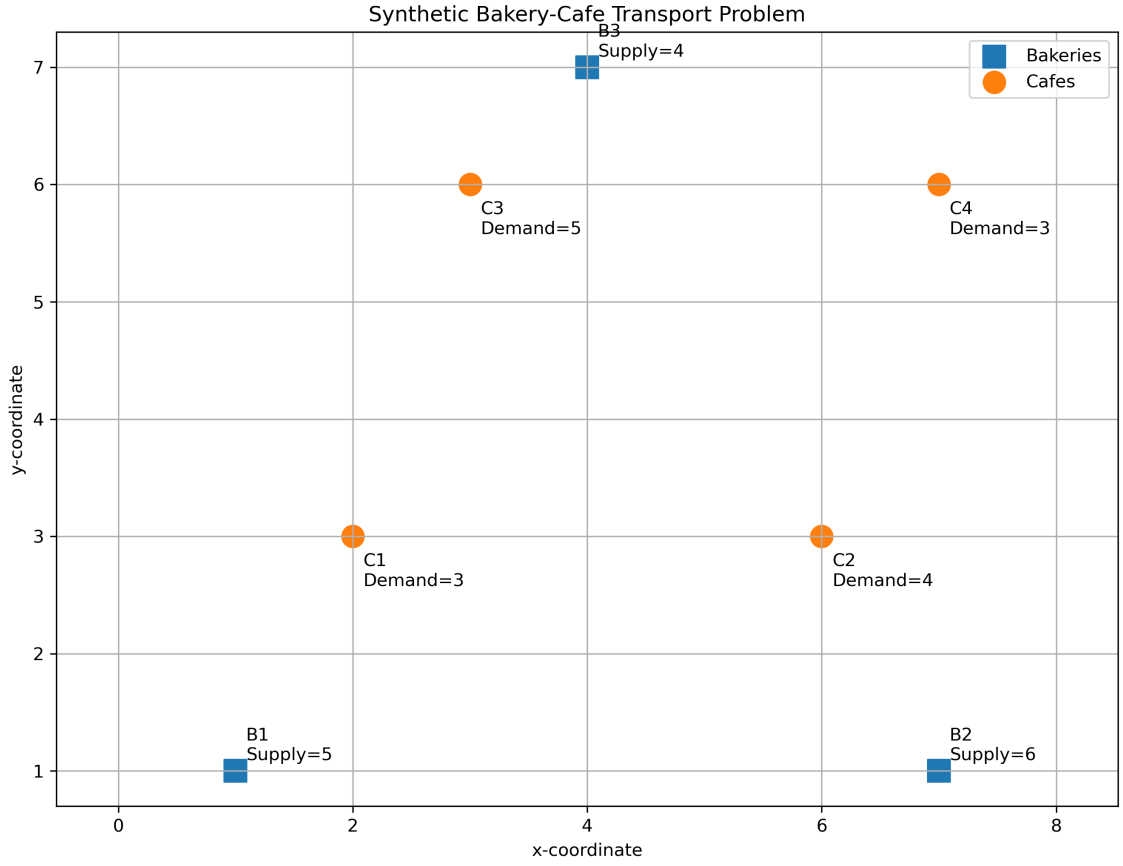
Thus,

$$\sum_{i=1}^{3} a_i = \sum_{j=1}^{4} b_j.$$

The transport problem is balanced.

Data Summary

Location	Coordinates	Supply or Demand
Bakery 1	$(1, 1)$	Supply 5
Bakery 2	$(7, 1)$	Supply 6
Bakery 3	$(4, 7)$	Supply 4
Café 1	$(2, 3)$	Demand 3
Café 2	$(6, 3)$	Demand 4
Café 3	$(3, 6)$	Demand 5
Café 4	$(7, 6)$	Demand 3



The Python verification confirmed that:

- all supplies are positive;
- all demands are positive;

- all locations lie in $\mathbb{R}^2$;
- total supply equals total demand.

Transportation Cost Matrix

The first transportation cost is Euclidean distance:

$$c_{ij} = |x_i - y_j|.$$

For points

$$x_i = (x_i^{(1)}, x_i^{(2)})$$

and

$$y_j = (y_j^{(1)}, y_j^{(2)}),$$

this is

$$c_{ij} = \sqrt{(x_i^{(1)} - y_j^{(1)})^2 + (x_i^{(2)} - y_j^{(2)})^2}.$$

The Euclidean cost matrix is

$$C = \begin{pmatrix} 2.2361 & 5.3852 & 5.3852 & 7.8102 \\ 5.3852 & 2.2361 & 6.4031 & 5.0000 \\ 4.4721 & 4.4721 & 1.4142 & 3.1623 \end{pmatrix}.$$

For example,

$$c_{11} = |(1,1) - (2,3)|.$$

Therefore,

$$c_{11} = \sqrt{(1-2)^2 + (1-3)^2} = \sqrt{5} \approx 2.2361.$$

The Python program independently calculated

$$c_{11} = 2.2360679775,$$

which exactly matched the manual calculation.

Squared Euclidean Cost

I also considered

$$c_{ij}^{(2)} = |x_i - y_j|^2.$$

The squared-distance cost matrix is

$$C^{(2)} = \begin{pmatrix} 5 & 29 & 29 & 61 \\ 29 & 5 & 41 & 25 \\ 20 & 20 & 2 & 10 \end{pmatrix}.$$

The program verified that

$$c_{ij}^{(2)} = (c_{ij})^2$$

for every pair of bakery and café locations.

Part II: The Kantorovich Problem

In the Kantorovich formulation, define

$$\gamma_{ij} \geq 0$$

to be the amount of bread sent from Bakery i to Café j.

The transport matrix is

$$\gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \gamma_{24} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & \gamma_{34} \end{pmatrix}.$$

Supply Constraints

Each bakery must ship exactly all of its supply:

$$\sum_{j=1}^{4} \gamma_{ij} = a_i.$$

Therefore,

$$\gamma_{11} + \gamma_{12} + \gamma_{13} + \gamma_{14} = 5,$$

$$\gamma_{21} + \gamma_{22} + \gamma_{23} + \gamma_{24} = 6,$$

and

$$\gamma_{31} + \gamma_{32} + \gamma_{33} + \gamma_{34} = 4.$$

Demand Constraints

Each café must receive exactly its demand:

$$\sum_{i=1}^{3} \gamma_{ij} = b_j.$$

Therefore,

$$\gamma_{11} + \gamma_{21} + \gamma_{31} = 3,$$
$$\gamma_{12} + \gamma_{22} + \gamma_{32} = 4,$$
$$\gamma_{13} + \gamma_{23} + \gamma_{33} = 5,$$

and

$$\gamma_{14} + \gamma_{24} + \gamma_{34} = 3.$$

Objective

The Kantorovich problem is

$$\min_{\gamma_{ij} \geq 0} \sum_{i=1}^{3} \sum_{j=1}^{4} c_{ij} \gamma_{ij}.$$

This is a linear programming problem.

I solved it using `scipy.optimize.linprog`.

Optimal Kantorovich Solution

For Euclidean cost, the optimal transport matrix was

$$\boxed{\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}}.$$

This means:

- Bakery 1 sends 3 units to Café 1 and 2 units to Café 3;

- Bakery 2 sends 4 units to Café 2 and 2 units to Café 4;
- Bakery 3 sends 3 units to Café 3 and 1 unit to Café 4.

Verification of Supply Constraints

The row sums are

$$3 + 2 = 5,$$
$$4 + 2 = 6,$$

and

$$3 + 1 = 4.$$

Thus,

$$(5, 6, 4)$$

is recovered exactly.

Verification of Demand Constraints

The column sums are

$$3,$$
$$4,$$
$$2 + 3 = 5,$$

and

$$2 + 1 = 3.$$

Thus,

$$(3, 4, 5, 3)$$

is also recovered exactly.

The program additionally verified nonnegativity and independently recomputed the objective value.

Optimal Euclidean Transportation Cost

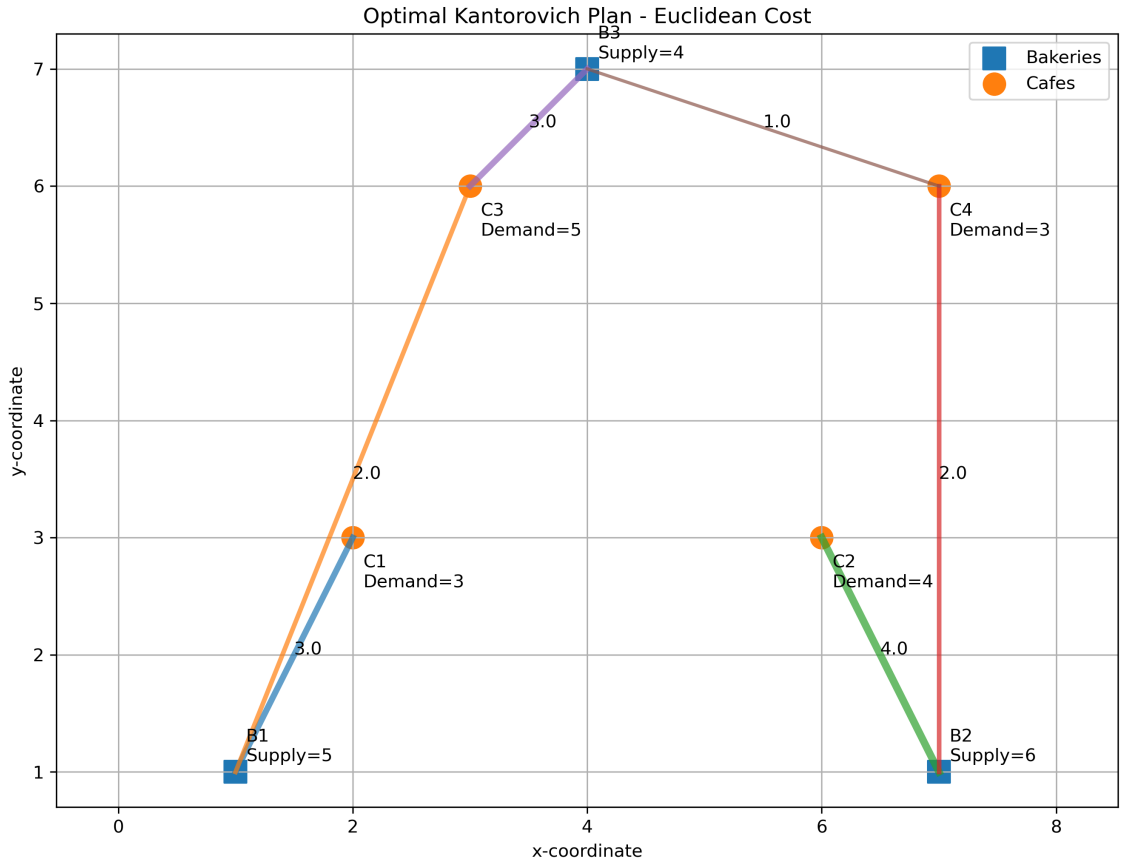
Only six entries of γ^* are nonzero.

Therefore,

$$J_K = 3\sqrt{5} + 2\sqrt{29} + 4\sqrt{5} + 2(5) + 3\sqrt{2} + \sqrt{10}.$$

Numerically,

$$J_K \approx 43.827724.$$



Squared Euclidean Kantorovich Solution

The squared-distance optimization produced the same transport matrix:

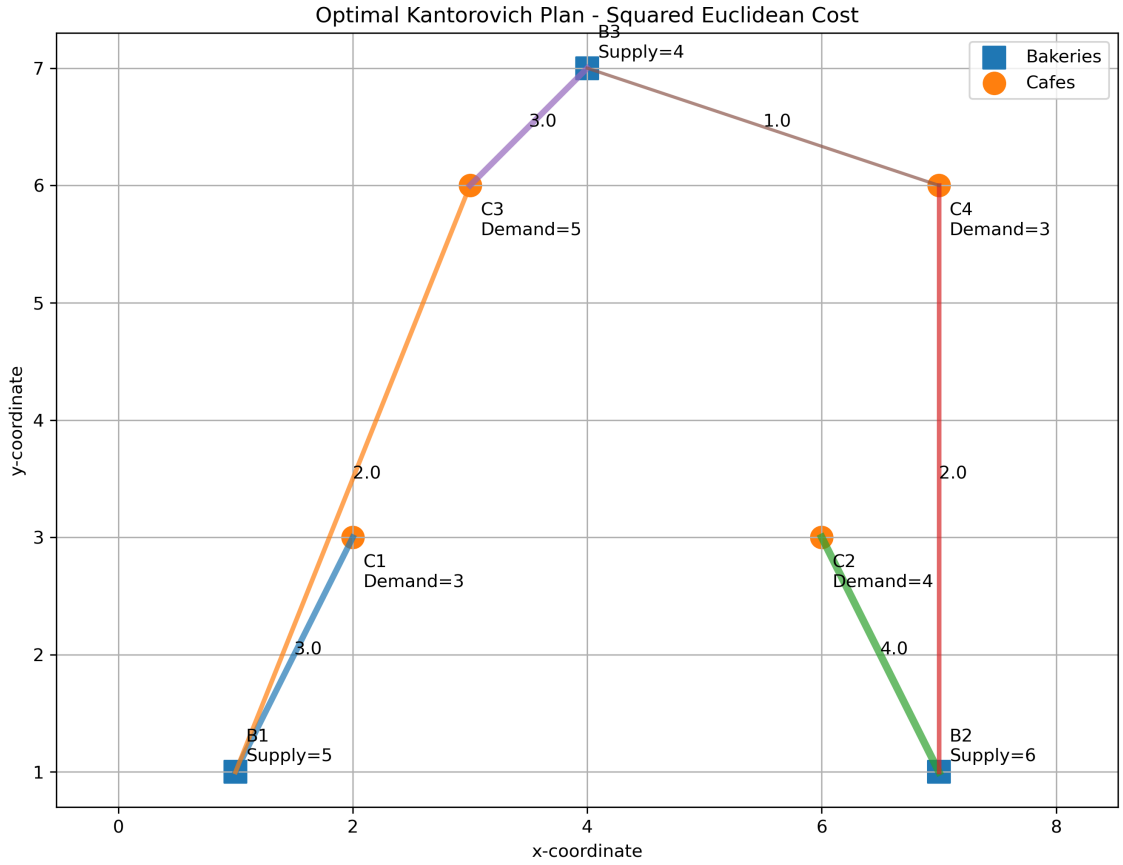
$$\gamma^{*(2)} = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Its cost is

$$3(5) + 2(29) + 4(5) + 2(25) + 3(2) + 1(10).$$

Therefore,

$$\boxed{J_K^{(2)} = 159}.$$



Thus both cost functions happened to produce the same optimal plan in this example, although this should not be expected for every transport problem.

Part III: The Monge Problem

In the Monge formulation, every bakery must send all of its supply to exactly one café.

A Monge map has the form

$$T : \{1, 2, 3\} \to \{1, 2, 3, 4\}.$$

The café constraints require

$$\sum_{i:T(i)=j} a_i = b_j.$$

The Monge cost would be

$$J_M(T) = \sum_{i=1}^{3} a_i c_{i,T(i)}.$$

Number of Possible Maps

Each of the three bakeries has four possible destinations.

Therefore,

$$4^3 = 64$$

Monge maps are possible.

The Python program exhaustively searched all 64 maps.

It found

$$0$$

admissible Monge maps.

Thus,

$$\boxed{\text{the Monge problem is infeasible for this city.}}$$

Why No Monge Map Exists

There are

$$N = 3$$

bakeries but

$$M = 4$$

cafés.

Every café has positive demand.

Because one bakery can serve only one café under Monge transport, three bakeries can serve at most three distinct cafés.

But four cafés require positive mass.

Therefore, at least one café would necessarily receive zero bread.

Thus no Monge map can satisfy the demand constraints.

There is also a second obstruction.

The possible sums formed from the indivisible bakery supplies

$$5,\ 6,\ 4$$

are

$$0,\ 4,\ 5,\ 6,\ 9,\ 10,\ 11,\ 15.$$

However, two cafés require only

$$3$$

units.

No subset of whole bakery supplies sums to 3.

Thus the specific supply and demand values also prevent a Monge solution.

Comparison with Kantorovich

The Kantorovich solution was

$$\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Each bakery splits its supply.

For example,

$$5 = 3 + 2,$$

$$6 = 4 + 2,$$

and

$$4 = 3 + 1.$$

This splitting is exactly what allows Kantorovich transport to satisfy all four cafés.

Therefore,

$$\boxed{\text{Kantorovich is feasible}}$$

while

$$\boxed{\text{Monge is infeasible}}.$$

Part IV: Comparison of $N > M$ and $N < M$

Two additional cases were compared.

Case 1: More Bakeries Than Cafés

For Case 1,

$$N = 4, \qquad M = 3.$$

The supplies are

$$a = (2, 2, 2, 2),$$

and demands are

$$b = (4, 2, 2).$$

The bakery locations are

$$(0, 0), \quad (0, 6), \quad (8, 0), \quad (8, 6),$$

and the café locations are

$$(0, 3), \quad (7, 0), \quad (7, 6).$$

The Euclidean cost matrix is

$$C \approx \begin{pmatrix} 3.0000 & 7.0000 & 9.2195 \\ 3.0000 & 9.2195 & 7.0000 \\ 8.5440 & 1.0000 & 6.0828 \\ 8.5440 & 6.0828 & 1.0000 \end{pmatrix}.$$

Case 1 Monge Solution

There are

$$3^4 = 81$$

possible Monge maps.

The exhaustive search found

$$12$$

feasible maps.

The optimal map was

$$T(1) = 1, \qquad T(2) = 1, \qquad T(3) = 2, \qquad T(4) = 3.$$

Its cost is

$$J_M = 2(3) + 2(3) + 2(1) + 2(1).$$

Thus,

$$J_M^* = 16.$$

Case 1 Kantorovich Solution

The optimal Kantorovich matrix was

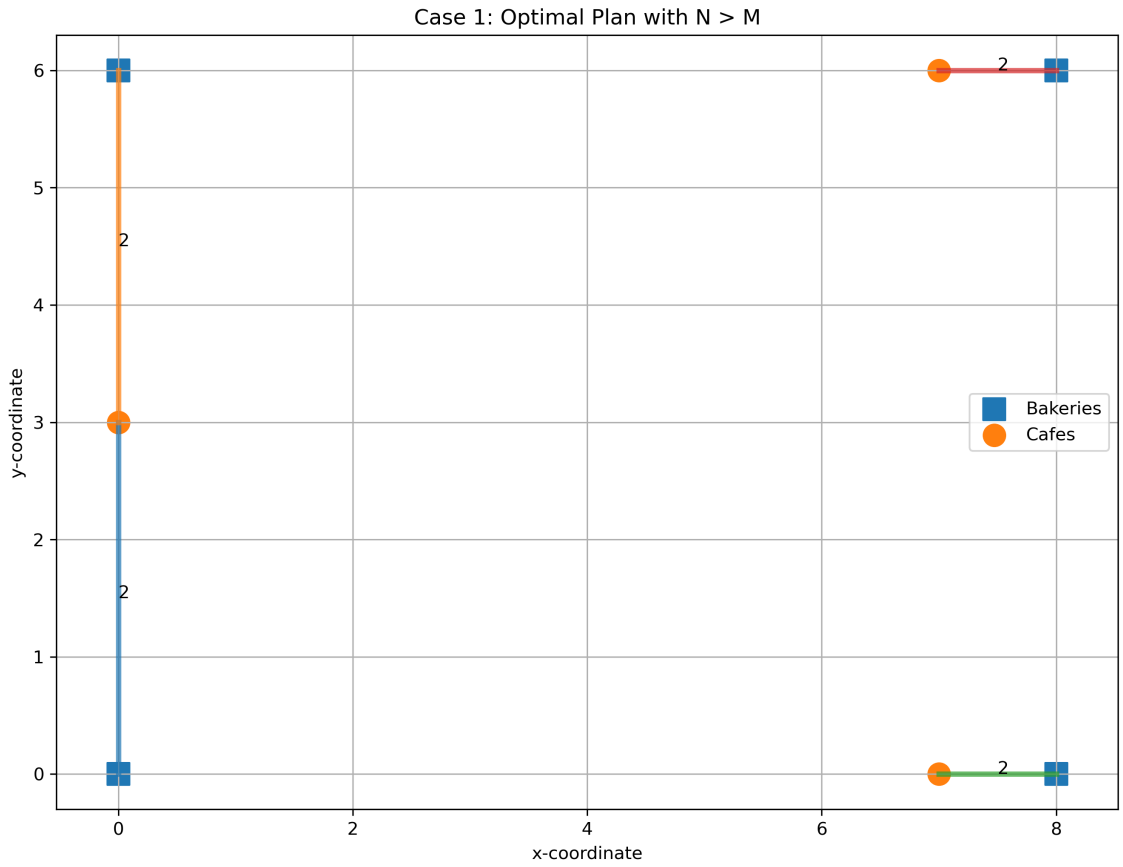
$$\gamma^* = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Its cost was also

$$J_K^* = 16.$$

Each bakery has exactly one positive destination.

Therefore, no mass splitting is required, and the displayed optimal Kantorovich plan is induced by the optimal Monge map.



Thus,

$$J_M^* = J_K^* = 16.$$

Case 2: Fewer Bakeries Than Cafés

For Case 2, the original city was reused:

$$N = 3, \qquad M = 4.$$

The exhaustive Monge search found

$$0$$

feasible maps.

Thus,

$$\boxed{\text{Monge is infeasible.}}$$

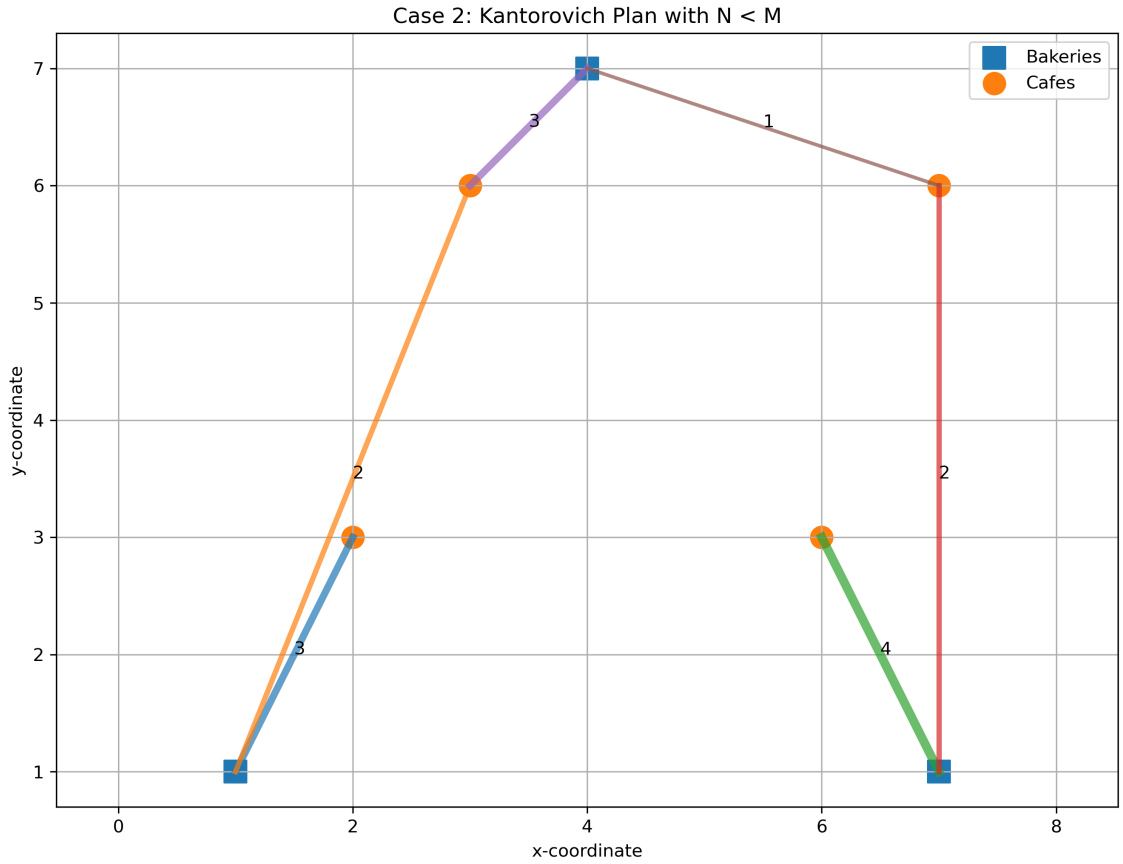
The Kantorovich problem remained feasible and produced

$$\gamma^* = \begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}.$$

Its Euclidean cost was

$$J_K^* \approx 43.827724.$$

Every bakery had two positive destinations, so every bakery split its supply.



Summary of the Two Cases

Property	**Case 1:** $N > M$	**Case 2:** $N < M$
Number of bakeries	4	3
Number of cafés	3	4

Property	Case 1: $N > M$	Case 2: $N < M$
Possible Monge maps	81	64
Feasible Monge maps	12	0
Monge feasible?	Yes	No
Optimal Monge cost	16	Does not exist
Kantorovich feasible?	Yes	Yes
Optimal Kantorovich cost	16	43.827724
Kantorovich requires splitting?	No	Yes

The experiments show that

$$N > M$$

does not automatically guarantee a Monge solution.

The supplies must also be compatible with the café demands.

In Case 1,

$$4 = 2 + 2,$$

so the café demands can be constructed from whole bakery supplies.

By contrast, if

$$N < M$$

and every café has positive demand, then a discrete Monge map is impossible because each bakery has only one destination.

Thus,

$$N < M \text{ and all } b_j > 0 \Longrightarrow \text{no Monge map exists.}$$

Kantorovich avoids this obstruction by permitting mass splitting.

Part V: LLM Prompts and Critique

Prompt for Generating the Synthetic City

> Generate a small discrete optimal transport example representing bakeries and cafés in a two-dimensional city. Use 3 bakeries and 4 cafés, assign positive supplies and demands, and make sure total supply equals total demand. Choose simple coordinates so that Euclidean transportation costs can be calculated and visualized easily.

Prompt for the Monge and Kantorovich Critique

> Explain the difference between Monge and Kantorovich optimal transport using a discrete bakery-café problem. Provide the mathematical formulations for both approaches and generate Python code to solve them using the same transportation example.

The second LLM generated mathematical formulations and Python code using:

- `scipy.optimize.linprog` for Kantorovich transport;
- `scipy.optimize.milp` for Monge transport.

I checked the mathematics and code carefully.

---

1. Supply Constraints

For Kantorovich transport, the LLM correctly imposed

$$\sum_j x_{ij} = a_i.$$

Its code constructs one equality constraint for each bakery.

For Monge transport, it introduces binary variables

$$z_{ij} \in \{0, 1\},$$

with

$$\sum_j z_{ij} = 1.$$

The resulting shipment is

$$x_{ij} = a_i z_{ij}.$$

Thus every bakery sends all of its supply to exactly one café.

The supply constraints are correct.

---

2. Demand Constraints

For Kantorovich transport, the LLM correctly imposed

$$\sum_i x_{ij} = b_j.$$

For Monge transport, it correctly imposed

$$\sum_i a_i z_{ij} = b_j.$$

Thus the demand constraints are also mathematically correct.

There was one notation typo in the written explanation.

It wrote

$$\sum_{i(i)=j} a_i = b_j,$$

when it should have written

$$\sum_{i:T(i)=j} a_i = b_j.$$

The implementation itself used the correct condition.

---

3. Transport Map Versus Transport Plan

The LLM correctly distinguished a Monge transport map from a Kantorovich transport plan.

Monge uses

$$T(i) = j,$$

meaning that the entire source mass a_i is assigned to one destination.

Kantorovich instead uses the matrix

$$(x_{ij}),$$

and multiple entries in one row may be positive.

Every feasible Monge map induces a Kantorovich plan through

$$x_{ij} = \begin{cases} a_i, & j = T(i), \\ 0, & j \neq T(i). \end{cases}$$

However, not every Kantorovich plan comes from a Monge map.

Thus,

$$\{\text{Monge-induced plans}\} \subseteq \{\text{Kantorovich plans}\}.$$

The LLM handled this distinction correctly.

---

4. Does a Monge Map Always Exist?

The LLM correctly stated that equal total supply and demand do not guarantee a Monge solution.

Its second example used

$$a = (5, 5)$$

and

$$b = (6, 4).$$

Total supply and demand are both

$$10,$$

but no bakery can split its supply.

Therefore, the demands 6 and 4 cannot be formed from two indivisible masses of size 5.

The LLM correctly reported that the Monge problem is infeasible.

Kantorovich can instead use

$$X = \begin{pmatrix} 5 & 0 \\ 1 & 4 \end{pmatrix},$$

which has the correct row and column sums.

The transportation cost is

$$5(2) + 1(3) + 4(4) = 29.$$

Thus this example is mathematically correct.

---

5. Handling the Case $N < M$

This was the main weakness of the LLM response.

The generated solver is general enough to handle

$$N < M.$$

However, neither original example actually tested this case.

Example 1 had

$$N = 3 > M = 2,$$

while Example 2 had

$$N = M = 2.$$

Therefore, the second example showed Monge infeasibility but did not directly test the structural case required in the homework.

I added an additional test using the original synthetic city:

$$N = 3 < M = 4.$$

The supplies were

$$(5, 6, 4),$$

and demands were

$$(3, 4, 5, 3).$$

The same LLM-generated solver returned the Kantorovich plan

$$\begin{pmatrix} 3 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 3 & 1 \end{pmatrix}$$

with cost

$$43.827724.$$

The Monge solver correctly reported that no feasible Monge solution exists.

Thus the generated code handles $N < M$ correctly even though the LLM did not initially include an example testing it.

Verification of the LLM's First Example

The LLM's first example used

$$a = (2, 3, 5),$$

$$b = (5, 5),$$

and

$$C = \begin{pmatrix} 2 & 6 \\ 3 & 4 \\ 6 & 1 \end{pmatrix}.$$

The Monge assignment

$$T(1) = 1, \qquad T(2) = 1, \qquad T(3) = 2$$

induces

$$X = \begin{pmatrix} 2 & 0 \\ 3 & 0 \\ 0 & 5 \end{pmatrix}.$$

Its cost is

$$2(2) + 3(3) + 5(1) = 18.$$

Both the Monge and Kantorovich optimizers give

$$18.$$

Thus the LLM's first example is correct.

Overall LLM Reliability

The LLM performed well on the central mathematics.

It correctly:

- formulated the Kantorovich supply constraints;
- formulated the Kantorovich demand constraints;

- implemented the Monge one-destination constraint;
- implemented the Monge demand constraints;
- distinguished a transport map from a transport plan;
- recognized that Monge feasibility is not guaranteed;
- recognized Kantorovich as a relaxation of Monge;
- correctly concluded that

$$C_K^* \leq C_M^*$$

whenever the Monge problem is feasible.

The main weakness was completeness rather than mathematical correctness.

The LLM was explicitly supposed to handle the case

$$N < M,$$

but its original examples did not test that case.

After adding the missing test, its code produced the mathematically correct result.

Thus the LLM was useful for generating the formulation and code, but its response still required independent checking to ensure that all requested cases had actually been tested.

Final Conclusion

The experiments demonstrate the central difference between Monge and Kantorovich optimal transport.

Monge transport requires a deterministic map:

$$T : \{1, \ldots, N\} \to \{1, \ldots, M\},$$

where each bakery sends its entire mass to one café.

Kantorovich transport instead allows a transport matrix

$$\gamma = (\gamma_{ij}),$$

where mass may be divided among multiple destinations.

Every Monge solution induces a Kantorovich plan, so whenever Monge is feasible,

$$C_K^* \leq C_M^*.$$

In the $N > M$ example, a compatible Monge map existed and both formulations had the

same optimal cost:

$$C_M^* = C_K^* = 16.$$

In the $N < M$ example, no Monge map existed, while Kantorovich remained feasible through mass splitting:

$$C_K^* \approx 43.827724.$$

Therefore, Kantorovich transport is more general than Monge transport.

The computational experiments also show why LLM-generated mathematics should be verified. The LLM's formulation and code were largely correct, but it initially failed to demonstrate one of the specific cases requested by the assignment.

Homework 3.2: Benamou-Brenier in One Dimension

Introduction

This homework studies the dynamic optimal transport interpretation of the Benamou-Brenier formula in one dimension.

Given two probability densities

$$\rho_0$$

and

$$\rho_1,$$

with cumulative distribution functions

$$F_0$$

and

$$F_1,$$

the monotone one-dimensional optimal transport map is

$$T(x) = F_1^{-1}(F_0(x)).$$

The corresponding Wasserstein displacement interpolation is

$$X_t(x) = (1-t)x + tT(x), \qquad 0 \le t \le 1,$$

and the intermediate probability density is

$$\rho_t = (X_t)_{\#}\rho_0.$$

The goals of the experiment were to:

- compute and verify the one-dimensional optimal transport map;
- visualize the displacement interpolation;
- estimate the squared Wasserstein distance;
- numerically evaluate the Benamou-Brenier action;
- compare compactly supported and Gaussian target distributions;
- evaluate the mathematical reliability of LLM-generated reasoning and code.

Part I: LLM Plan and Preliminary Verification

Prompt Given to the LLM

> Given two one-dimensional probability densities ρ_0 and ρ_1, how can I compute the optimal transport map using CDFs and quantile functions? How can I use this map to animate the Benamou-Brenier displacement interpolation?

LLM Plan

The LLM proposed computing

$$F_0(x) = \int_{-\infty}^{x} \rho_0(s) \, ds$$

and

$$F_1(x) = \int_{-\infty}^{x} \rho_1(s) \, ds.$$

The optimal transport map is then

$$T(x) = F_1^{-1}(F_0(x)).$$

Once the map is known, particles move according to

$$X_t(x) = (1-t)x + tT(x).$$

The density is transported through

$$\rho_t = (X_t)_{\#}\rho_0.$$

The proposed numerical procedure was:

1. evaluate the source and target densities on a grid;
2. normalize the densities;
3. compute their CDFs;
4. construct the target quantile function;
5. calculate $T(x)$;

6. calculate $X_t(x)$;

7. reconstruct the intermediate densities;

8. plot the resulting transport.

Preliminary Sanity Check

Before applying the method to the homework densities, I tested the implementation on two Gaussian distributions:

$$\rho_0 \approx N(-2, 1)$$

and

$$\rho_1 \approx N(2, 1).$$

Since the target is exactly a translation of the source by four units, the exact optimal map is

$$T(x) = x + 4.$$

The numerical calculation gave an average displacement

$$3.9999999642,$$

with maximum interior error approximately

$$2.93 \times 10^{-7}.$$

The defining quantile identity

$$F_1(T(x)) = F_0(x)$$

had maximum numerical error

$$1.11 \times 10^{-16}.$$

The squared Wasserstein distance should be

$$W_2^2 = 16,$$

and the numerical calculation gave

$$W_2^2 = 15.9999999028.$$

Thus the CDF, quantile, and displacement-interpolation implementation was verified before being used for the main experiment.

Part II: Half-Circle Density to Uniform

Density

The initial density is

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1,$$

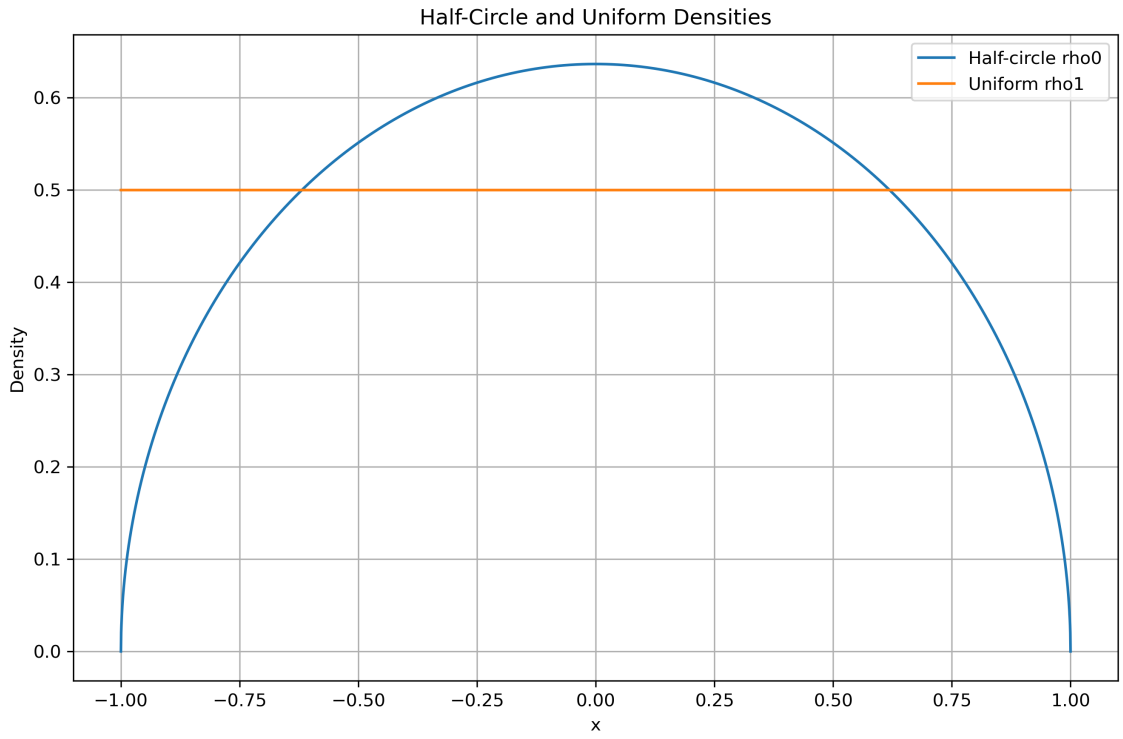
and zero outside $[-1, 1]$.

The target density is uniform:

$$\rho_1(x) = \frac{1}{2}, \qquad -1 \le x \le 1,$$

and zero outside $[-1, 1]$.

Source and Target Densities



The half-circle density is concentrated more strongly around the origin, whereas the uniform target distributes the probability mass evenly across the interval.

CDFs

For the source,

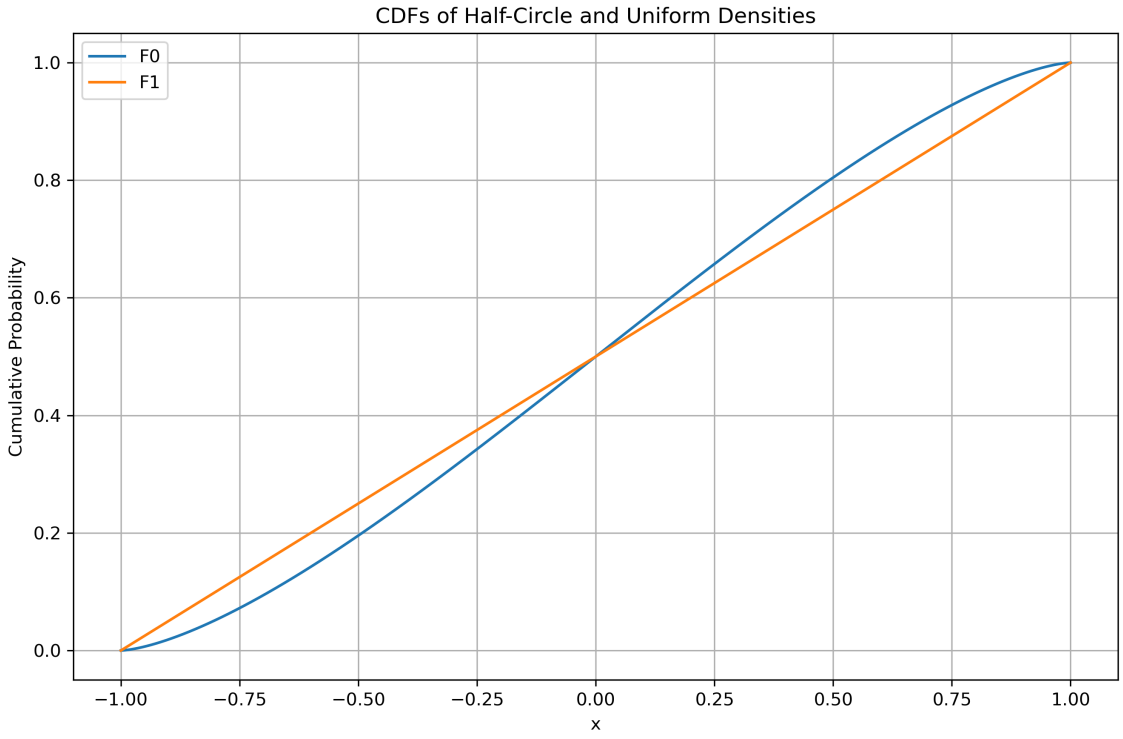
$$F_0(x) = \frac{1}{2} + \frac{1}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right), \qquad -1 \le x \le 1.$$

For the target,

$$F_1(x) = \frac{x+1}{2}, \qquad -1 \le x \le 1.$$

Thus

$$F_1^{-1}(u) = 2u - 1.$$



The numerical and analytic CDFs agreed closely.

The maximum error for the numerical source CDF was approximately

$$2.03 \times 10^{-6}.$$

Optimal Transport Map

The optimal map is

$$T(x) = F_1^{-1}(F_0(x)).$$

Therefore,

$$T(x) = 2F_0(x) - 1,$$

which gives

$$T(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The map satisfies

$$T(-1) = -1,$$

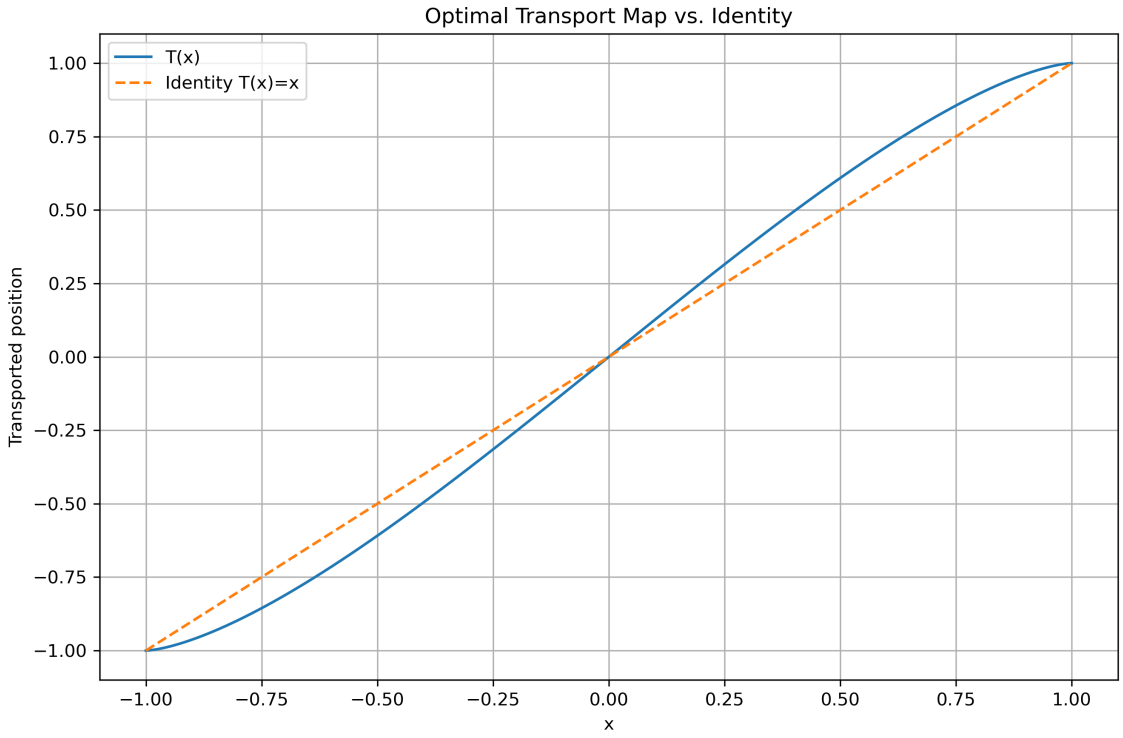
$$T(0) = 0,$$

and

$$T(1) = 1.$$

The numerical and analytic transport maps differed by at most approximately

$$4.05 \times 10^{-6}.$$



For

$$x > 0,$$

we have

$$T(x) > x,$$

so particles move outward to the right.

For

$$x < 0,$$

we have

$$T(x) < x,$$

so particles move outward to the left.

This is exactly what is required to flatten the initially peaked density.

Displacement Interpolation

The Wasserstein geodesic is

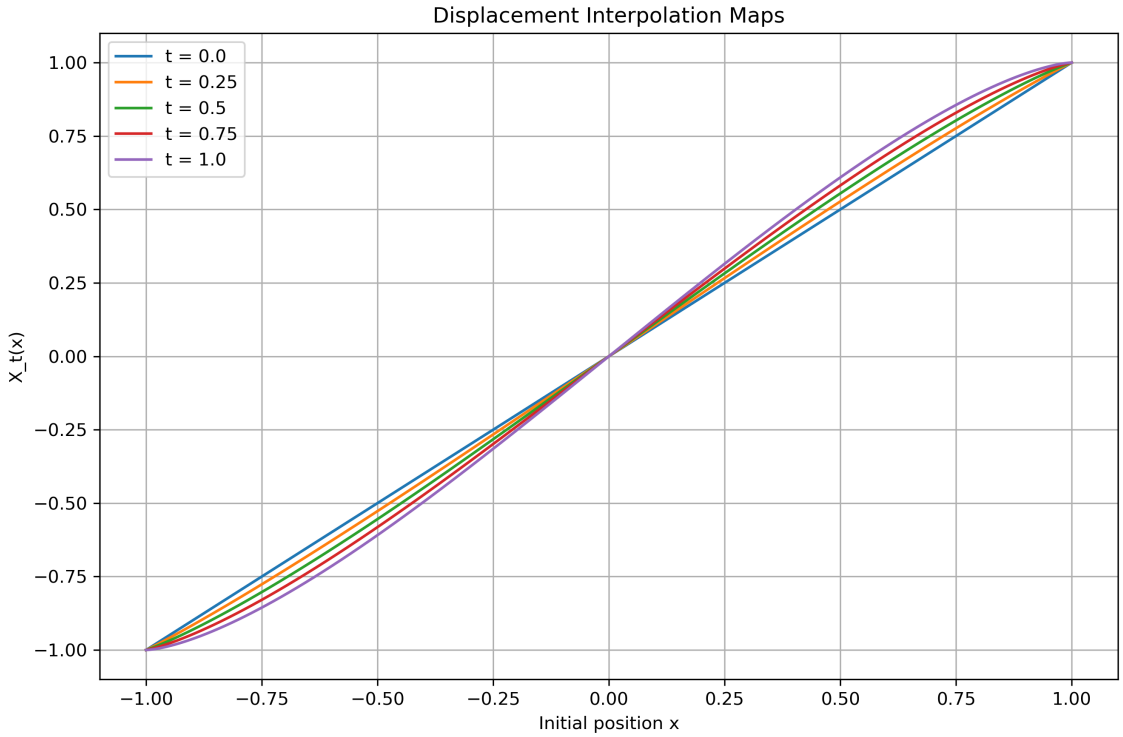
$$X_t(x) = (1-t)x + tT(x).$$

Substituting the transport map,

$$X_t(x) = (1-t)x + \frac{2t}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The maps were evaluated at

$$t = 0,\ 0.25,\ 0.5,\ 0.75,\ 1.$$



The interpolation begins with the identity map and continuously approaches the optimal transport map.

Interpolated Densities

Differentiating the transport map gives

$$T'(x) = \frac{4}{\pi}\sqrt{1-x^2}.$$

Therefore,

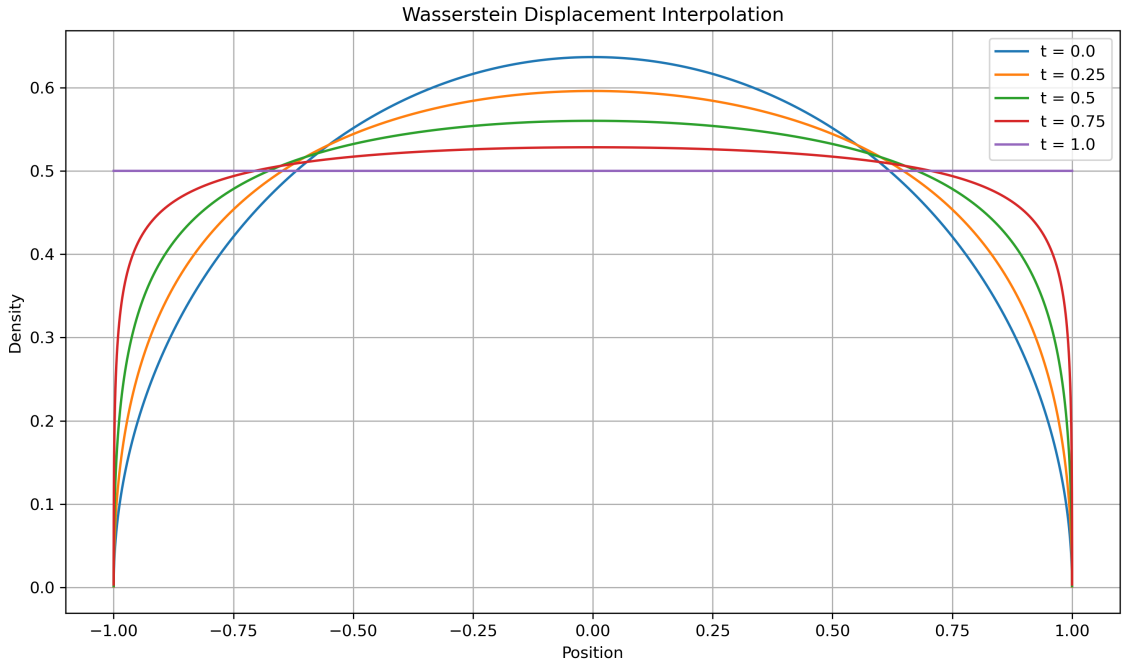
$$X_t'(x) = (1-t) + \frac{4t}{\pi}\sqrt{1-x^2}.$$

Conservation of mass gives

$$\rho_t(X_t(x))X_t'(x) = \rho_0(x).$$

Hence,

$$\rho_t(X_t(x)) = \frac{\frac{2}{\pi}\sqrt{1-x^2}}{(1-t) + \frac{4t}{\pi}\sqrt{1-x^2}}.$$



At

$$t = 0,$$

the density is the half-circle.

As t increases, the central peak decreases and probability mass moves outward.

At

$$t = 1,$$

the density becomes

$$\rho_1(x) = \frac{1}{2}.$$

The total mass remained approximately one for every tested time.

Velocity Along Particle Paths

The particle trajectory is

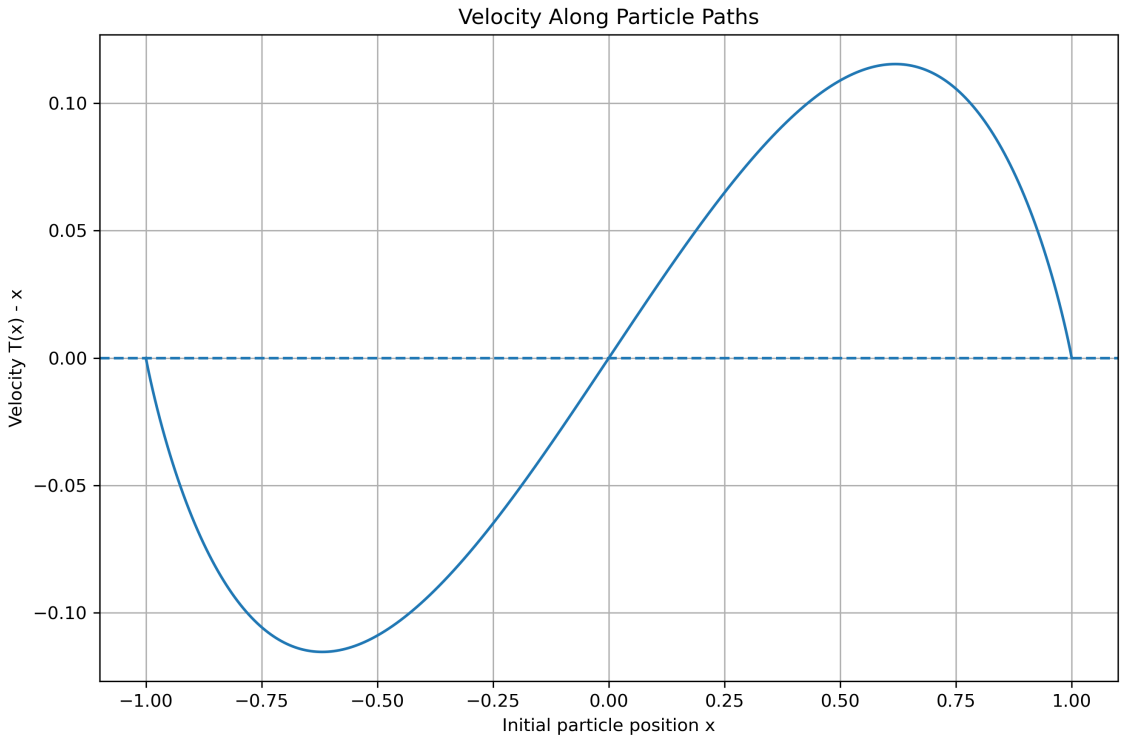
$$X_t(x) = (1-t)x + tT(x).$$

Differentiating with respect to time gives

$$\dot{X}_t(x) = T(x) - x.$$

Therefore,

$$\dot{X}_t(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right) - x.$$



The velocity is negative on the left side and positive on the right side.

Thus particles move outward from the center.

The velocity is zero at

$$x = -1, \quad 0, \quad 1.$$

Each particle has constant **Lagrangian** velocity along its own displacement-interpolation path. The corresponding Eulerian velocity field is

$$v_t(y) = T(x) - x \quad \text{where } y = X_t(x),$$

so at a fixed spatial location y the velocity field can depend on time through the inverse map $x = X_t^{-1}(y)$.

Part III: Numerical Wasserstein Distance and Benamou-Brenier Action

The squared Wasserstein distance is

$$W_2^2(\rho_0, \rho_1) = \int_{-1}^{1} |T(x) - x|^2 \rho_0(x) \, dx.$$

For this problem,

$$W_2^2 = \int_{-1}^{1} \left[\frac{2}{\pi} \left(\arcsin x + x\sqrt{1 - x^2} \right) - x \right]^2 \frac{2}{\pi} \sqrt{1 - x^2} \, dx.$$

The numerical estimate was approximately

$$W_2^2 \approx 0.00692838.$$

As an additional check, the analytic benchmark is

$$W_2^2 = \frac{7}{12} - \frac{256}{45\pi^2},$$

which is also approximately

$$0.00692838.$$

Benamou-Brenier Action

The dynamic action is

$$\mathcal{A} = \int_0^1 \int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y) \, dy \, dt.$$

Along the particle path,

$$v_t(X_t(x)) = T(x) - x.$$

Using

$$y = X_t(x)$$

and conservation of probability mass,

$$\rho_t(X_t(x))X_t'(x) = \rho_0(x),$$

we obtain

$$\int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y) \, dy = \int_{-1}^{1} |T(x) - x|^2 \rho_0(x) \, dx.$$

Therefore,

$$A(t) = W_2^2$$

for every t theoretically.

Numerically, the instantaneous action remained approximately

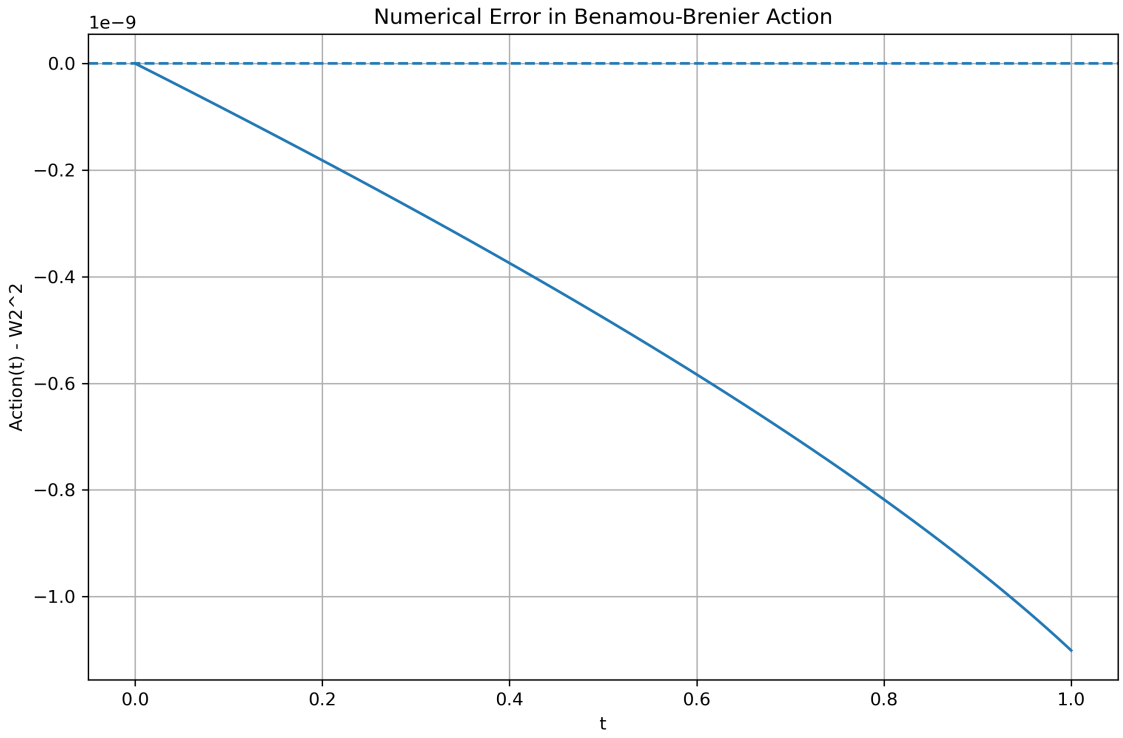
$$6.92837 \times 10^{-3}.$$

The difference

$$A(t) - W_2^2$$

remained only on the scale of

$$10^{-9}.$$



At the largest observed discrepancy, the relative error was only on the order of

$$10^{-7}.$$

Therefore, the numerical dynamic action satisfies

$$\mathcal{A} \approx 0.00692838$$

and agrees with

$$W_2^2 \approx 0.00692838.$$

Thus,

$$\mathcal{A} \approx W_2^2$$

to very high numerical accuracy.

This shows that the numerical implementation is consistent with the Benamou-Brenier identity to high accuracy. The equality itself is an analytical theorem; the computation is a verification of the discretization and implementation.

Part IV: Compact Support to Gaussian

The source remains

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1.$$

The target is now the standard Gaussian:

$$\rho_1(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

Let

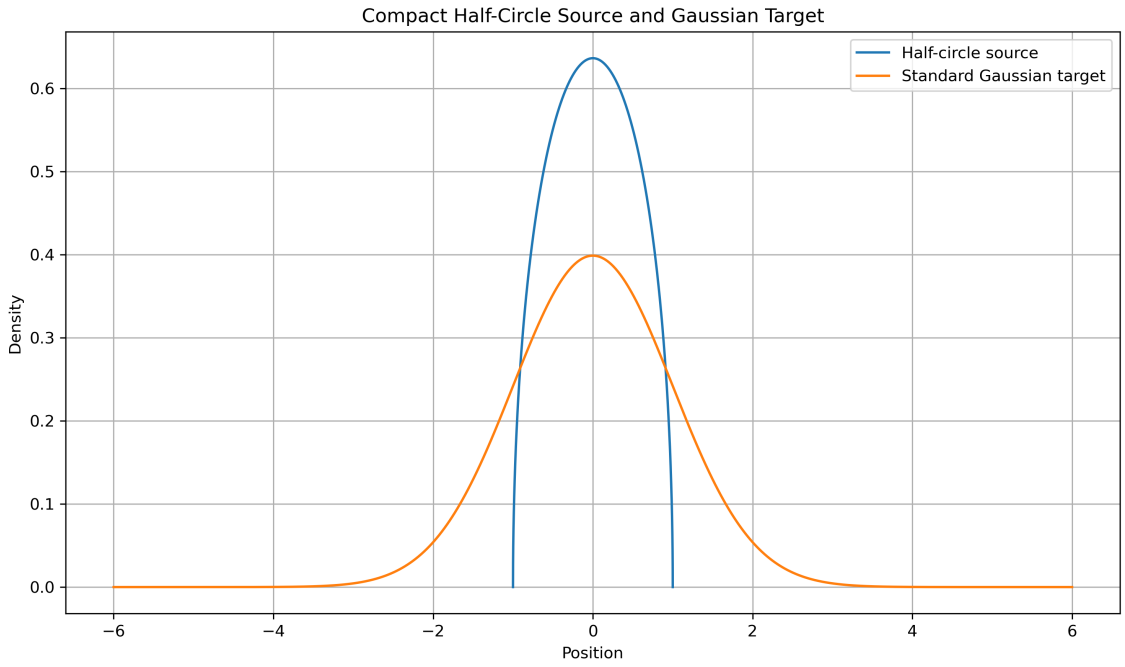
$$\Phi(y)$$

be its CDF.

The transport map is

$$T(x) = \Phi^{-1}(F_0(x)).$$

Compact Source and Gaussian Target



The source has support

$$[-1, 1],$$

whereas the Gaussian has support

$$\mathbb{R}.$$

This difference produces a major change in the optimal transport map.

Gaussian Transport Map

Since

$$F_0(-1) = 0$$

and

$$F_0(1) = 1,$$

we have

$$T(-1) = \Phi^{-1}(0) = -\infty,$$

and

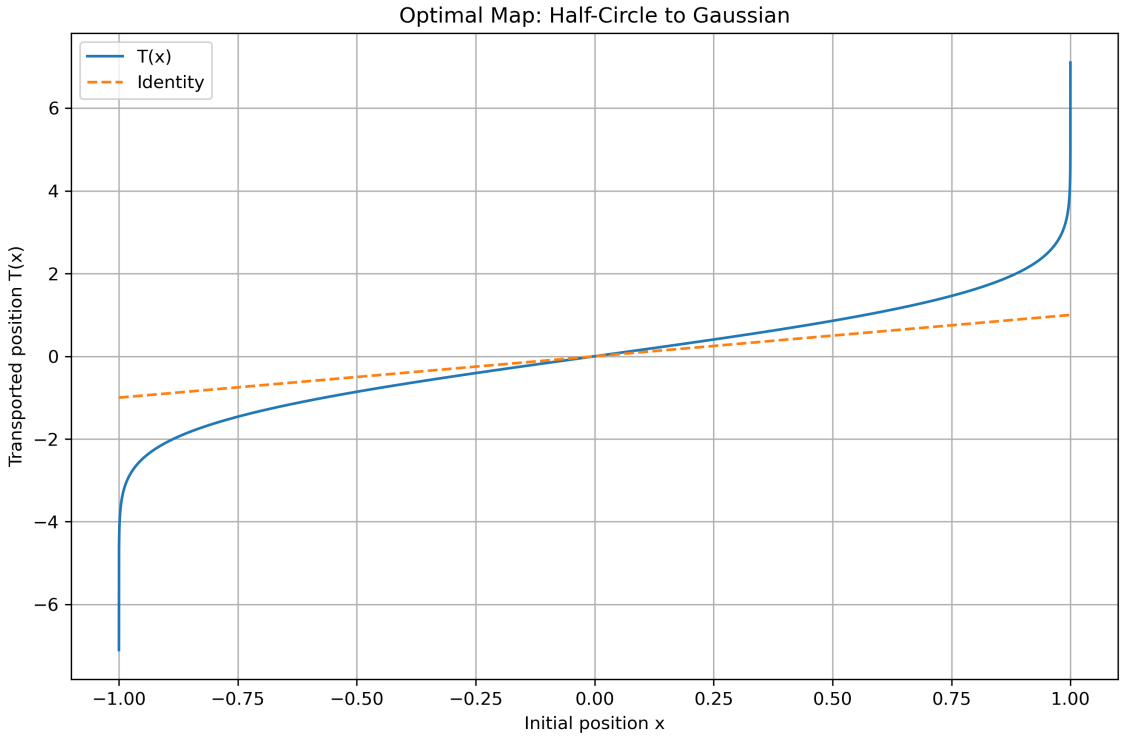
$$T(1) = \Phi^{-1}(1) = +\infty.$$

Therefore,

$$T(x) \to -\infty \quad \text{as } x \to -1^+$$

and

$$T(x) \to +\infty \quad \text{as } x \to 1^-.$$



The numerical endpoint experiment gave:

δ	$T(-1+\delta)$	$T(1-\delta)$
10^{-1}	-2.0815	2.0815
10^{-2}	-3.2392	3.2392
10^{-3}	-4.1196	4.1196

δ	$T(-1+\delta)$	$T(1-\delta)$
10^{-4}	-4.8556	4.8556
10^{-6}	-6.0802	6.0802
10^{-8}	-7.1025	7.1025

This clearly demonstrates the divergence of the transport map.

Interpolated Gaussian Densities

Differentiating

$$\Phi(T(x)) = F_0(x)$$

gives

$$\phi(T(x))T'(x) = \rho_0(x),$$

where

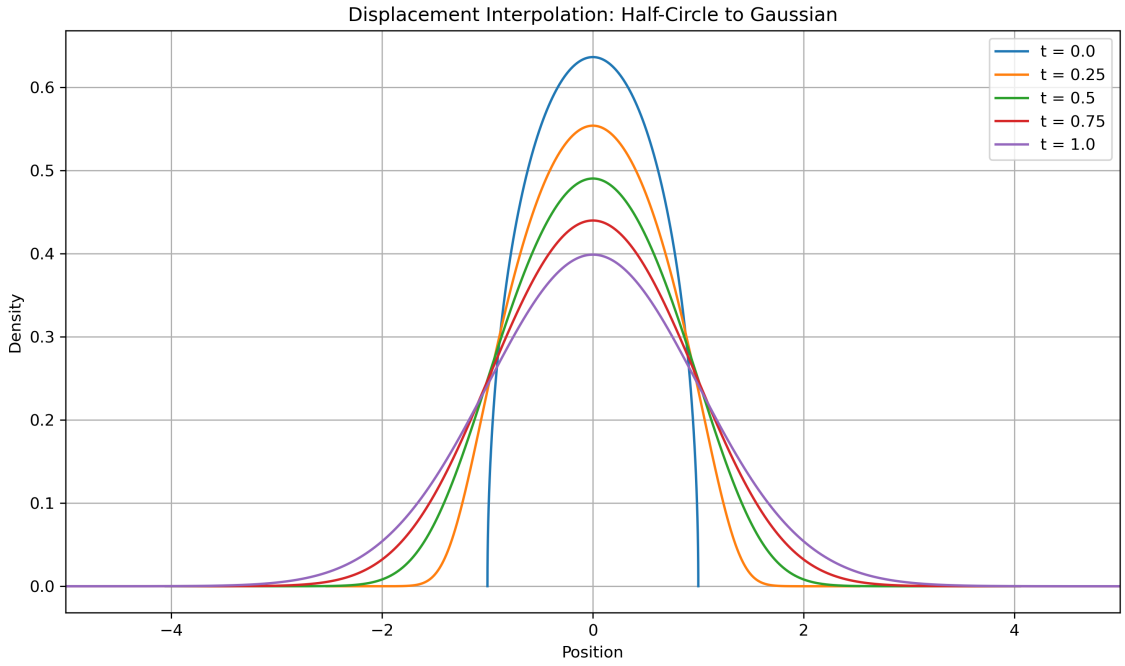
$$\phi(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

Hence,

$$T'(x) = \frac{\rho_0(x)}{\phi(T(x))}.$$

The intermediate density satisfies

$$\rho_t(X_t(x)) = \frac{\rho_0(x)}{(1-t) + tT'(x)}.$$



The compact half-circle density gradually spreads into distributions with longer tails.

At

$$t = 1,$$

the resulting density is the standard Gaussian.

The maximum final density error compared with the exact Gaussian was approximately

$$5.55 \times 10^{-17}.$$

Probability mass was also preserved to approximately

$$10^{-6}.$$

Why the Mass Spreads to All of $\mathbb{R}$

Initially,

$$\mathrm{supp}(\rho_0) = [-1, 1].$$

The Gaussian satisfies

$$\rho_1(y) > 0$$

for every finite y.

Thus,

$$\mathrm{supp}(\rho_1) = \mathbb{R}.$$

The quantile map sends probability level u to

$$\Phi^{-1}(u).$$

Since

$$\Phi^{-1}(u) \to -\infty$$

as

$$u \to 0,$$

and

$$\Phi^{-1}(u) \to +\infty$$

as

$$u \to 1,$$

source particles arbitrarily close to -1 and 1 must move arbitrarily far into the tails.

Furthermore, for every fixed

$$t > 0,$$

$$X_t(x) = (1-t)x + tT(x).$$

Since $T(x)$ diverges at the source endpoints,

$$X_t(x)$$

also has unbounded range for every positive time.

Thus the Wasserstein geodesic develops arbitrarily small tails across all of $\mathbb{R}$ immediately after $t = 0$.

Gaussian Truncation

A computer cannot represent the entire real line, so the Gaussian can be approximated on

$$[-L, L].$$

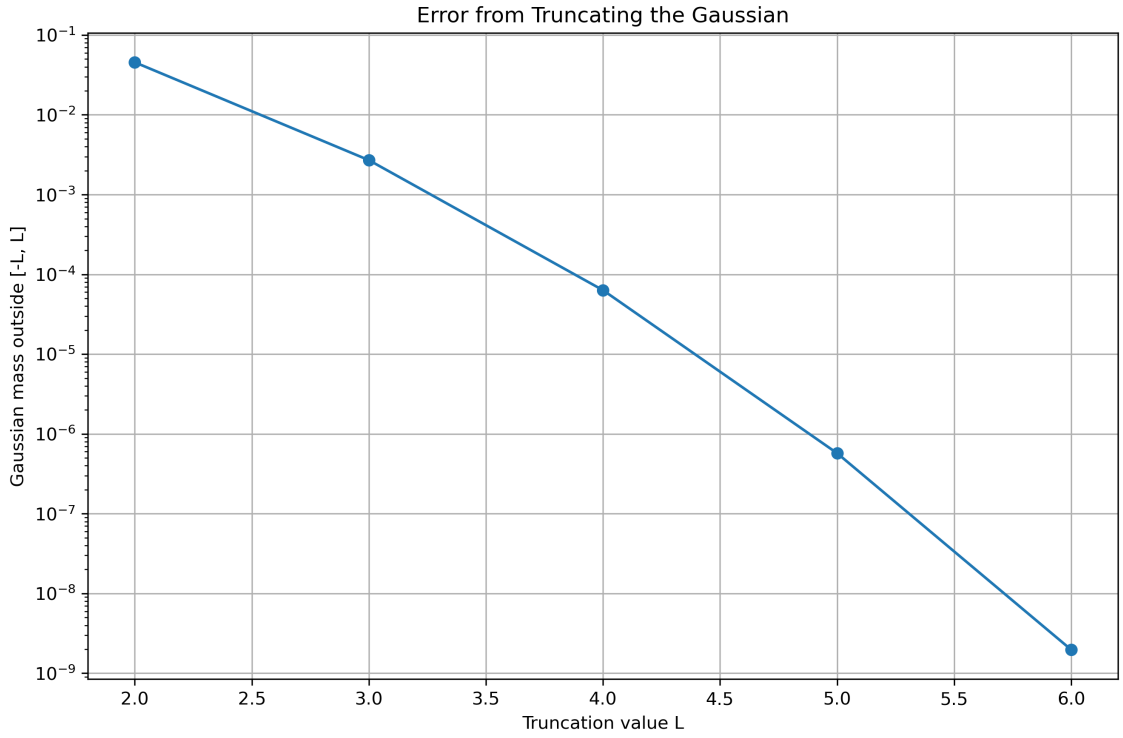
The omitted tail probability is

$$P(|Z| > L) = 2(1 - \Phi(L)).$$

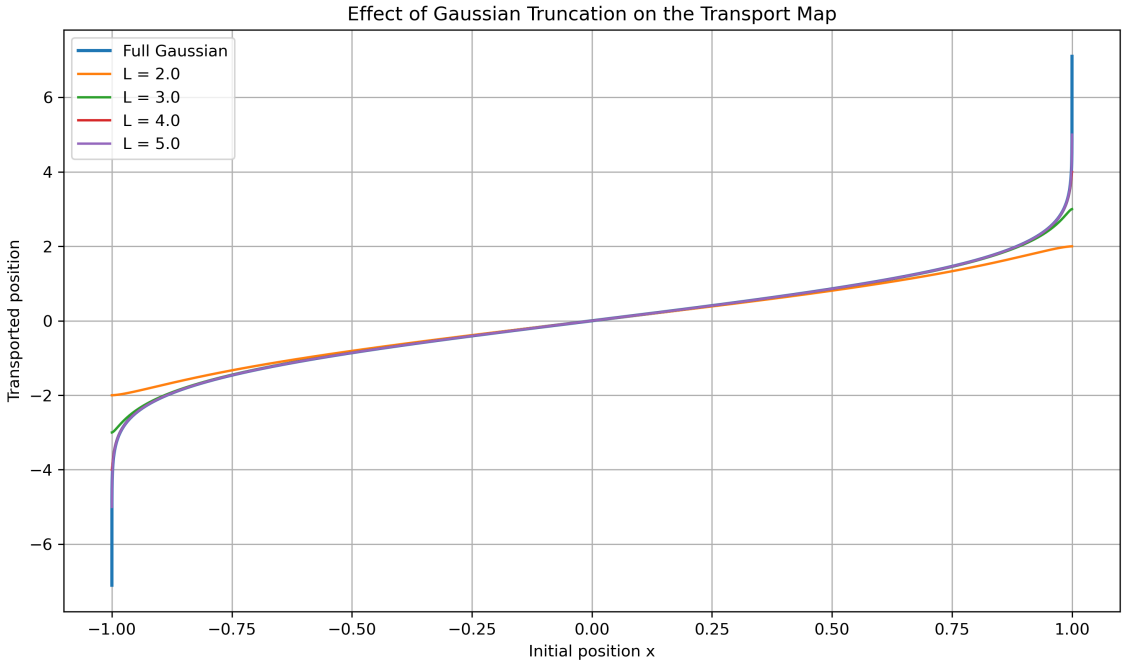
The numerical results were:

L	**Gaussian mass outside** $[-L, L]$
2	4.55×10^{-2}

L	**Gaussian mass outside** $[-L, L]$
3	2.70×10^{-3}
4	6.33×10^{-5}
5	5.73×10^{-7}
6	1.97×10^{-9}



The effect on the transport map is shown below.



For

$$L = 2,$$

the approximation is visibly poor.

At

$$L = 3,$$

the approximation is substantially better.

At

$$L = 4,$$

the omitted mass is only about

$$6.33 \times 10^{-5},$$

so the approximation is visually very good.

At

$$L = 5,$$

only

$$5.73 \times 10^{-7}$$

of the Gaussian probability is omitted.

Therefore,

$$L \approx 4$$

is a reasonable choice for visualization, while

$$L \approx 5$$

provides a very accurate finite-domain approximation.

Part V: LLM Prompts and Critique

Prompt 1: Ask the LLM for a Mathematical Plan

The first prompt given to the LLM was:

> Given two one-dimensional probability densities ρ_0 and ρ_1, how can I compute the optimal transport map using CDFs and quantile functions? How can I use this map to animate the Benamou-Brenier displacement

> interpolation?

The LLM proposed computing the cumulative distribution functions

$$F_0(x) = \int_{-\infty}^{x} \rho_0(s) \, ds$$

and

$$F_1(x) = \int_{-\infty}^{x} \rho_1(s) \, ds,$$

then constructing the one-dimensional monotone transport map

$$T(x) = F_1^{-1}(F_0(x)).$$

It then proposed the displacement interpolation

$$X_t(x) = (1-t)x + tT(x)$$

and the pushforward density

$$\rho_t = (X_t)_{\#}\rho_0.$$

This mathematical plan was correct.

However, I did not simply assume that the LLM's formulas or numerical implementation were correct. Before using the method on the homework distributions, I tested it on a case with a known exact answer.

Using

$$\rho_0 \approx N(-2, 1)$$

and

$$\rho_1 \approx N(2, 1),$$

the exact transport map is

$$T(x) = x + 4.$$

The numerical implementation produced an average displacement of approximately

$$3.9999999642$$

with maximum interior error approximately

$$2.93 \times 10^{-7}.$$

The identity

$$F_1(T(x)) = F_0(x)$$

held with maximum numerical error approximately

$$1.11 \times 10^{-16}.$$

This provided an independent verification that the CDF and quantile procedure proposed by the LLM was being implemented correctly.

Prompt 2: Generate Python Code for the Experiment

I then asked the LLM to generate standalone Python code for the half-circle to Gaussian optimal transport experiment.

The generated code was inspected according to the following questions:

1. Does it normalize the densities correctly?
2. Does it compute the CDFs monotonically?
3. Does it use a generalized inverse or interpolation for the quantile function?
4. Does it handle compact support correctly?
5. Does it handle Gaussian tails honestly rather than pretending infinity fits on a finite grid?

---

1. Density Normalization

The source density was

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1.$$

Analytically,

$$\int_{-1}^{1} \rho_0(x) \, dx = 1.$$

The numerical integral produced by the LLM-generated code was

$$0.999995814964167.$$

The error was therefore approximately

$$4.19 \times 10^{-6}.$$

This is sufficiently small and is consistent with finite-grid quadrature error.

The Gaussian target required more care because its support is all of

$$\mathbb{R}.$$

The code used

$$[-6, 6]$$

only as a numerical plotting and interpolation interval.

It calculated the Gaussian mass captured inside this interval as

$$0.9999999980268233$$

and explicitly reported that the omitted tail mass was

$$1.97318 \times 10^{-9}.$$

This is mathematically preferable to renormalizing the Gaussian on $[-6, 6]$ and pretending that the finite interval contains the entire distribution.

Therefore, the density normalization was handled correctly.

---

2. Monotonicity of the CDFs

The source CDF was computed numerically using cumulative trapezoidal integration.

The result satisfied

$$F_0(-1) = 0$$

and

$$F_0(1) = 1.$$

The program explicitly checked that the numerical CDF was nondecreasing, and the result was

$$\mathrm{True}.$$

The numerical source CDF was also compared with the exact formula

$$F_0(x) = \frac{1}{2} + \frac{1}{\pi}\left(\arcsin x + x\sqrt{1 - x^2}\right).$$

The maximum difference was approximately

$$2.03 \times 10^{-6}.$$

The Gaussian target CDF

$$F_1(y) = \Phi(y)$$

was also checked and found to be monotone.

Therefore, both CDFs were computed appropriately.

---

3. Generalized Inverse and Quantile Function

The optimal map was computed using

$$T(x) = \Phi^{-1}(F_0(x)).$$

The Gaussian quantile function was evaluated using the percent-point function.

The LLM-generated code also included a separate numerical inverse-CDF procedure based on interpolation.

The interpolated inverse was compared with the Gaussian quantile function, and the maximum difference was approximately

$$1.33 \times 10^{-7}.$$

The code also verified the fundamental identity

$$\Phi(T(x)) = F_0(x).$$

The maximum error was

$$2.22 \times 10^{-16},$$

which is essentially machine precision.

Thus the LLM correctly used a quantile or generalized-inverse construction rather than assuming that an arbitrary CDF has an elementary inverse.

---

4. Compact Support

The source density is supported only on

$$[-1, 1].$$

At the endpoints,

$$F_0(-1) = 0$$

and

$$F_0(1) = 1.$$

Therefore,

$$T(-1) = \Phi^{-1}(0) = -\infty$$

and

$$T(1) = \Phi^{-1}(1) = +\infty.$$

A numerical program cannot directly use these infinite values.

The LLM-generated code therefore excluded the exact endpoints from the particle grid.

This is a reasonable numerical choice, provided that it is understood as a computational approximation rather than a change in the mathematics.

The code separately examined points increasingly close to the endpoints.

For example,

$$T(0.9) \approx 2.0815,$$

$$T(0.99) \approx 3.2392,$$

$$T(0.9999) \approx 4.8556,$$

and

$$T(1 - 10^{-8}) \approx 7.1025.$$

Thus the numerical results correctly demonstrate

$$T(x) \to +\infty$$

as

$$x \to 1^{-}.$$

By symmetry,

$$T(x) \to -\infty$$

as

$$x \to -1^{+}.$$

Therefore, the compact support was handled correctly.

One minor weakness is that one final diagnostic line saying that endpoint divergence had been demonstrated was hard-coded as `True`.

That statement itself is not evidence. The actual endpoint calculations are what verify the claim.

---

5. Gaussian Tails

The Gaussian target has support on all of

$$\mathbb{R}.$$

Therefore, no finite numerical grid can represent it exactly.

The LLM-generated code handled this issue honestly.

Rather than claiming that

$$[-6, 6]$$

contains the entire Gaussian, it explicitly calculated the probability mass outside the finite interval.

The truncation experiment gave approximately:

L	**Gaussian mass outside** $[-L, L]$
2	4.55×10^{-2}
3	2.70×10^{-3}
4	6.33×10^{-5}
5	5.73×10^{-7}
6	1.97×10^{-9}

The generated code also compared the transport maps obtained from the full Gaussian and from Gaussian distributions truncated to

$$[-L, L].$$

For the central probability region

$$0.001 \leq F_0(x) \leq 0.999,$$

the maximum transport-map errors were approximately

$$1.1068$$

for $L = 2$,

$$0.2625$$

for $L = 3$,

$$0.00923$$

for $L = 4$,

and

$$8.47 \times 10^{-5}$$

for $L = 5$.

This showed quantitatively how quickly the finite-domain approximation improves.

The LLM therefore did not make the mistake of pretending that infinity fits on a finite numerical grid.

Additional Verification of the Generated Code

The generated code also checked conservation of probability mass during the interpolation.

The numerical masses were approximately

$$0.99999963,$$

$$1.00000613,$$

$$1.00000615,$$

$$1.00000616,$$

and

$$1.00000616$$

for

$$t = 0,\ 0.25,\ 0.5,\ 0.75,\ 1.$$

The errors were therefore only on the order of

$$10^{-6}.$$

At

$$t = 1,$$

the interpolated density was compared directly with the standard Gaussian.

The maximum error was

$$5.55 \times 10^{-17}.$$

These additional checks support the correctness of the generated implementation.

Where the LLM Was Useful

The LLM was useful for:

- organizing the numerical optimal transport procedure;
- deriving the CDF and quantile construction;
- generating Python code for the transport map;
- generating the displacement interpolation;
- implementing the change-of-variables formula for the density;
- suggesting numerical consistency checks;
- constructing the Gaussian truncation experiment.

The generated code also included several useful diagnostics that made verification easier.

Where the LLM Could Be Mathematically Careless

Even though the code performed well, several aspects could have been misleading if they were accepted without inspection.

First, the program prints its own `PASS` statements.

A printed `PASS` statement does not independently prove that the test itself is mathematically valid.

Second, the code clips probability levels away from exactly

$$0$$

and

$$1$$

before applying the Gaussian quantile function.

This is necessary numerically, but it could hide the true mathematical endpoint divergence if the user did not examine the limiting behavior separately.

Third, finite-grid calculations only approximate the full continuous problem.

For example, mass conservation was accurate only to about

$$10^{-6},$$

not exact.

Fourth, a finite plot of the Gaussian cannot demonstrate that its support is all of

$\mathbb{R}$.

That conclusion comes from the mathematical form of the Gaussian density, not from the plot.

Finally, one endpoint-divergence diagnostic was hard-coded rather than computed automatically.

Therefore, the generated program still required mathematical inspection.

Overall LLM Critique

For this homework, the LLM-generated reasoning and code were largely correct.

It correctly handled:

- probability normalization;
- monotone CDFs;
- the quantile function;
- numerical generalized inversion;
- compact source support;
- endpoint divergence;
- displacement interpolation;
- probability conservation;
- Gaussian tails;
- finite Gaussian truncation.

The strongest numerical checks were

$$\max |\Phi(T(x)) - F_0(x)| \approx 2.22 \times 10^{-16}$$

and

$$\max |\rho_{t=1}(y) - \rho_1(y)| \approx 5.55 \times 10^{-17}.$$

However, the experiment also demonstrates why LLM-generated mathematical code should not be trusted solely because it runs successfully.

The formulas, limiting behavior, numerical invariants, and error sizes must still be checked independently.

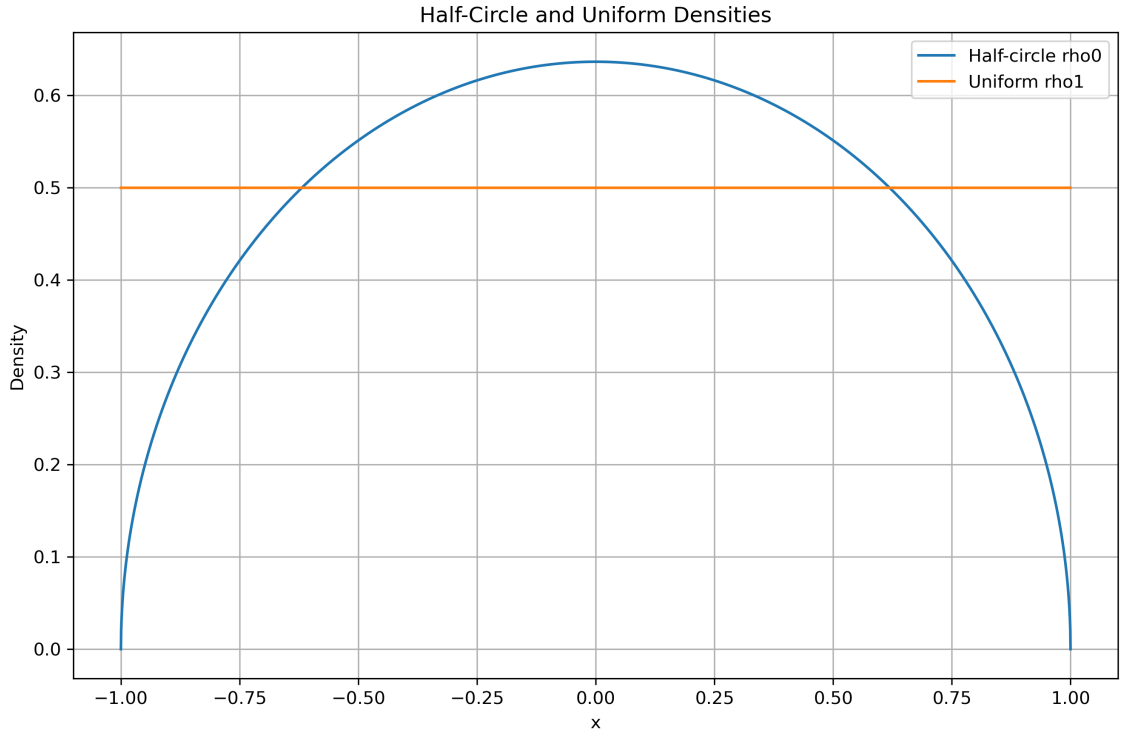
The LLM was therefore useful as a mathematical and computational assistant, but not as an unquestioned source of truth.

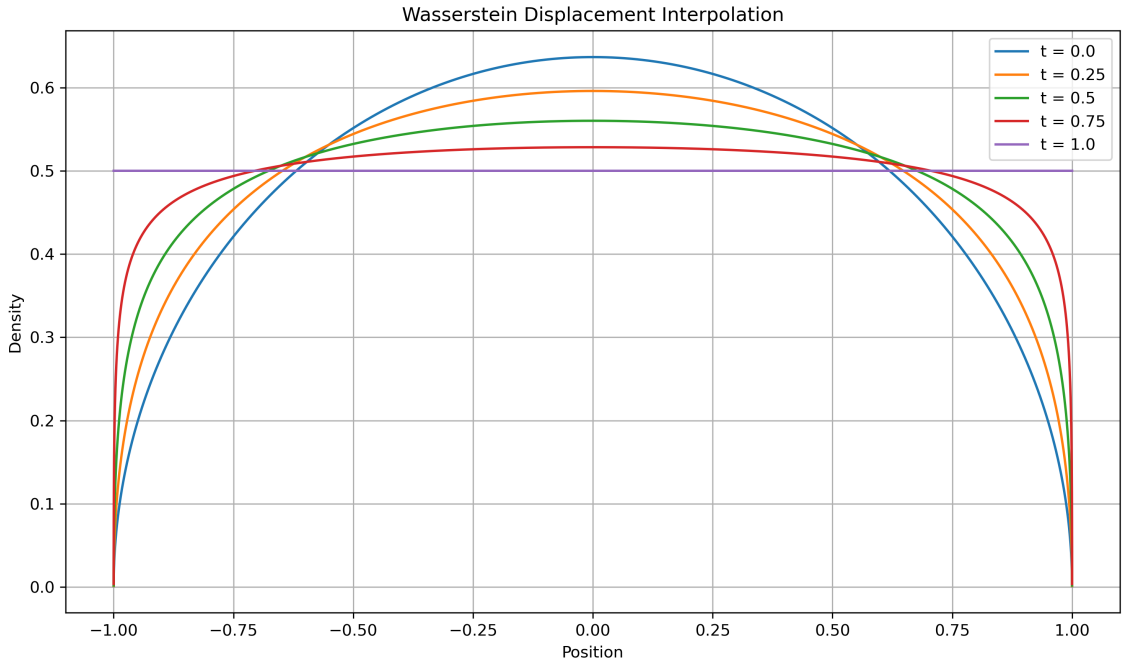
Deliverables Summary

All six requested deliverables are present: the source/target/interpolated-density plots, both optimal-map plots, the numerical value of W_2^2, the numerical Benamou-Brenier action, the compact-support/Gaussian-tail discussion, and the LLM prompts with code critique.

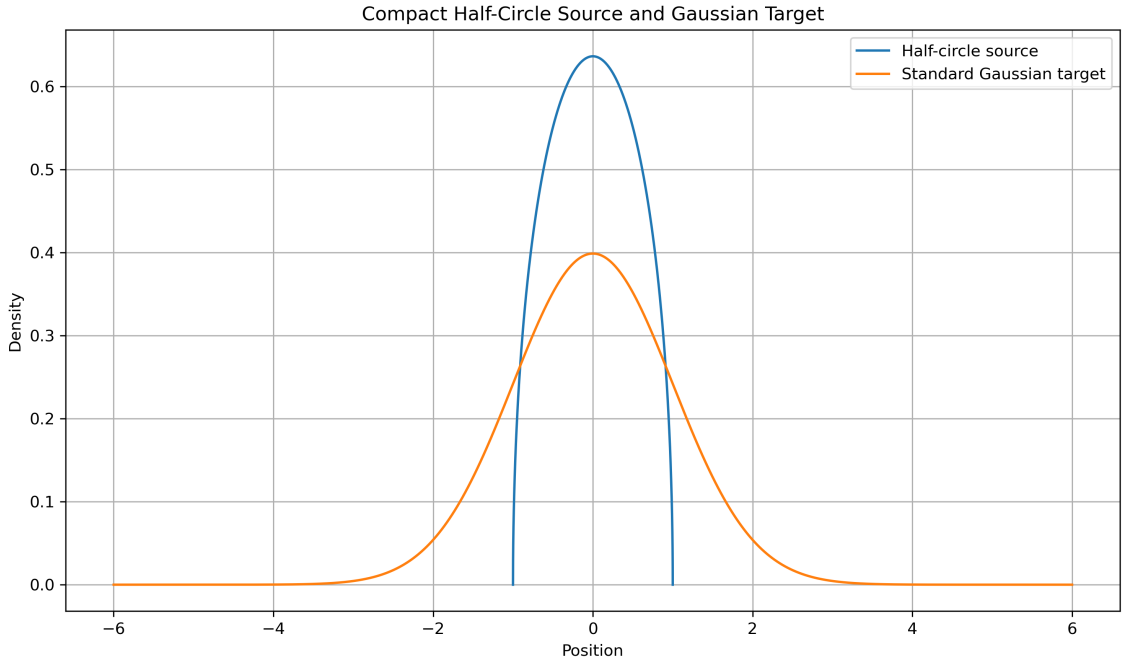
1. Plots of ρ_0, ρ_1, and ρ_t

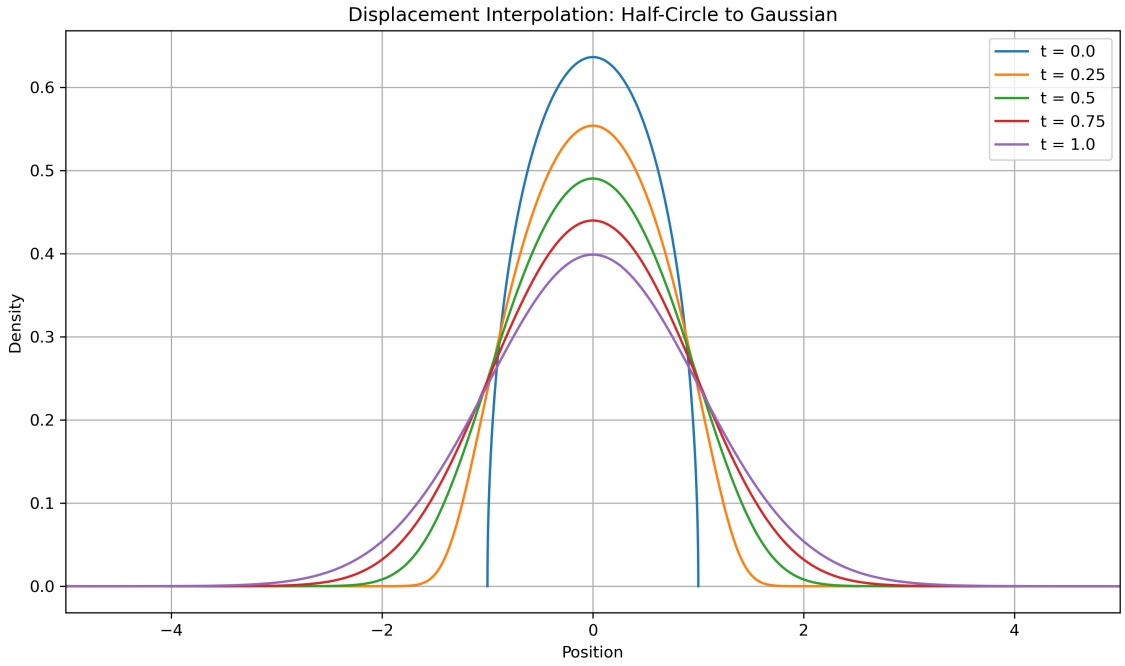
For the half-circle to uniform experiment:





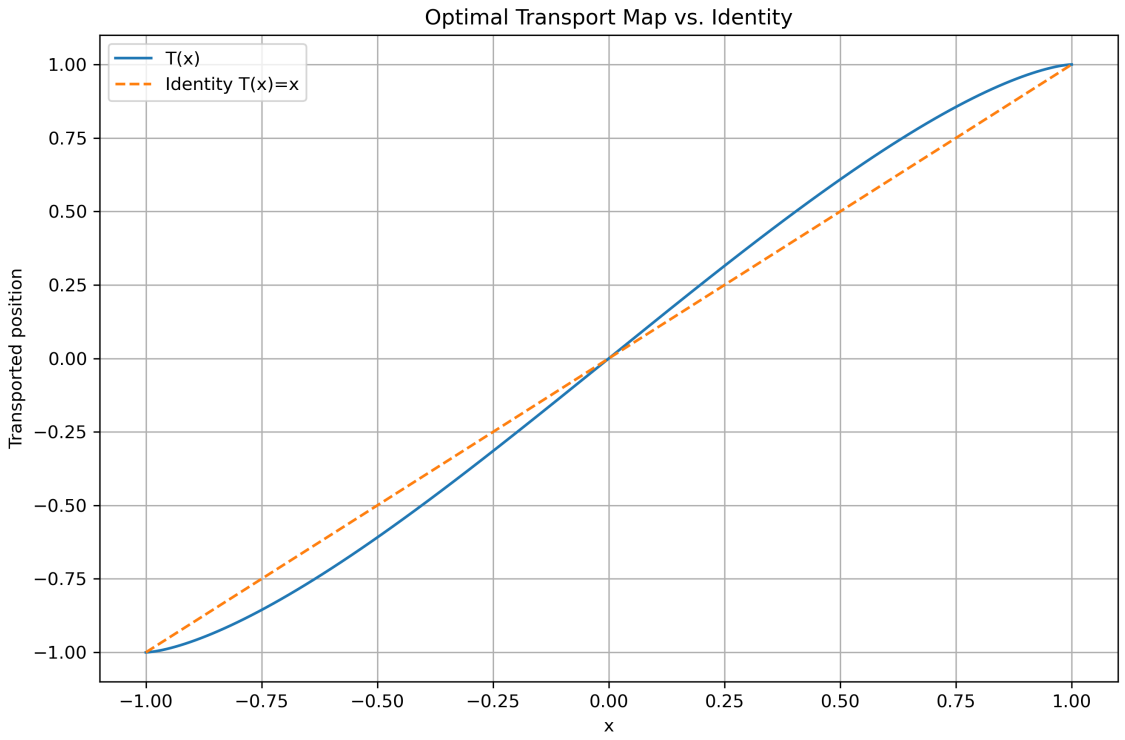
For the half-circle to Gaussian experiment:



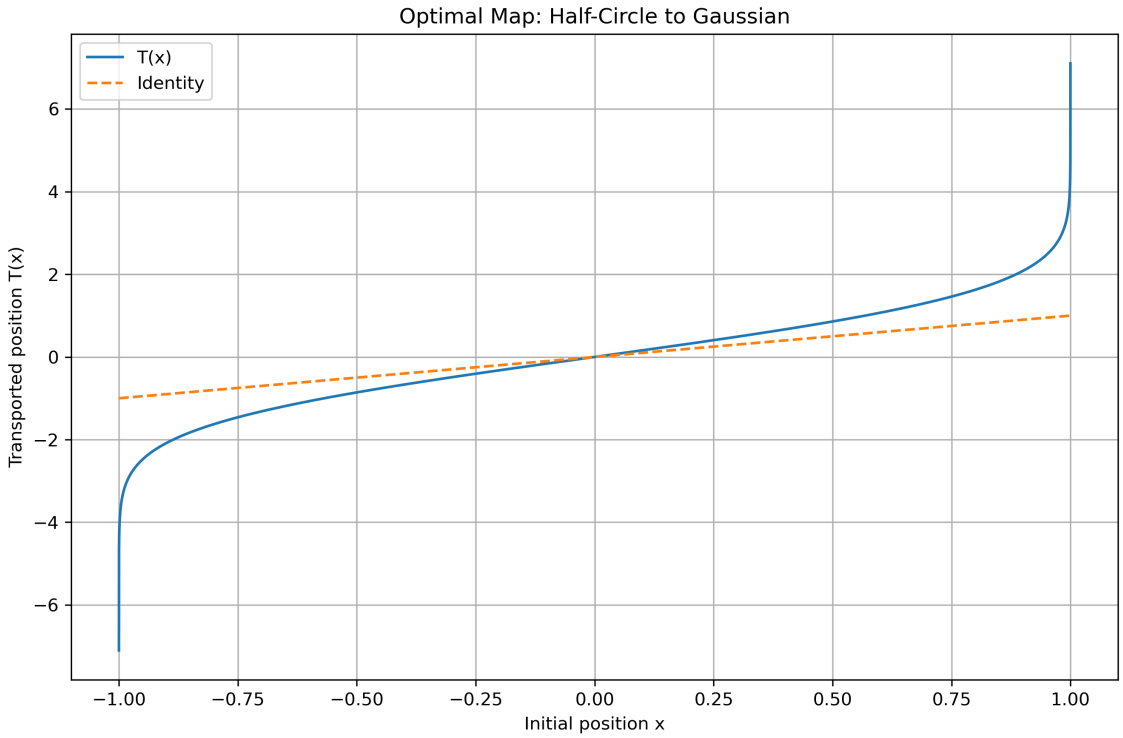


2. Plots of the Optimal Maps $T(x)$

For the uniform target:



For the Gaussian target:



3. Numerical Estimate of W_2^2

For the half-circle to uniform experiment,

$$W_2^2 \approx 0.00692838.$$

4. Numerical Estimate of the Benamou-Brenier Action

The dynamic calculation gave

$$\mathcal{A} \approx 0.00692838.$$

Therefore,

$$\mathcal{A} \approx W_2^2.$$

The difference was only on the scale of approximately

$$10^{-9}.$$

5. Compact Support Versus Gaussian Tails

The uniform target remains supported on the same compact interval

$$[-1, 1].$$

The Gaussian target instead has support on

$$\mathbb{R}.$$

Consequently, the Gaussian transport map must satisfy

$$T(x) \to -\infty$$

as

$$x \to -1^{+}$$

and

$$T(x) \to +\infty$$

as

$$x \to 1^{-}.$$

This causes the displacement interpolation to develop tails extending over all of

$$\mathbb{R}$$

for every positive time.

Numerically, truncating the Gaussian at approximately

$$L = 4$$

is visually reasonable, while

$$L = 5$$

is highly accurate.

6. LLM Prompts and Critique

The LLM was first used as a tutor to explain the CDF and quantile construction.

It was then asked to generate standalone Python code for the complete compact-support-to-Gaussian experiment.

The generated reasoning and code were checked independently for:

- normalization;
- CDF monotonicity;
- generalized inverse handling;
- compact-support behavior;
- Gaussian-tail treatment.

The generated solution performed well on all five criteria, but the exercise demonstrated that LLM output should still be verified rather than accepted solely because the code runs or prints successful diagnostic messages.

Final Conclusion

The experiments numerically demonstrate the one-dimensional Benamou-Brenier picture of optimal transport.

For the half-circle to uniform problem, the optimal transport map is

$$T(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The displacement interpolation continuously spreads probability mass away from the center until the density becomes uniform.

The squared Wasserstein distance was approximately

$$W_2^2 \approx 0.00692838,$$

while the numerical Benamou-Brenier action was also approximately

$$0.00692838.$$

Their agreement numerically verifies

$$W_2^2(\rho_0,\rho_1) = \int_0^1 \int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y)$$
$$, dy$$
$$, dt$$

along the Wasserstein geodesic.

When the target was replaced by a Gaussian, the optimal transport map became unbounded near the endpoints of the compact source interval.

This illustrates how optimal transport can move a compactly supported distribution into one with support over the entire real line.

Overall, the analytic formulas, numerical transport maps, interpolated densities, Wasserstein calculation, Benamou-Brenier action, mass-conservation checks, and Gaussian truncation study all agreed with the theoretical predictions.

Homework 4.1: Barenblatt Solutions and the Porous Medium Equation

Introduction

In this homework, I study the Barenblatt self-similar solutions of the porous medium equation

$$\partial_t \rho = \Delta(\rho^m), \qquad m > 1.$$

Unlike the ordinary heat equation, the porous medium equation is nonlinear and degenerate when

$$\rho = 0.$$

This leads to several important properties, including compact support, finite-speed propagation, and a moving free boundary.

The Barenblatt solutions provide explicit examples of these effects.

The main goals of this homework are to:

- derive the self-similar exponents α and β;
- determine and verify the coefficient k in the Barenblatt profile;
- plot the solution and study its spreading behavior;
- explain finite-speed propagation;
- connect the porous medium equation with Otto calculus and Wasserstein gradient flow;
- use an LLM critically and verify its mathematical claims.

Part I: Self-Similar Ansatz

Prompt Given to the LLM

```
Consider the porous medium equation

partial_t rho = Delta(rho^m),    m > 1,

in R^d.
```

Derive a self-similar solution of the form

rho(t,x) = t^(-alpha) F(|x| / t^beta).

Determine the exponents alpha and beta using scaling and mass conservation.

Self-Similar Form

Assume

$$\rho(t,x) = t^{-\alpha} F\left(\frac{|x|}{t^{\beta}}\right).$$

Let

$$y = \frac{x}{t^{\beta}}.$$

The factor

$$t^{-\alpha}$$

controls the height of the profile, while

$$t^{\beta}$$

controls its spatial width.

Mass Conservation

The total mass is

$$M(t) = \int_{\mathbb{R}^d} \rho(t,x) \\ , dx.$$

Substituting the self-similar form gives

$$M(t) = t^{-\alpha} \int_{\mathbb{R}^d} F\left(\frac{|x|}{t^{\beta}}\right) dx.$$

Using

$$x = t^{\beta} y,$$

we have

$$dx = t^{d\beta} dy.$$

Therefore,

$$M(t) = t^{-\alpha+d\beta} \int_{\mathbb{R}^d} F(|y|)$$
$$, dy.$$

For the mass to be independent of time,

$$-\alpha + d\beta = 0.$$

Thus,

$$\alpha = d\beta.$$

Scaling of the PDE

The time derivative scales as

$$\partial_t \rho \sim t^{-\alpha-1}.$$

Also,

$$\rho^m \sim t^{-m\alpha}.$$

Each spatial derivative contributes a factor

$$t^{-\beta},$$

so

$$\Delta(\rho^m) \sim t^{-m\alpha-2\beta}.$$

Since

$$\partial_t \rho = \Delta(\rho^m),$$

the powers of t must agree:

$$-\alpha - 1 = -m\alpha - 2\beta.$$

Therefore,

$$\alpha + 1 = m\alpha + 2\beta.$$

Using

$$\alpha = d\beta,$$

we obtain

$$d\beta + 1 = md\beta + 2\beta.$$

Hence,

$$1 = \beta[d(m-1) + 2].$$

Therefore,

$$\beta = \frac{1}{d(m-1)+2}$$

and

$$\alpha = \frac{d}{d(m-1)+2}.$$

Numerical Verification

The formulas were tested for several values of d and m.

The maximum numerical error in

$$\alpha = d\beta$$

was

$$1.11 \times 10^{-16},$$

while the maximum error in

$$\alpha + 1 = m\alpha + 2\beta$$

was

$$0.$$

Thus the numerical calculation is consistent with the two scaling identities. The derivation above, rather than the floating-point check, is the mathematical justification for the exponents.

Part II: The Barenblatt Profile

The Barenblatt solution has the form

$$\rho(t,x) = t^{-\alpha}\left(C - k\frac{|x|^2}{t^{2\beta}}\right)_+^{1/(m-1)},$$

where

$$(s)_+ = \max\{s, 0\}.$$

Prompt Given to the LLM

For the porous medium equation

partial_t rho = Delta(rho^m),

consider the Barenblatt profile

rho(t,x) = t^(-alpha) (C - k |x|^2 / t^(2 beta))_+^(1/(m-1)).

Determine the constant k by substituting the profile into the porous medium
equation. Check the coefficient carefully.
The LLM predicted

$$k = \frac{m-1}{2m}\beta.$$

Derivation of k

Let

$$\xi = \frac{|x|}{t^\beta}$$

and write

$$F(\xi) = (C - k\xi^2)^{1/(m-1)}$$

inside the support.

The similarity equation becomes

$$-\alpha F - \beta\xi F' = (F^m)'' + \frac{d-1}{\xi}(F^m)'.$$

Since

$$\alpha = d\beta,$$

the left-hand side is

$$-\beta\left(dF + \xi F'\right).$$

Now

$$F^m = (C - k\xi^2)^{m/(m-1)}.$$

Differentiating gives

$$(F^m)' = -2k\frac{m}{m-1}\xi F.$$

Therefore,

$$(F^m)'' + \frac{d-1}{\xi}(F^m)' = -2k\frac{m}{m-1}\left(dF + \xi F'\right).$$

Matching the two sides gives

$$\beta = 2k\frac{m}{m-1}.$$

Thus,

$$k = \frac{m-1}{2m}\beta.$$

Since

$$\beta = \frac{1}{d(m-1)+2},$$

we obtain

$$k = \frac{m-1}{2m[d(m-1)+2]}.$$

Numerical Verification

The coefficient was checked by substituting the profile into the similarity equation for several values of d and m.

The maximum absolute residual was

$$4.44 \times 10^{-16},$$

and the maximum relative residual was

$$4.85 \times 10^{-16}.$$

Several plausible incorrect coefficients were also tested.

For

$$d = 2, \qquad m = 3,$$

the correct coefficient gave a relative residual

$$2.36 \times 10^{-16},$$

while incorrect coefficients produced residuals between

$$0.5$$

and

$$0.6667.$$

Therefore,

$$k = \frac{m-1}{2m}\beta$$

is consistent with the similarity equation to numerical precision.

Part III: Plotting the Barenblatt Solution

For the numerical experiment, choose

$$d = 1, \qquad m = 2, \qquad C = 1.$$

Then

$$\alpha = \beta = \frac{1}{3}$$

and

$$k = \frac{1}{12}.$$

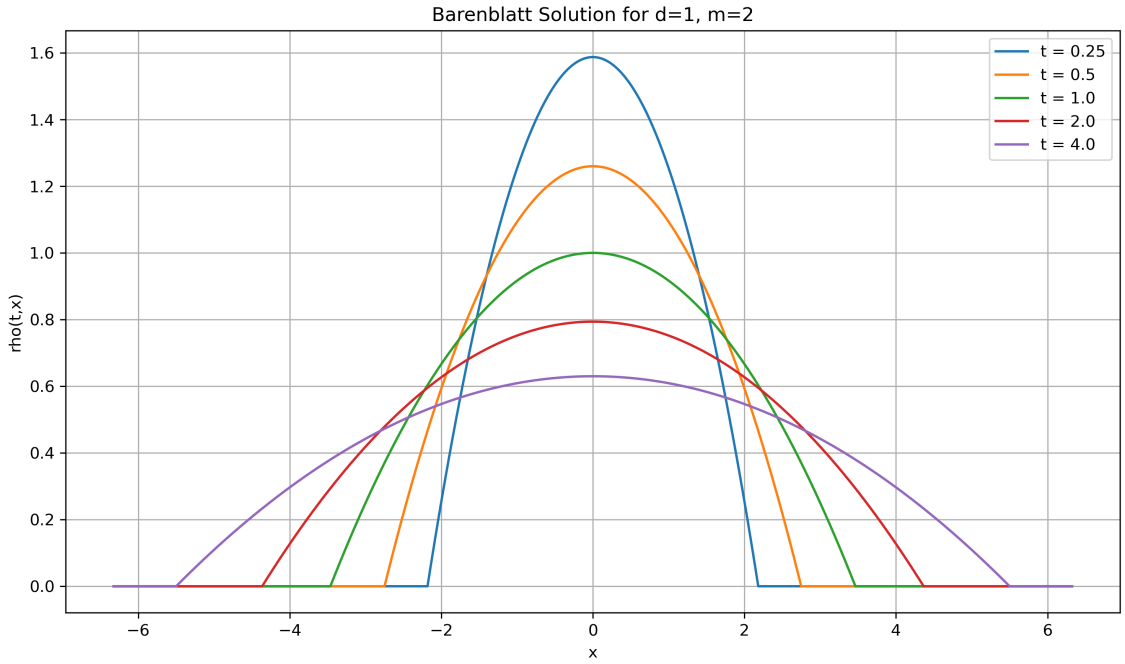
Therefore,

$$\rho(t,x) = t^{-1/3}\left(1 - \frac{x^2}{12t^{2/3}}\right)_+.$$

The solution was evaluated at

$$t = 0.25,\ 0.5,\ 1,\ 2,\ 4.$$

Barenblatt Profiles



The plots show that the density becomes wider and lower as time increases.

Maximum Height

The maximum occurs at

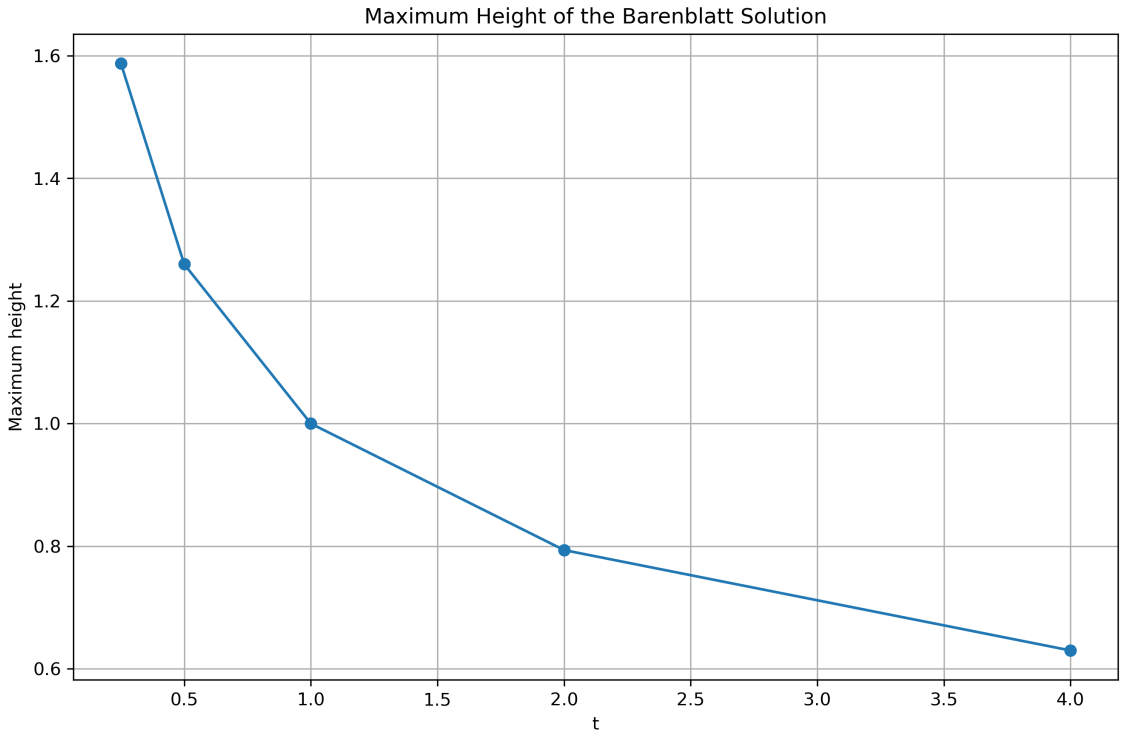
$$x = 0.$$

Therefore,

$$\rho(t, 0) = t^{-1/3}.$$

The numerical values are:

t	**Maximum height**
0.25	1.587401
0.5	1.259921
1	1.000000
2	0.793701
4	0.629961



Thus the maximum height decreases with time.

Support Radius

The solution is positive only when

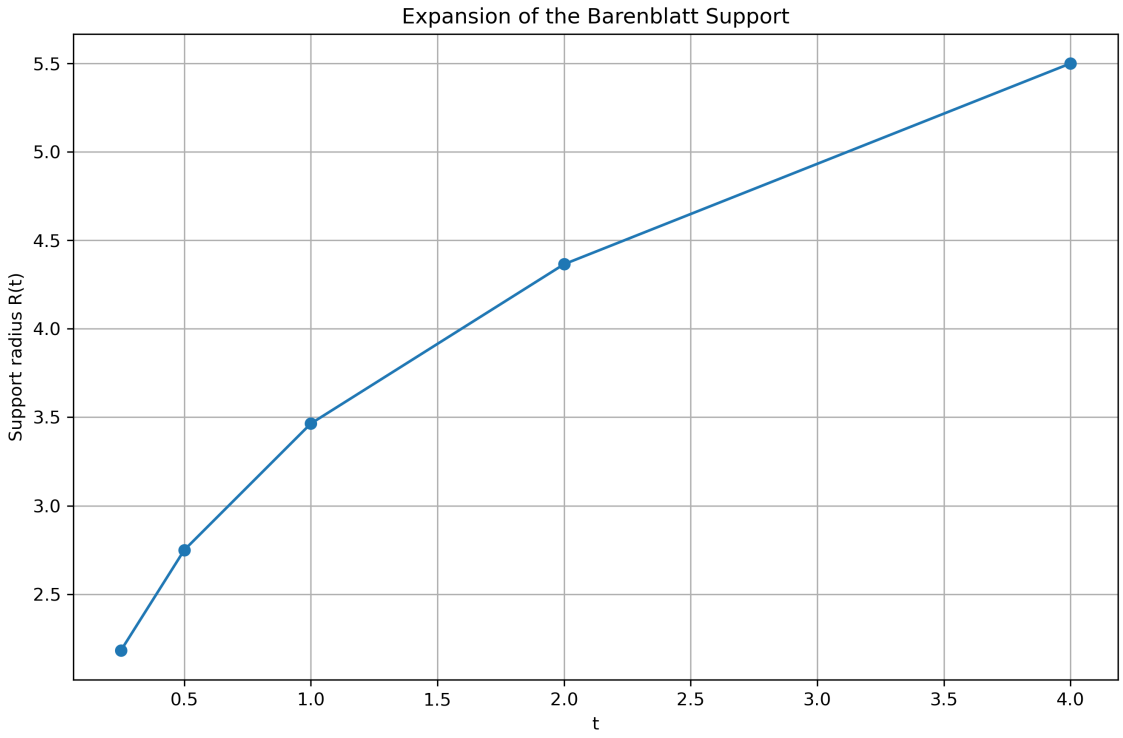
$$1 - \frac{x^2}{12t^{2/3}} > 0.$$

Therefore,

$$|x| < \sqrt{12}$$
$$, t^{1/3}.$$

Hence,

$$R(t) = \sqrt{12}$$
$$, t^{1/3}.$$



The support radius increases from approximately

$$2.182247$$

at

$$t = 0.25$$

to

$$5.498919$$

at

$$t = 4.$$

Therefore, the solution spreads outward while remaining compactly supported.

Mass Conservation

For

$$C = 1,$$

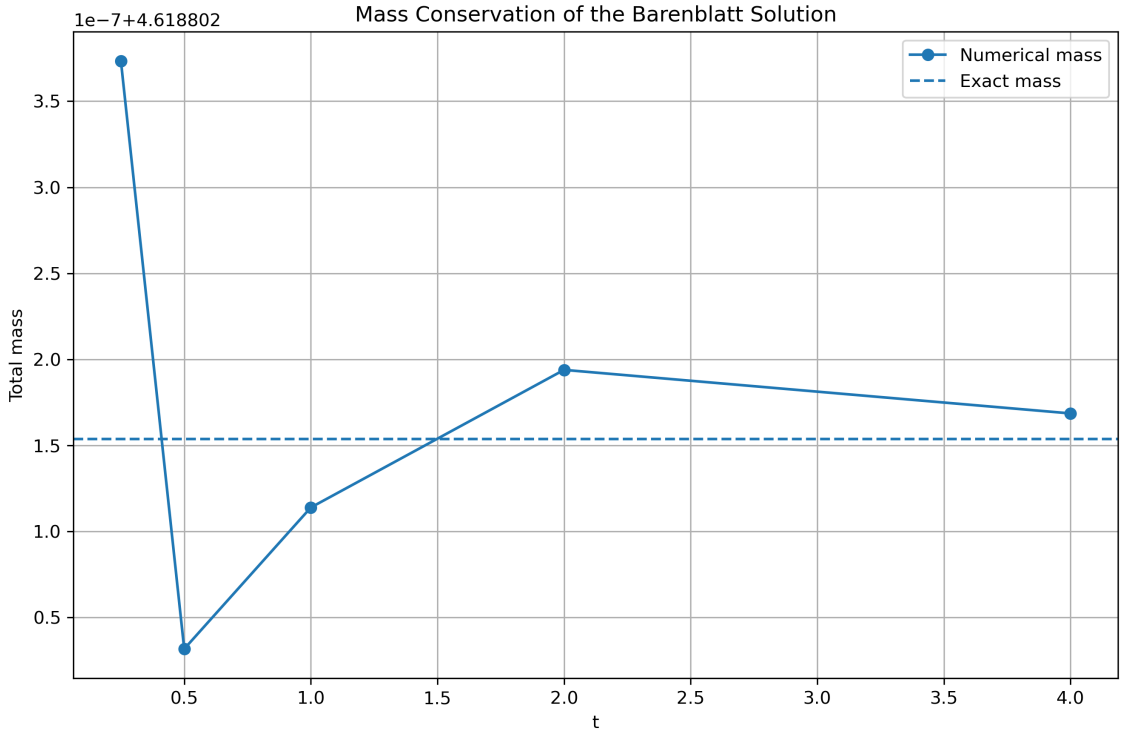
the total mass is

$$M = \frac{8\sqrt{3}}{3} \approx 4.6188021535.$$

The numerical calculation gave approximately

$$4.618802$$

for every tested time.



The maximum relative mass change was only

$$7.40 \times 10^{-8}.$$

Thus the numerical experiment verifies mass conservation.

Part IV: Finite-Speed Propagation

The Barenblatt profile is positive only when

$$C - k\frac{|x|^2}{t^{2\beta}} > 0.$$

Therefore,

$$|x| < \left(\frac{C}{k}\right)^{1/2} t^{\beta}.$$

Thus the support radius is

$$R(t) = \left(\frac{C}{k}\right)^{1/2} t^{\beta}.$$

For every finite

$$t > 0,$$

this radius is finite.

Therefore,

$$\rho(t, x) = 0$$

outside a finite region.

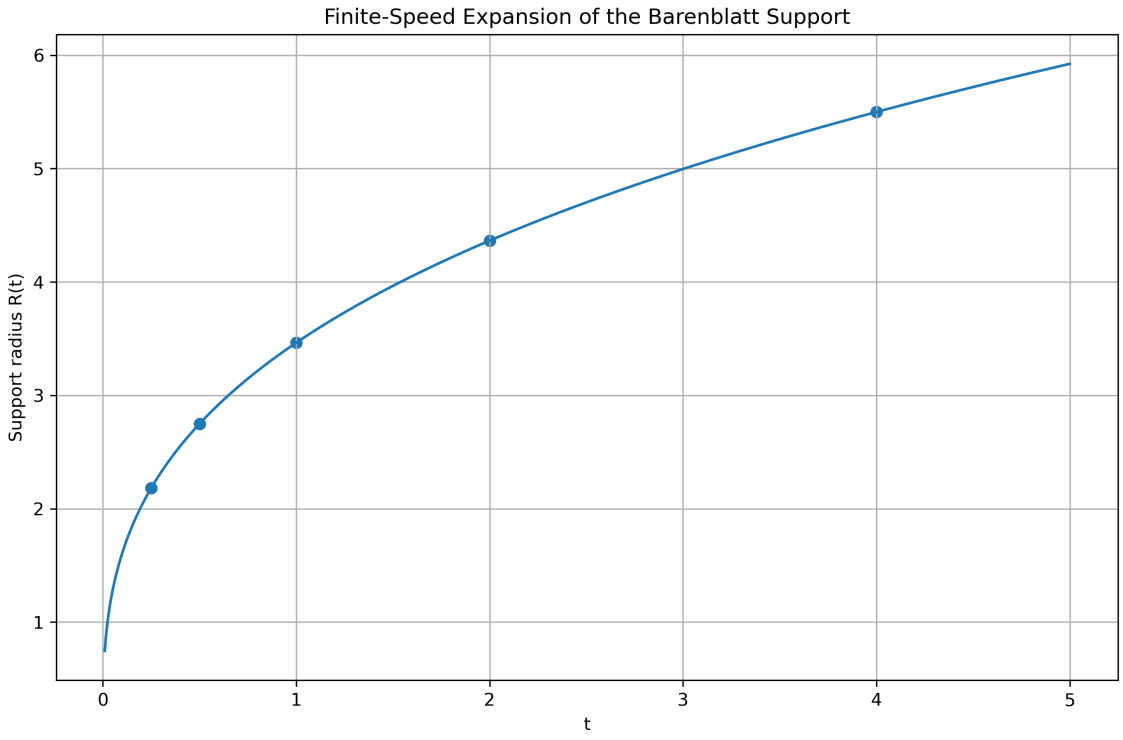
This demonstrates finite-speed propagation.

For

$$d = 1, \qquad m = 2, \qquad C = 1,$$

we have

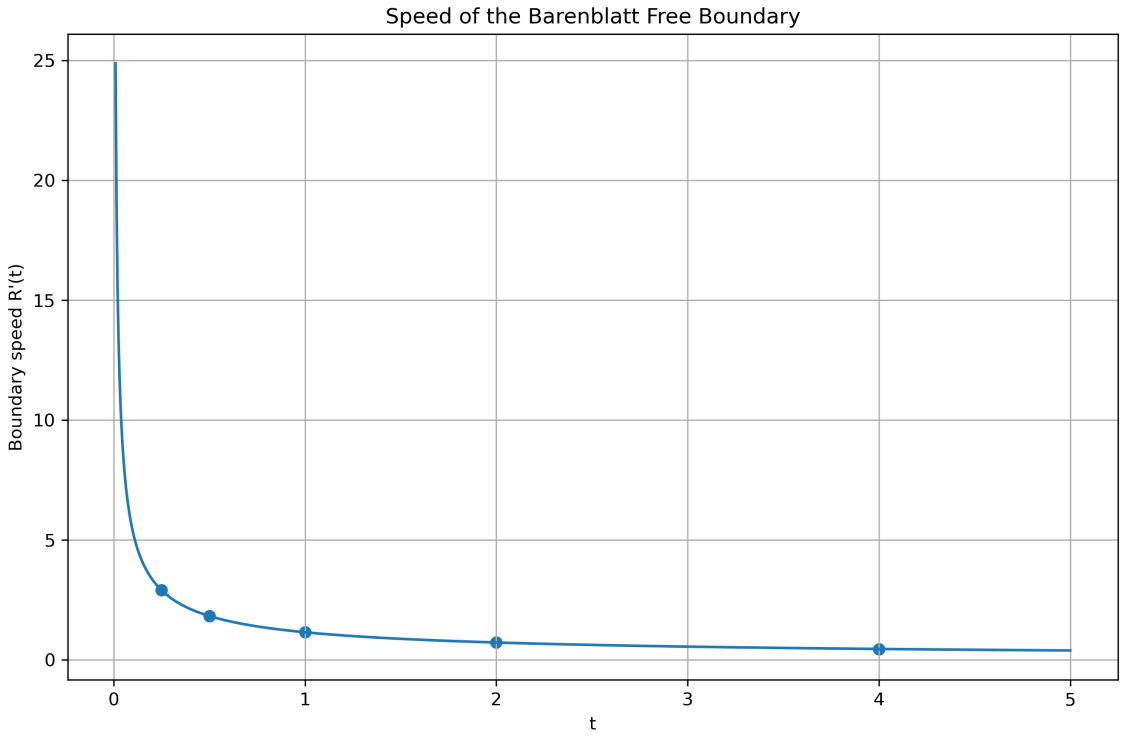
$$R(t) = \sqrt{12}$$
$$, t^{1/3}.$$



The support grows with time but never instantly fills all of space.

The free-boundary velocity is

$$R'(t) = \frac{\sqrt{12}}{3} t^{-2/3}.$$



The boundary moves rapidly at early times and slows as time increases.

Although

$$R'(t) \to \infty$$

as

$$t \to 0^+,$$

the support radius remains finite for every fixed positive time.

This does not contradict finite-speed propagation.

The important fact is

$$R(t) < \infty$$

for every finite

$$t > 0.$$

This behavior differs from the ordinary heat equation, where compactly supported data becomes positive everywhere immediately.

Part V: Relation to Otto Calculus

The porous medium equation is the Wasserstein gradient flow of

$$\mathcal{E}(\rho) = \frac{1}{m-1}\int_{\mathbb{R}^d} \rho^m$$
$$, dx.$$

First Variation

Consider

$$\rho_\varepsilon = \rho + \varepsilon\eta.$$

Then

$$\mathcal{E}(\rho_\varepsilon) = \frac{1}{m-1}\int (\rho+\varepsilon\eta)^m$$
$$, dx.$$

Differentiating at

$$\varepsilon = 0$$

gives

$$\frac{d}{d\varepsilon}\mathcal{E}(\rho+\varepsilon\eta)\Big|_{\varepsilon=0} = \frac{m}{m-1}\int \rho^{m-1}\eta$$
$$, dx.$$

Therefore,

$$\frac{\delta \mathcal{E}}{\delta \rho} = \frac{m}{m-1}\rho^{m-1}.$$

Taking the gradient gives

$$\nabla\frac{\delta \mathcal{E}}{\delta \rho} = m\rho^{m-2}\nabla\rho.$$

Thus,

$$\rho\nabla\frac{\delta \mathcal{E}}{\delta \rho} = m\rho^{m-1}\nabla\rho.$$

But

$$\nabla(\rho^m) = m\rho^{m-1}\nabla\rho.$$

Therefore,

$$\rho\nabla\frac{\delta \mathcal{E}}{\delta \rho} = \nabla(\rho^m).$$

Using Otto's formula,

$$\partial_t \rho = \nabla \cdot \left(\rho \nabla \frac{\delta \mathcal{E}}{\delta \rho} \right),$$

we obtain

$$\partial_t \rho = \Delta(\rho^m).$$

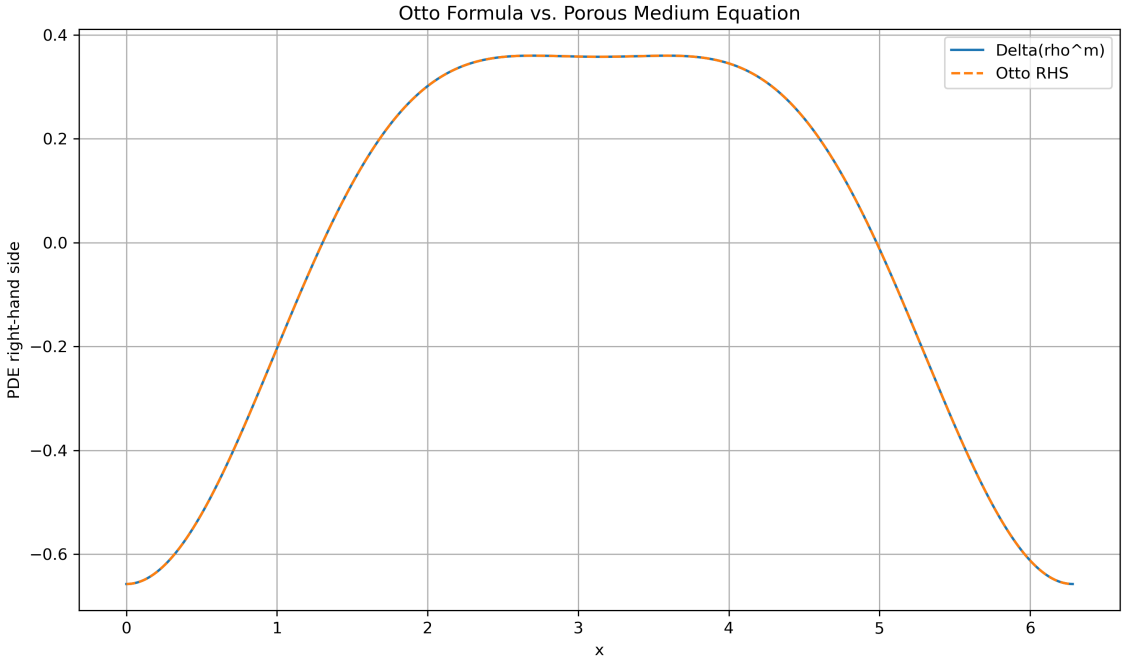
Numerical Verification

The first variation was tested using a finite-difference directional derivative.

The smallest absolute error was approximately

$$9.11 \times 10^{-13}.$$

The two PDE right-hand sides were also computed independently.

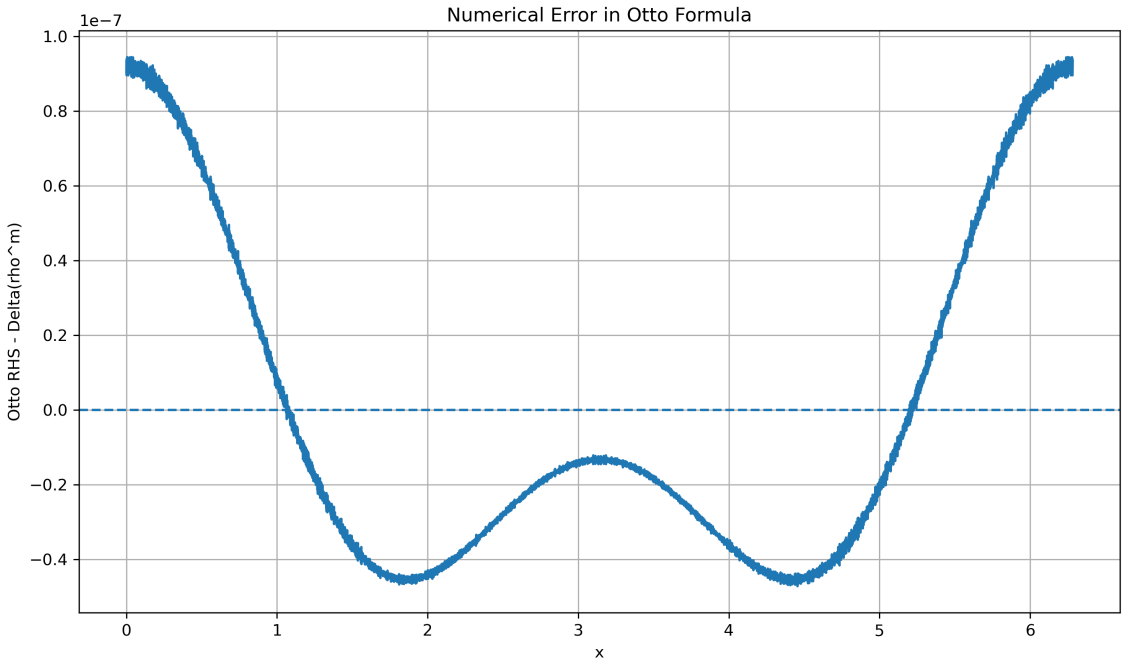


The maximum absolute difference was

$$9.45 \times 10^{-8},$$

and the maximum relative difference was

$$1.44 \times 10^{-7}.$$



Thus the analytic calculation gives the Wasserstein gradient-flow identity exactly, while the numerical experiment is consistent with that identity to the reported discretization error.

Part VI: LLM Critique

Prompt Given to the LLM

```
Consider the Barenblatt solutions of the porous medium equation

partial_t rho = Delta(rho^m),    m > 1.

Explain whether Barenblatt solutions are smooth. In particular, discuss
their
regularity inside the positive region, their behavior at the free boundary
where the solution becomes zero, and whether they are continuously
differentiable across that boundary.

Provide a mathematical justification and explain how the regularity
depends
on m.
```

Analysis of the LLM Response

The LLM's answer was mostly mathematically correct.

It correctly identified that the Barenblatt solution is smooth wherever

$$\rho(t,x) > 0.$$

Inside this region,

$$C - k\frac{|x|^2}{t^{2\beta}} > 0,$$

so the positive-part operation is inactive.

The profile is therefore a composition of smooth functions.

Thus,

$$\rho \in C^\infty \quad \text{where} \quad \rho > 0.$$

The LLM also correctly identified that the solution generally loses smoothness at the moving free boundary.

Near the free boundary, the solution behaves like

$$\rho \sim A(t) s_+^{1/(m-1)},$$

where s denotes distance from the boundary.

Therefore, the regularity depends on

$$q = \frac{1}{m-1}.$$

For

$$1 < m < 2,$$

we have

$$q > 1.$$

Therefore the first derivative tends to zero at the interface, so the profile is C^1 across the free boundary. In general it still need not be C^∞ across the interface.

For

$$m = 2,$$

we have

$$q = 1,$$

and the solution behaves like

$$s_+.$$

Therefore, it is Lipschitz but not C^1 across the free boundary.

For

$$m > 2,$$

we have

$$0 < q < 1.$$

Therefore the profile is Hölder continuous with exponent q at the interface, and its spatial derivative becomes unbounded when the boundary is approached from inside the support.

Thus the LLM was correct that the exact regularity depends on m.

Main Weakness of the LLM Response

The main omission was that the response did not clearly emphasize that the Barenblatt solution should be interpreted as a weak solution.

At the moving boundary, classical derivatives may fail to exist or fail to be continuous.

Therefore, the equation

$$\partial_t \rho = \Delta(\rho^m)$$

does not need to hold pointwise in the classical sense at every point of the free boundary.

Instead, the solution satisfies the porous medium equation in the weak sense.

This is an important mathematical distinction.

Why the Lack of Smoothness Is Not a Defect

The porous medium equation can be written as

$$\partial_t \rho = \nabla \cdot \left(m \rho^{m-1} \nabla \rho \right).$$

The effective diffusion coefficient is

$$D(\rho) = m \rho^{m-1}.$$

Since

$$m > 1,$$

we have

$$D(\rho) \to 0$$

as

$$\rho \to 0^+.$$

Therefore, the equation becomes degenerate near the free boundary.

This degeneracy is precisely what allows the solution to have:

- compact support;
- a moving free boundary;
- finite-speed propagation;
- limited regularity at the interface.

Therefore, the loss of smoothness is not a defect.

It is a characteristic feature of nonlinear porous-medium diffusion.

Overall LLM Reliability

The LLM was useful for generating the scaling derivation, the coefficient k, numerical verification code, and the regularity discussion.

However, the mathematical claims still required independent verification.

The exponents α and β were checked using both scaling identities.

The coefficient k was verified using the PDE residual, and deliberately incorrect coefficients were shown to fail.

The numerical Barenblatt solution was checked for mass conservation, decreasing height, and expanding compact support.

Otto's formula was verified independently using finite differences.

Finally, the LLM's smoothness discussion required an important correction: the Barenblatt profile should be understood as a weak solution across the moving free boundary.

Overall, the LLM was a useful mathematical assistant, but its answers were reliable only after checking the formulas, numerical residuals, limiting behavior, and weak-solution interpretation.

Deliverables Summary

All seven requested deliverables are present in the report:

1. the LLM prompts used in Parts I, II, and VI;
2. the derivation of α and β;

3. the derivation and verification of k;
4. plots of the Barenblatt profiles, height, support radius, and mass;
5. a discussion of finite-speed propagation;
6. the Otto-calculus first variation and gradient-flow calculation;
7. a critique of the LLM's regularity discussion and mathematical reliability.

Conclusion

The Barenblatt solution of the porous medium equation

$$\partial_t \rho = \Delta(\rho^m), \qquad m > 1,$$

has the form

$$\rho(t,x) = t^{-\alpha}\left(C - k\frac{|x|^2}{t^{2\beta}}\right)_+^{1/(m-1)}$$

with

$$\alpha = \frac{d}{d(m-1)+2},$$

$$\beta = \frac{1}{d(m-1)+2},$$

and

$$k = \frac{m-1}{2m}\beta.$$

The numerical experiments were consistent with the self-similar scaling, the coefficient k, conservation of mass, the decreasing maximum height, and the expanding compact support.

The support radius satisfies

$$R(t) = \left(\frac{C}{k}\right)^{1/2} t^{\beta},$$

which remains finite for every finite positive time.

Therefore, the porous medium equation exhibits finite-speed propagation.

The equation is also the Wasserstein gradient flow of

$$\mathcal{E}(\rho) = \frac{1}{m-1}\int \rho^m$$
$$, dx.$$

Finally, the Barenblatt solution is smooth inside its positive region but generally loses smoothness at its moving free boundary.

It should therefore be interpreted as a weak solution, and this limited regularity is a natural feature of nonlinear degenerate diffusion rather than a defect of the solution.

Homework 4.2: An Explicit JKO Step for the Heat Equation

Introduction

In this homework, I study the Jordan-Kinderlehrer-Otto (JKO) scheme for the heat equation.

Consider the entropy functional

$$\mathcal{E}(\rho) = \int_{\mathbb{R}} \rho(x) \log \rho(x)\, dx.$$

Its Wasserstein gradient flow is the heat equation

$$\partial_t \rho = \partial_{xx} \rho.$$

Assume that the initial density is a centered Gaussian

$$\rho^0 = \mathcal{N}(0, \sigma_0^2).$$

Since the heat equation preserves Gaussianity, I restrict the JKO scheme to centered Gaussian densities

$$\rho = \mathcal{N}(0, \sigma^2).$$

This reduces the infinite-dimensional Wasserstein minimization problem to a scalar optimization problem involving the standard deviation σ.

The main goals of this homework are to:

- compute the entropy of a centered Gaussian;
- compute the Wasserstein distance between centered Gaussians;
- derive an explicit JKO update for σ;
- compare the discrete JKO evolution with the exact heat equation;
- investigate convergence as the time step τ decreases;
- evaluate the mathematical reliability of an LLM-generated solution.

Part I: Entropy of a Gaussian

Let

$$\rho = \mathcal{N}(0, \sigma^2).$$

Its density is

$$\rho(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

The entropy functional used in the JKO scheme is

$$\mathcal{E}(\rho) = \int_{\mathbb{R}} \rho(x) \log \rho(x)\, dx.$$

Derivation

Taking the logarithm of the Gaussian density gives

$$\log \rho(x) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{x^2}{2\sigma^2}.$$

Therefore,

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi\sigma^2) \int_{\mathbb{R}} \rho(x)\, dx - \frac{1}{2\sigma^2} \int_{\mathbb{R}} x^2 \rho(x)\, dx.$$

Since ρ is a probability density,

$$\int_{\mathbb{R}} \rho(x)\, dx = 1.$$

Also,

$$\int_{\mathbb{R}} x^2 \rho(x)\, dx = \sigma^2.$$

Thus,

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2}.$$

Since

$$\frac{1}{2} = \frac{1}{2}\log e,$$

we obtain

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e\sigma^2)}.$$

Expanding the logarithm gives

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e) - \log \sigma.$$

Therefore, up to constants independent of σ,

$$\boxed{\mathcal{E}(\rho) = -\log \sigma.}$$

Numerical Verification

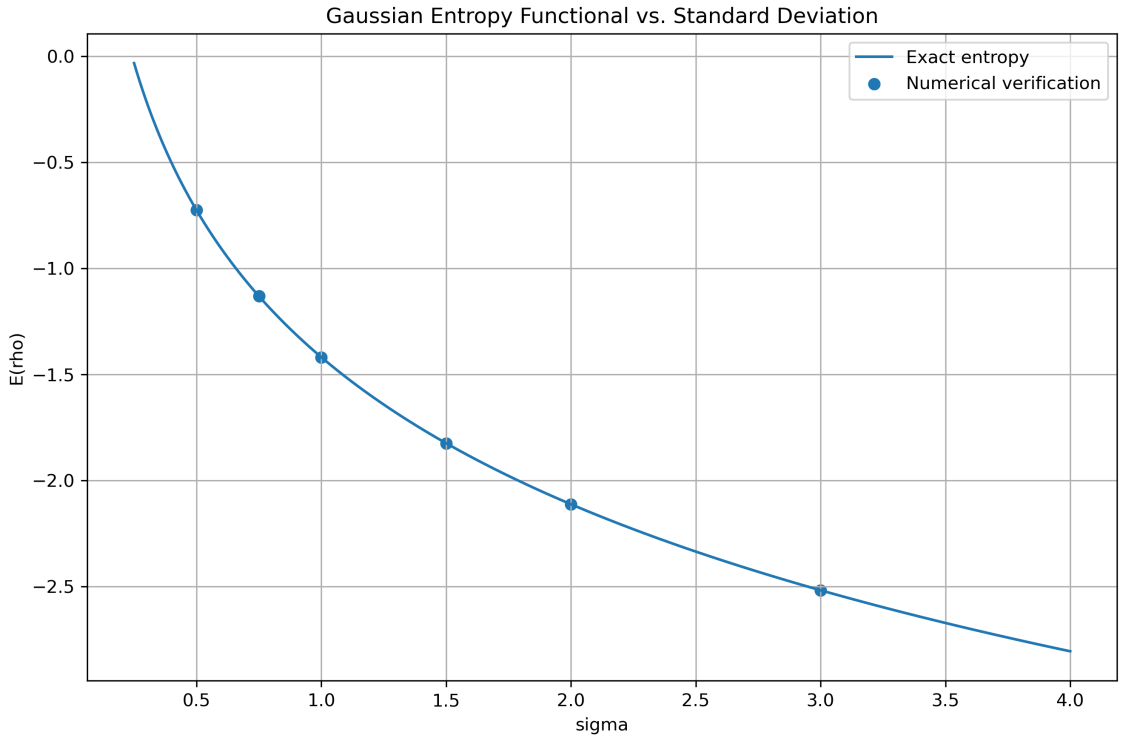
The entropy formula was verified numerically for several values of σ.

For each Gaussian, I checked:

- normalization;
- the second moment;
- the full entropy functional;
- the reduced entropy after removing the constant independent of σ.

The numerical entropy values lie on the theoretical curve

$$-\frac{1}{2}\log(2\pi e\sigma^2).$$

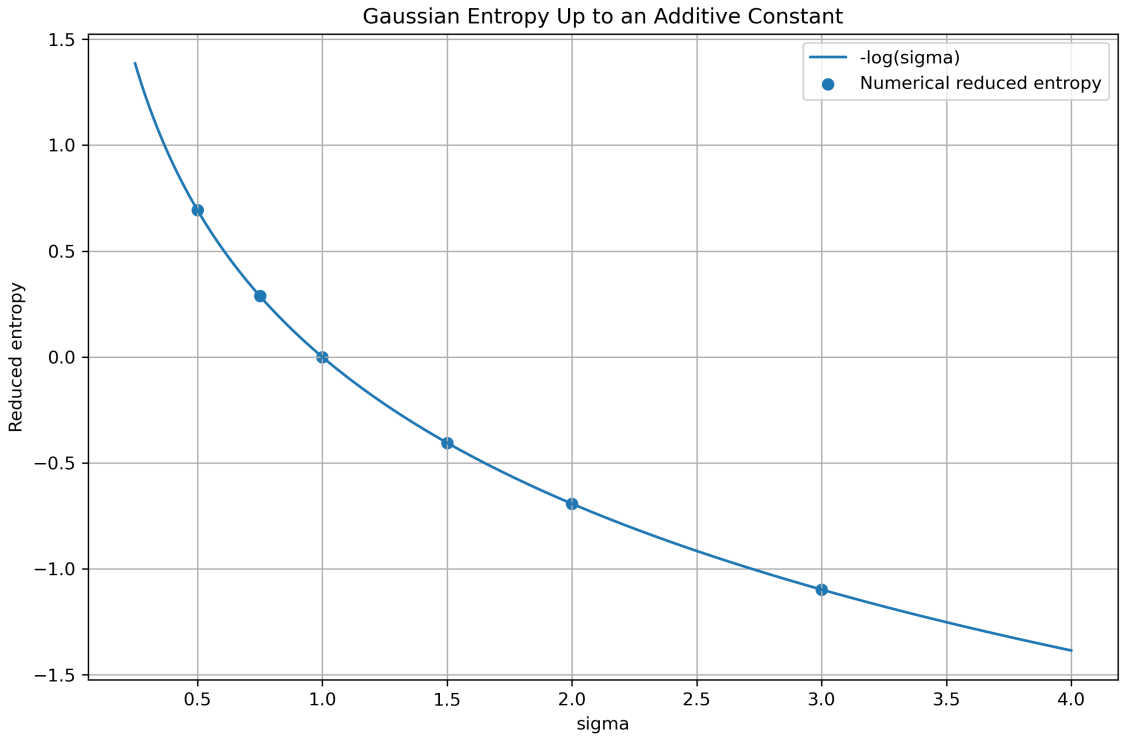


After removing the constant

$$-\frac{1}{2}\log(2\pi e),$$

the remaining quantity agrees with

$$-\log \sigma.$$



Since

$$\frac{d\mathcal{E}}{d\sigma} = -\frac{1}{\sigma} < 0,$$

the functional decreases as the Gaussian spreads.

The functional

$$\int \rho \log \rho \, dx$$

is the negative of the usual differential entropy.

Part II: Wasserstein Distance Between Centered Gaussians

Consider

$$\mu = \mathcal{N}(0, \sigma^2)$$

and

$$\nu = \mathcal{N}(0, \sigma_k^2).$$

The goal is to compute

$$W_2^2(\mu, \nu).$$

Quantile Derivation

In one dimension,

$$W_2^2(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(u) - F_\nu^{-1}(u) \right|^2 du.$$

Let Φ denote the standard Gaussian CDF.

For a centered Gaussian,

$$F_\mu^{-1}(u) = \sigma \Phi^{-1}(u)$$

and

$$F_\nu^{-1}(u) = \sigma_k \Phi^{-1}(u).$$

Therefore,

$$W_2^2(\mu, \nu) = (\sigma - \sigma_k)^2 \int_0^1 [\Phi^{-1}(u)]^2 du.$$

If

$$U \sim \text{Uniform}(0, 1),$$

then

$$\Phi^{-1}(U) \sim \mathcal{N}(0, 1).$$

Hence,

$$\int_0^1 [\Phi^{-1}(u)]^2 du = 1.$$

Thus,

$$\boxed{W_2^2 \left(\mathcal{N}(0, \sigma^2), \mathcal{N}(0, \sigma_k^2) \right) = (\sigma - \sigma_k)^2.}$$

Equivalently,

$$\boxed{W_2 = |\sigma - \sigma_k|.}$$

The formula depends on the difference between the standard deviations, not the difference between the variances.

Optimal Transport Map

The monotone optimal map from

$$\mathcal{N}(0, \sigma_k^2)$$

to

$$\mathcal{N}(0, \sigma^2)$$

is

$$T(x) = \frac{\sigma}{\sigma_k} x.$$

Then

$$W_2^2 = \int_{\mathbb{R}} |T(x) - x|^2 \rho_k(x)\, dx.$$

Since

$$T(x) - x = \left(\frac{\sigma}{\sigma_k} - 1 \right) x,$$

we get

$$W_2^2 = \left(\frac{\sigma - \sigma_k}{\sigma_k} \right)^2 \int_{\mathbb{R}} x^2 \rho_k(x)\, dx.$$

Using

$$\int_{\mathbb{R}} x^2 \rho_k(x)\, dx = \sigma_k^2,$$

we again obtain

$$\boxed{W_2^2 = (\sigma - \sigma_k)^2.}$$

Numerical Verification

The formula was checked numerically using both:

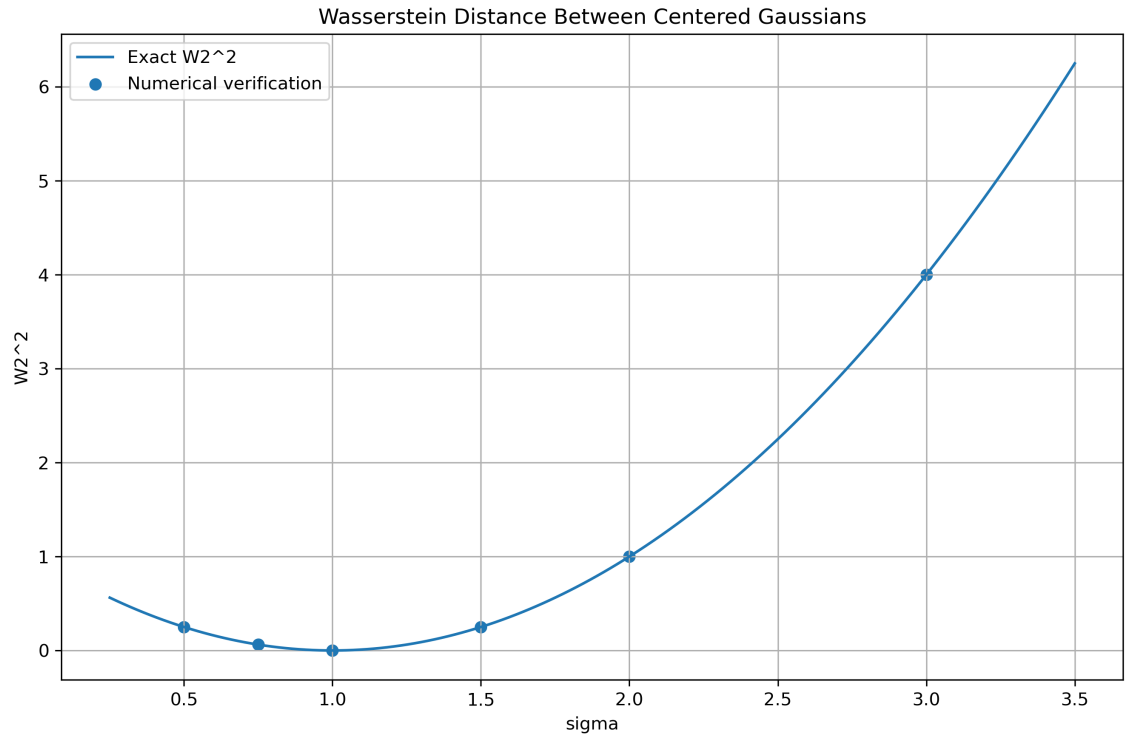
- the optimal transport map;
- the quantile representation.

For fixed

$$\sigma_k = 1,$$

the theoretical curve is

$$W_2^2 = (\sigma - 1)^2.$$



The numerical values lie directly on the theoretical parabola.

Thus the numerical computation confirms

$$\boxed{W_2^2 = (\sigma - \sigma_k)^2.}$$

Part III: The JKO Minimization

From Parts I and II, the JKO step becomes

$$\sigma_{k+1} = \arg\min_{\sigma>0} \left\{ -\log\sigma + \frac{1}{2\tau}(\sigma - \sigma_k)^2 \right\}.$$

Define

$$J(\sigma) = -\log\sigma + \frac{1}{2\tau}(\sigma - \sigma_k)^2.$$

First-Order Condition

Differentiate:

$$J'(\sigma) = -\frac{1}{\sigma} + \frac{\sigma - \sigma_k}{\tau}.$$

At the minimizer,

$$J'(\sigma) = 0.$$

Therefore,

$$-\frac{1}{\sigma} + \frac{\sigma - \sigma_k}{\tau} = 0.$$

Multiplying by $\tau\sigma$ gives

$$\sigma(\sigma - \sigma_k) = \tau.$$

Thus,

$$\sigma^2 - \sigma_k\sigma - \tau = 0.$$

Using the quadratic formula,

$$\sigma = \frac{\sigma_k \pm \sqrt{\sigma_k^2 + 4\tau}}{2}.$$

Because

$$\sqrt{\sigma_k^2 + 4\tau} > \sigma_k,$$

the negative root satisfies

$$\frac{\sigma_k - \sqrt{\sigma_k^2 + 4\tau}}{2} < 0.$$

Since σ is a standard deviation,

$$\sigma > 0.$$

Therefore, the negative root must be discarded.

The JKO update is

$$\boxed{\sigma_{k+1} = \frac{\sigma_k + \sqrt{\sigma_k^2 + 4\tau}}{2}.}$$

Convexity

The second derivative is

$$J''(\sigma) = \frac{1}{\sigma^2} + \frac{1}{\tau}.$$

For

$$\sigma > 0, \qquad \tau > 0,$$

we have

$$J''(\sigma) > 0.$$

Therefore, the objective is strictly convex, and the positive critical point is the unique global minimizer.

Numerical Verification

The explicit formula was compared with an independent numerical optimization for several values of σ_k and τ.

The maximum difference between the explicit formula and the numerical optimizer was

$$2.39 \times 10^{-9}.$$

The largest value of

$$|J'(\sigma_{k+1})|$$

was

$$1.45 \times 10^{-14}.$$

The maximum residual in

$$\sigma_{k+1}^2 - \sigma_k \sigma_{k+1} - \tau = 0$$

was

$$5.27 \times 10^{-16}.$$

The smallest computed value of

$$J''(\sigma_{k+1})$$

was

$$1.09167 > 0.$$

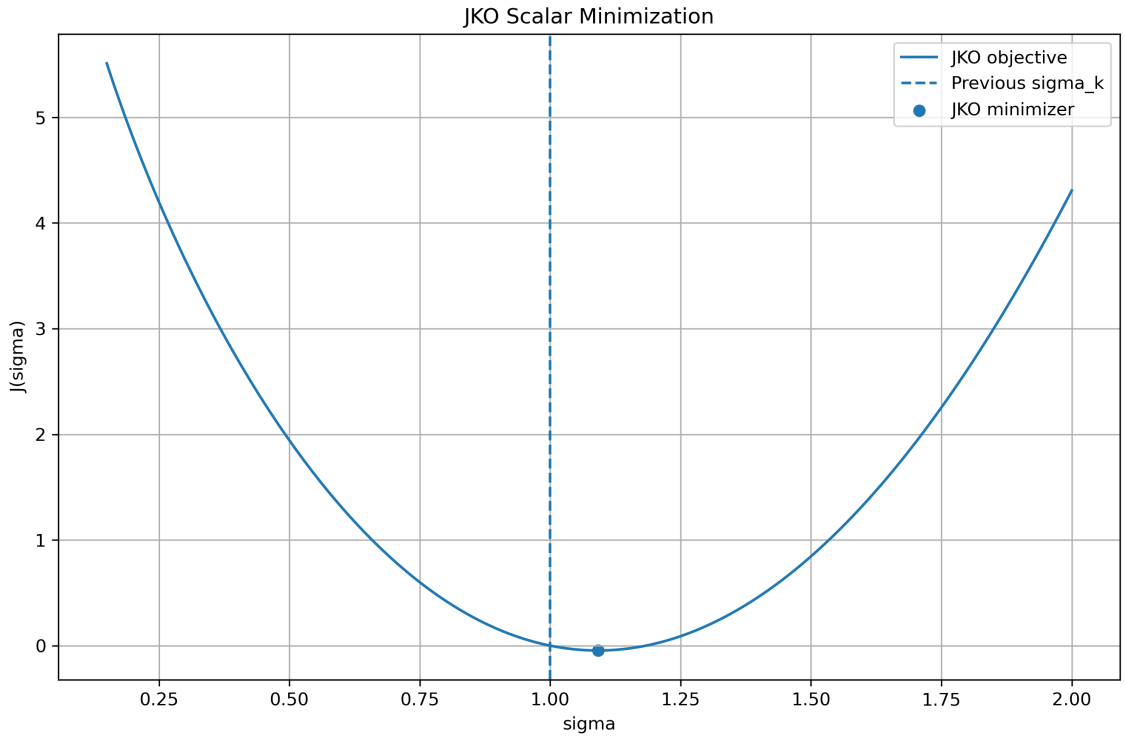
Thus the analytic minimizer is strongly verified.

For the example

$$\sigma_k = 1, \qquad \tau = 0.1,$$

the minimizer is

$$\sigma_{k+1} \approx 1.091608.$$

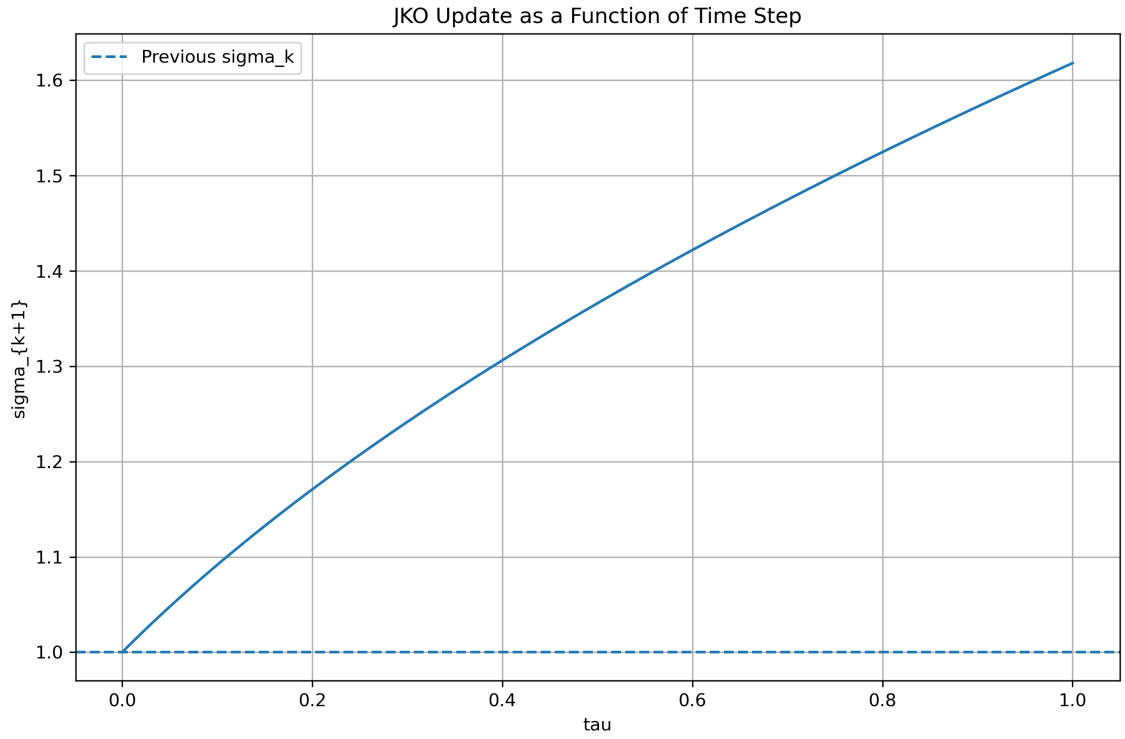


The minimum lies to the right of

$$\sigma_k = 1,$$

showing that the Gaussian spreads after one JKO step.

For fixed $\sigma_k = 1$, the update increases as τ increases.



The optimality condition can also be written as

$$\boxed{\sigma_{k+1} - \sigma_k = \frac{\tau}{\sigma_{k+1}}.}$$

Thus every JKO step satisfies

$$\sigma_{k+1} > \sigma_k.$$

Part IV: Compare with the Heat Equation

The exact heat equation solution starting from

$$\rho^0 = \mathcal{N}(0, \sigma_0^2)$$

is

$$\rho(t) = \mathcal{N}\left(0, \sigma_0^2 + 2t\right).$$

Therefore,

$$\boxed{\sigma^2_{\text{exact}}(t) = \sigma_0^2 + 2t.}$$

At the JKO time levels

$$t_k = k\tau,$$

the exact variance is

$$\boxed{\sigma_0^2 + 2k\tau.}$$

Variance Increment of the JKO Scheme

From Part III,

$$\sigma_{k+1} - \sigma_k = \frac{\tau}{\sigma_{k+1}}.$$

Then

$$\sigma_{k+1}^2 - \sigma_k^2 = (\sigma_{k+1} - \sigma_k)(\sigma_{k+1} + \sigma_k).$$

Therefore,

$$\sigma_{k+1}^2 - \sigma_k^2 = \frac{\tau}{\sigma_{k+1}}(\sigma_{k+1} + \sigma_k).$$

Since

$$\sigma_k = \sigma_{k+1} - \frac{\tau}{\sigma_{k+1}},$$

we obtain

$$\boxed{\sigma_{k+1}^2 - \sigma_k^2 = 2\tau - \frac{\tau^2}{\sigma_{k+1}^2}.}$$

The exact heat equation increases the variance by

$$2\tau.$$

Therefore,

$$\sigma_{k+1}^2 - \sigma_k^2 < 2\tau.$$

This shows that the finite-step JKO scheme slightly underestimates the exact variance growth.

Numerical Comparison

For the numerical experiment, I chose

$$\sigma_0 = 1$$

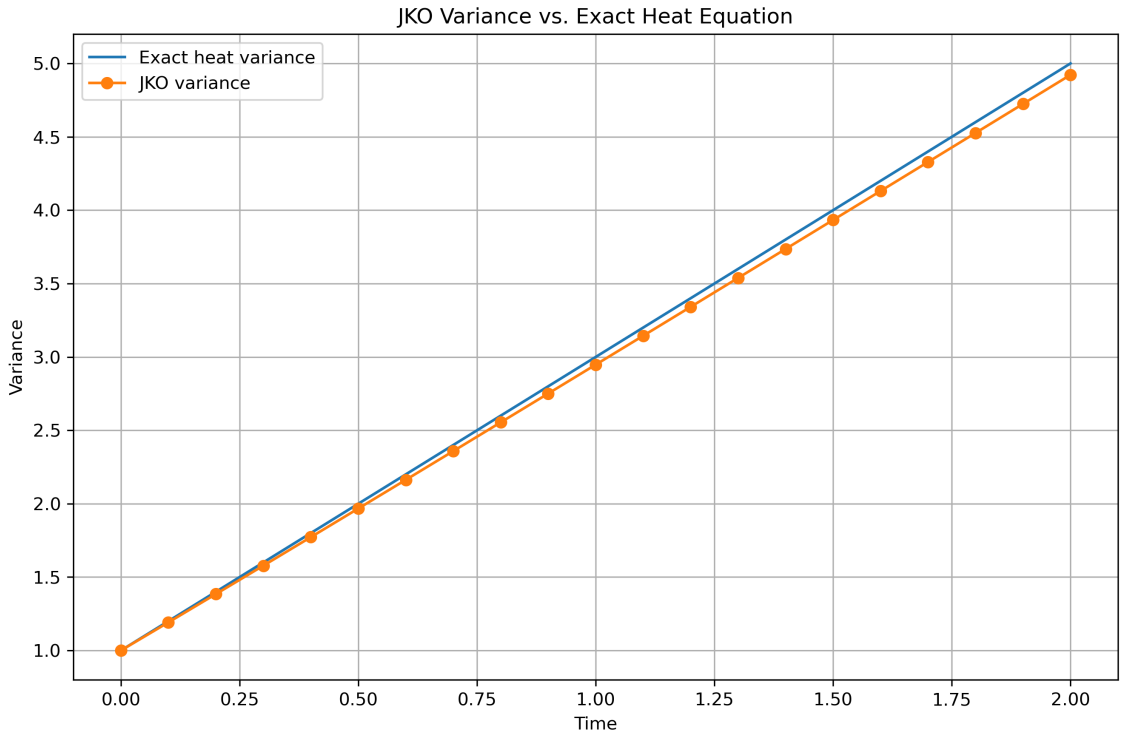
and

$$T = 2.$$

For

$$\tau = 0.1,$$

the JKO variance closely follows the exact heat equation.



The JKO curve remains slightly below the exact variance, as predicted analytically.

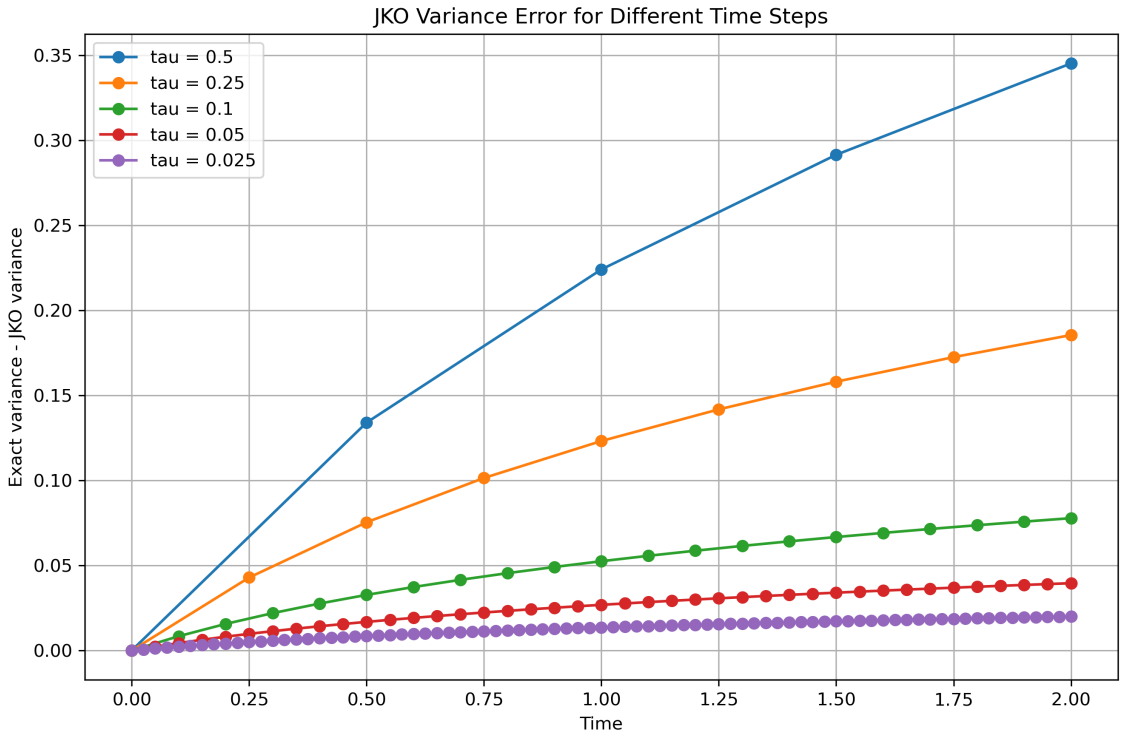
The experiment was repeated for

$$\tau = 0.5,\ 0.25,\ 0.1,\ 0.05,\ 0.025.$$

The final-time results were:

τ	Steps	JKO variance	Exact variance	Error
0.500	4	4.654848	5.000000	0.345152
0.250	8	4.814548	5.000000	0.185452
0.100	20	4.922209	5.000000	0.077791
0.050	40	4.960449	5.000000	0.039551
0.025	80	4.980055	5.000000	0.019945

The error decreases consistently as τ becomes smaller.



The numerical variance-increment identity was verified with maximum residual

$$2.11 \times 10^{-15}.$$

Convergence Rate

The observed convergence orders were

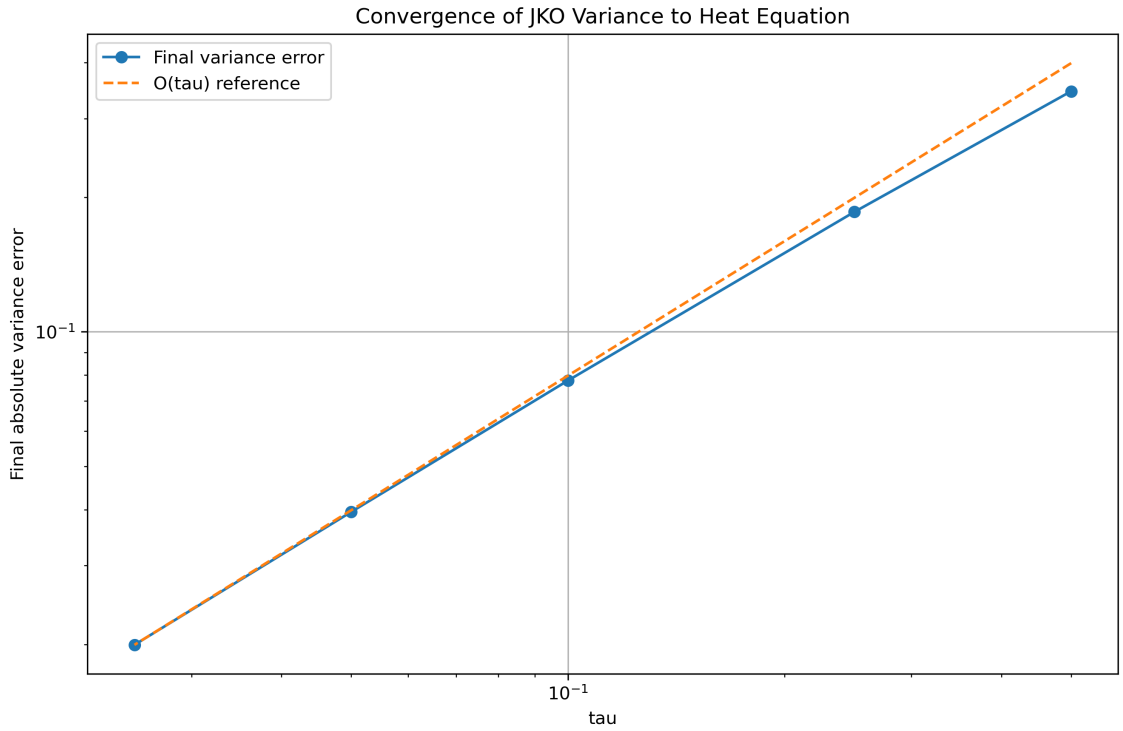
$$0.896, \qquad 0.948, \qquad 0.976, \qquad 0.988.$$

These approach

$$1$$

as

$$\tau \to 0.$$



Therefore, the numerical experiment indicates approximately first-order convergence:

$$\boxed{\text{error} = O(\tau).}$$

This is also consistent with the optimality condition

$$\frac{\sigma_{k+1} - \sigma_k}{\tau} = \frac{1}{\sigma_{k+1}},$$

which is the backward Euler discretization of

$$\frac{d\sigma}{dt} = \frac{1}{\sigma}.$$

Multiplying by 2σ gives

$$\frac{d}{dt}\sigma^2 = 2,$$

so

$$\sigma^2(t) = \sigma_0^2 + 2t.$$

Thus the JKO iteration converges to the exact Gaussian heat flow as

$$\tau \to 0.$$

Part V: LLM Critique

Prompt Given to the LLM

> Solve Homework 4.2 for a centered Gaussian.
>
> Compute the Gaussian entropy functional, derive the Wasserstein distance between centered Gaussians, solve the scalar JKO minimization explicitly, and compare the resulting iteration with the exact heat equation.
>
> Be careful to distinguish between the standard deviation σ and variance σ^2.

The LLM-generated response was checked according to the criteria in the assignment.

1. Gaussian Entropy

The LLM correctly obtained

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e\sigma^2)}.$$

It also correctly simplified this, up to a constant independent of σ, to

$$\boxed{\mathcal{E}(\rho) = -\log\sigma.}$$

This agrees with both the analytic derivation and numerical verification from Part I.

Therefore, the Gaussian entropy calculation is correct.

2. Wasserstein Distance

The LLM correctly used the one-dimensional quantile formula and obtained

$$\boxed{W_2^2\left(\mathcal{N}(0,\sigma^2),\mathcal{N}(0,\sigma_k^2)\right) = (\sigma-\sigma_k)^2.}$$

This agrees with the independent optimal-map and quantile calculations from Part II.

Therefore, the Wasserstein calculation is correct.

3. Positive Root for σ_{k+1}

The LLM correctly derived

$$\sigma^2 - \sigma_k\sigma - \tau = 0.$$

It then obtained

$$\sigma = \frac{\sigma_k \pm \sqrt{\sigma_k^2 + 4\tau}}{2}.$$

The negative root is less than zero and therefore cannot represent a standard deviation.

The LLM correctly selected

$$\boxed{\sigma_{k+1} = \frac{\sigma_k + \sqrt{\sigma_k^2 + 4\tau}}{2}.}$$

It also checked

$$J''(\sigma) > 0,$$

showing that this is the unique minimizer.

Therefore, the JKO minimization is correct.

4. Distinction Between σ and σ^2

The LLM does not confuse the standard deviation

$$\sigma$$

with the variance

$$\sigma^2.$$

It correctly uses

$$(\sigma - \sigma_k)^2$$

for the squared Wasserstein distance.

It then correctly compares

$$\sigma_k^2$$

with the exact heat-equation variance

$$\sigma_0^2 + 2k\tau.$$

Thus the distinction between standard deviation and variance is maintained throughout the derivation.

5. Sign and Behavior of the Entropy

The LLM correctly observes that

$$\frac{d\mathcal{E}}{d\sigma} = -\frac{1}{\sigma} < 0.$$

Therefore, the functional decreases as the Gaussian spreads.

The LLM also correctly notes that

$$\int \rho \log \rho \, dx$$

is the negative of the usual differential entropy.

One useful clarification made by the LLM is that this functional is not negative for every possible value of σ.

Indeed,

$$\mathcal{E}(\rho) < 0$$

when

$$2\pi e \sigma^2 > 1.$$

For sufficiently small σ, the functional can be positive.

Thus the most important general statement is not that it is always negative, but rather that

$$\boxed{\mathcal{E}(\rho) \text{ decreases as } \sigma \text{ increases.}}$$

Minor Issue in the LLM Response

The convergence statement

$$\sigma_k^2 \to \sigma_0^2 + 2k\tau \qquad \text{as } \tau \to 0$$

should be interpreted more carefully.

When taking

$$\tau \to 0,$$

the number of steps k should increase so that the physical time

$$t = k\tau$$

remains fixed.

A more precise statement is:

$$k\tau \to t$$

and

$$\sigma_k^2 \to \sigma_0^2 + 2t.$$

This is a minor issue of mathematical precision and does not affect the main derivation.

Overall Assessment of the LLM

The LLM performed well on this problem.

It correctly:

- computed the Gaussian entropy;
- computed the Wasserstein distance;
- derived the JKO minimizer;
- selected the physically meaningful positive root;
- distinguished standard deviation from variance;
- recognized that the entropy functional decreases as the Gaussian spreads;
- connected the JKO iteration to the exact heat equation.

The main formulas were also independently verified numerically.

Therefore, the LLM was mathematically reliable for this problem, although the convergence statement benefited from a small clarification concerning the limit

$$\tau \to 0.$$

Conclusion

For a centered Gaussian

$$\rho = \mathcal{N}(0, \sigma^2),$$

the entropy functional is

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e \sigma^2)}.$$

Up to constants independent of σ,

$$\boxed{\mathcal{E}(\rho) = -\log \sigma.}$$

The squared Wasserstein distance between two centered Gaussians is

$$\boxed{W_2^2\left(\mathcal{N}(0, \sigma^2), \mathcal{N}(0, \sigma_k^2)\right) = (\sigma - \sigma_k)^2.}$$

The JKO minimization therefore reduces to

$$\arg\min_{\sigma>0}\left\{-\log\sigma+\frac{1}{2\tau}(\sigma-\sigma_k)^2\right\},$$

whose unique minimizer is

$$\boxed{\sigma_{k+1}=\frac{\sigma_k+\sqrt{\sigma_k^2+4\tau}}{2}.}$$

The exact heat equation satisfies

$$\boxed{\sigma^2(t)=\sigma_0^2+2t.}$$

For finite τ, the JKO variance grows slightly more slowly than the exact heat-equation variance.

However, the numerical experiments show that the error decreases approximately linearly with τ.

The observed convergence orders approach

$$1,$$

confirming approximately first-order convergence.

Therefore, the explicit Gaussian JKO scheme provides a consistent discrete approximation of the Wasserstein gradient flow of entropy, and in the limit

$$\tau\to 0,$$

it reproduces the exact Gaussian solution of the heat equation.